



Review article

A thorough review of aircraft landing operation from practical and theoretical standpoints at an airport which may include a single or multiple runways

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ABSTRACT

The organization of landing operation and related issues within an Air Traffic Control (ATC) system in the airport Terminal Maneuvering Area (TMA) are hitherto performed by human air traffic controllers. These latter skilled buffers are responsible for aligning and controlling a continuous inbound flow of aircraft. These tasks are beginning from the time of notification of the getting of arrivals to the TMA entry fixes until the end of landings on the assigned runway(s). The whole duties carried out by air traffic controllers, such as arrival control, sequencing, scheduling, and assignment solutions are frequently implemented according to the information shown on various instruments, and some simple rules such as First Come First Served (FCFS) discipline. It is also worth noting that the training and past experience of the air traffic controllers are playing a leading role, and are essential part of decision-making processes. Since the demand for air traffic has continued to rise and with the emergence of congested airports, it became clear that the mission entrusted to the air traffic controller and its ability to undertake in its job in the right way has become increasingly more exacting. Consequently, there has been a greater demand to make the ATC more automated by improving aircraft scheduling/sequencing techniques, and developing further decision support systems, to which the challenge intends to alleviate the human intervention and not to replace the air traffic controller. Here of, the core goal is to provide a system review of past and the most up-to-date research related to the Aircraft Landing Problem (ALP) for an airport, which may include a single or multiple runways. Recognizing that the theoretical aspect relating to the ALP is dealt with this study, the evaluation of several solutions for that specific problem is furthermore covered. This evaluation will not succeed without explaining first the landing operation from a practical standpoint, which is a fundamental way to better understand the reasons why research effort has been devoted to suggest solutions for the overall arrival management. A comparative study which selects the distinguishing features that can be recognized in some research works against other related works is then proposed. In the last, a summary of the findings of published literature and recommendations for the future trend of ALP are pointed up as conclusions.

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1. Introduction

1.1. Research background and objective

The air traffic demand has seen a remarkable plethora in recent epoch. The introduction of low-cost carriers in Western Countries and remodeling of airport–airline relationships [1] count among the factors that contributed dramatically to its rising all over the world. In addition to this steep increase, their foreseeable growth recorded in the last decade remains positive, seeing that the traffic flows between airports are expected to boost between twice and triple in the prior years. The latter hypothesis is supported from various sources believed to be reliable. First and foremost, the Federal Aviation Administration (FAA) and aeronautical industries are predicting a growth between 150% and 250% over the next two decades [2]. Indeed, the further example presented by Bertsimas and Frankovich [3] attested that the total number of air traffic operations at combined FAA and contract toward airports, is estimated to increase from 6.1 million in 2006 to 81.1 million by 2020, and 95.9 million by 2030. Second, on its part, the Airport Council International (ACI) indicated that in 2012, over 6 billion passengers on domestic and international flights were reported worldwide. By 2025, it is anticipated that this number will increase by 50% at least [4]. Finally, *Alexander de Juniac*¹ confirmed the previous predictions, and announced in a recent television interview that the air passengers' number handled by airports worldwide is predicted to double to touch approximately 8.2 billion by 2037.

The growing traffic combined with the use of “Hub and Spoke” operations by air carriers' [5], on the one hand, and, the rapid

growth of Unmanned Aerial Vehicle² (UAV) traffic [6], on the other hand, causes many challenges. The highly congested airports (e.g., New York's *La Guardia airport* [7]) are classified as the most distinguished ones. Thereby, automatically, the capacity bottleneck in air traffic system will become airports, where their capacity is nearly saturated or even overloaded, as the demand rate approaches or may temporarily attain the service rate. This means that the demand in some airports attained the Airport Acceptance Rate (AAR) or the Operationally Acceptable Level of Traffic (OALT) sector. An interesting and highly symbolic example is that of *China*, when at peak traffic, the volume at their seven busiest airports exceeds their capacities by more than 30% [8]. It is argued that if the demand for air transport keep growing as predicted [9,10], and in the absence of effective policies, the air traffic congestion, which is currently a persistent phenomenon in ATC system, will worsen in the future. However, all signs, including the real-world statistical data, point to persisted strong growth of air traffic congestion.

To begin with, it is necessary to stress that the acute air traffic congestion is admitted as one of the prime trigger for delays that will have adversely impact on airline operation costs [11]. For example, ground and airborne delays cost the *United States* airline industry an estimated 5.5 billion dollars annually [12]. Moreover, such frequent and lengthy delays which spread over the network and make it vulnerable, have already placed passenger discomfort that leads to complaints toward airlines and airport staff [13]. We must not forget that delays bring adverse effects on the environment, pertaining to the extra fuel consumption burned by aircraft. Types of delay, included airborne delays that result from the congestion phenomena, were one of the most serious and widespread problems for the ATC system. About 60% of all

¹ Chief Executive Officer of the International Air Transport Association (IATA).

² UAV: aircraft without a human pilot aboard.

List of used abbreviations

AAR	Airport Acceptance Rate
ABC	Artificial Bee Colony
ABM	Agent-Based Model
ACA	Ant Colony Algorithm
ACDR	Aircraft Conflict Detection and Resolution
ACI	Airport Council International
ACO	Ant Colony Optimization
ACS	Ant Colony System
AGH	Arc Greedy Heuristic
ALP	Aircraft Landing Problem
ALS	Aircraft Landing Scheduling
AMAN	Arrival MANager
AMCC	Avoid Most Critical Completion Time
AMSP	Avoid Most Similar Pair
ANSP	Air Navigation Service Provider
ARTCC	Air Route Traffic Control Center
ASEP	Arrival SEquencing Problem
ASP	Arrival Scheduling Problem
ASS	Arrival Scheduling and Sequencing
ATC	Air Traffic Control
ATCC	Air Traffic Command Center
ATCT	Airport Traffic Control Tower
ATFM	Air Traffic Flow Management
ATM	Air Traffic Management
ATP	Aircraft Take-off Problem
ATS	Air Traffic Services
BA	Bionomic Algorithm
B&B	Branch and Bound
B&C	Branch and Cut
BCIA	Beijing Capital International Airport
CA	Cellular Automata
CAA	Civil Aviation Authority
CAO	Cellular Automata Optimization
CCG	Complementary Column Generation
CDA	Continuous Descent Approach
CDM	Collaborative Decisions Making
GDP	Ground Delays Program
CNS	Communication Navigation and Surveillance
COMPAS	Computer Oriented Metering Planning and Advisory System
COP	Combinatorial Optimization Problem
CPS	Constrained Position Shifting
CTAS	Center TRACON Automation System
DP	Dynamic Programming
DSSE	Dynamic Sequence Searching and Evaluation
DST	Decision Support Tool
EABC	Efficient Artificial Bee Colony
EDA	Estimation Distribution Algorithm
EF	Entry Fix
ELT	Earliest Landing Time
EO	Extremal Optimization
EPS	Explorative Perturbative Search
ETAs	Estimated Time of Arrivals

ETMA	Extended Terminal Manoeuvring Area
FAA	Federal Aviation Administration
FAF	Final Approach Fix
FAST	Final Approach Spacing Tool
FCFS	First Come First Served
FCFS-RW	First Come First Served-Runway
FCFS-SE	First Come First Served- System Entrance
FIFO	First In First Out
FLA	Failed to Land Aircraft
FMS	Flight Management System
FPA	Flower Pollination Algorithm
GA	Genetic Algorithm
GAs	Genetic Algorithms
GDP	Ground Delays Program
GH	Ground Holding
GLS	Genetic Local Search
HABC	Hybrid Artificial Bee Colony
HC	Hill Climbing
HNN	Hopfield Neural Network
HP	Holding Pattern
HPs	Holding Patterns
IATA	International Air Transport Association
IAF	Initial Approach Fix
ICAO	International Civil Aviation Organization
ID	Iterated Descent
IE	Implicit Enumeration
IFR	Instrument Flight Rules
ILS	Iterated Local Search
IMC	Instrument Meteorological Conditions
IP	Integer Programming
IPDS	Interior Point Dual Stabilization
IRHA	Iterative Rolling Horizon Approach
JPS	Jackson Preemptive Schedule
JGH	Job Greedy Heuristic
JSS	Job Shop Scheduling
LDD	Lagrangian Dual Decomposition
LFSS	Least Fuel First Service
LLT	Latest Landing Time
LMIS	Linear Matrix InequalityS
LP	Linear Programming
MABC	Modified Artificial Bee Colony
MAESTRO	Means to Aid Expedition and Sequencing of Traffic with Research of Optimization
MC	Monte Carlo
MHNN	Mutation Hopfield Neural Network
MILP	Mixed Integer Linear Program
MIP	Mixed Integer Programming
ML-PM	Multi Level-Point Merge
MOEAs	Multi-Objective Evolutionary Algorithms
MPS	Maximum Position Shifting
MTOW	Maximum Take-Off Weight
NATS	National Air Traffic Services
NN	Neural Network
NP	Nondeterministic Polynomial-time

NSGA- II	Non-dominated Sorting Genetic Algorithm II
OABC	Original Artificial Bee Colony
OALT	Operationally Acceptable Level of Traffic
OPD	Optimized Profile Descent
PLT	Preferred Landing Time
PMS	Point Merge System
PSO	Particle Swarm Optimization
RG	Randomized Greedy
RHA	Rolling Horizon Approach
RHC	Receding Horizon Control
RLT	Reformulation Linearization Technique
RPH	Rolling Planning Horizon
RPS	Relative Position Shifting
RSP	Runway Scheduling/Sequencing Problem
RT	Runway Threshold
SA	Simulated Annealing
SAA	Sample Average Approximation
SDP	Standard Descent Profile
SLT	Scheduled Landing Time
SS	Scatter Search
STA	Scheduled Time of Arrival
STAR	Standard Terminal Arrival Route
STARs	Standard Terminal Arrival Routes
STASS	Stochastic Terminal Area Scheduling Simulation
TA	Time Advance
TCA	Terminal Control Area
TDA	Time Decomposition Approach
TDTSP	Time-Dependent Traveling Salesman Problem
TIM	Time-Indexed Model
TLT	Target Landing Time
TLTs	Target Landing Times
TMA	Terminal Manoeuvring Area
TS	Tabu Search
TSP	Traveling Salesman Problem
TRACON	Terminal Radar Approach CONTROL
UAV	Unmanned Aerial Vehicle
VFS	Vector Flight Spacing
VFR	Visual Flight Rules
VI	Valid Inequalities
VMC	Visual Meteorological Conditions
VND	Variable Neighborhood Descent
VNS	Variable Neighborhood Search
VRP	Vehicle Routing Problem
WV	Wake Vortex
WVs	Wake Vortices

airborne delays occurred during the arrival phase [14]. These delays could reach record levels (i.e., several tens of minutes (min)) particularly at rush periods. In fact, they disturbed considerably the air transportation's order, and be considered the cause of significant economic losses from all the stakeholders' point of view. For instance, the costs of air traffic delays incurred by Air Traffic Flow Management (ATFM) (during take-off, flight, or landing) for all *European airports* was estimated to be as high as 1.25 billion euro in 2011 [15].

The first areas, casualties' of congestion and airborne delays within an Air Traffic Management (ATM) system, have been historically the capacity within a radius of about 50 nautical miles (NM) from an airport, namely the TMA surrounding the airport [16]. Indeed, this extremely complex airspace has become a sensitive area with an uncertain and dynamic environment [17]. This latter situation was mainly due to several aspects. It was basically up to the large amount of descending and ascending aircraft operated in this relatively small area in a brief period of time. Since the TMA has been highlighted as the outstanding key area, and remains the main bottleneck for the nation's ATC network in the airport [18], the aim of perfectly managing aircraft operations (i.e., landings and take-offs) on the TMA available resources, have become complex and difficult tasks in real-life. It is furthermore necessary to be reminded about the aforementioned negative impacts owing to air traffic congestion and airborne delays, which have become serious enough that they can no longer be ignored, mainly with the expected increase in the demand for civil aviation transportation. Thus, these facts stimulated the focus on the air transportation industry that is pushing for attempts to alleviate the near phenomena's. Improving aircraft operations within the TMA region around the busy airport has been proven among the promising solutions. Thus, the opportunity to modelize and optimize the landings and take-offs flows in the terminal airspace was tackled with proper emphasis by several researchers and engineers from different countries since the 1970s.

In this paper, we restrict our investigation to the aircraft landings at an airport which may include a single or multiple runways, referenced in the literature as the ALP. The aircraft landing operation, be considered as a crucial management issue in the terminal area of airports, have long been a hot issue and subject of many scientific publications up today (see. e.g., Hong et al. [17], Salhipour [19], Ben Messaoud and Ghedira [20], Zhang et al. [21]), confirming the importance of this aspect. Furthermore, on the one hand, the official regulations pays particular attention to that operation, which has gone from difficult to manage to a global problem in the air transport system. On the other hand, the aircraft landing operation has different importance from various researchers' perspectives. We also see that model and solution approaches provided in the literature differ from others in several respects. Accordingly, we present a detailed analysis of many studies that has mainly appeared during the past four decades. In the next content, we will deal with all of these points in greater detail.

1.2. Paper organization

The outline of this paper is as follows: a presentation of the aircraft landing management from practical standpoint, the landing operation identified hurdles and suggested solutions are explained in Section 2. The various techniques of modeling the problem, the objective functions, and the different operational constraints are described in the third part. The most distinguished classification for the ALP on a single (multiple) runway (s), and a review of the application of the different exact, approximate algorithms, and software systems are pointed out in Section 4. Then, a comparative study of related works is provided in Section 5. Finally, a discussion and the conclusion of the paper are raised in Section 6.

2. Aircraft landing management from a practical point of view

By establishing a set of services [22], the ATM system involves, organizes, and controls the flow of traffic on the ground and in the airspace around the airport [23], in a manner to ensure the safe and efficient movement of aircraft from the origin to

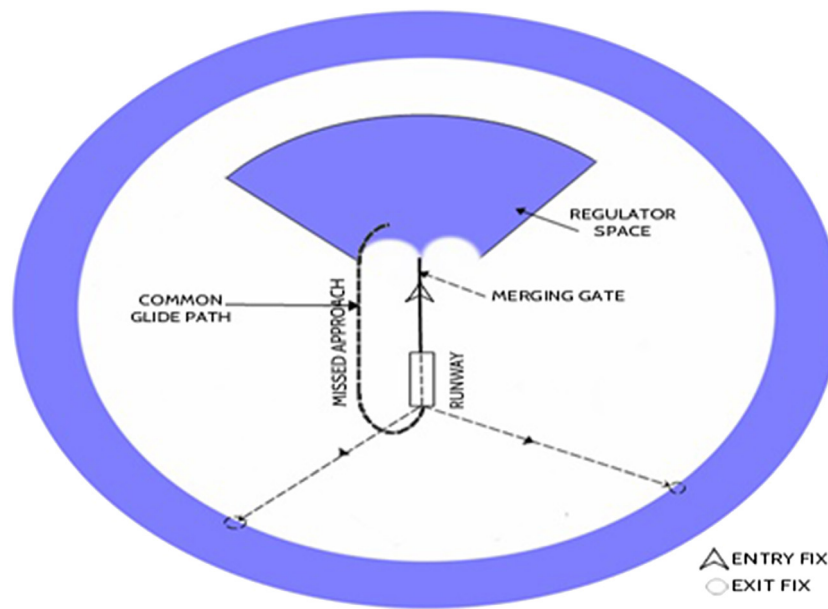


Fig. 1. Structure of the TMA.

the destination airport [24]. The current organization of the ATM system is viewed as consisting both tactical and strategic components [25]. The tactical component, ATC, is the complete control of air traffic, and is ultimately responsible for safety in particular the separation assurance between aircraft [26]. While the strategic component, ATFM, aims to ensure the overall unimpeded flow of aircraft through the airspace, so as to avoid congestion and delays. However, when delays are unavoidable, it reduces as much as possible their impact on airspace users [25].

2.1. Air traffic control system

Today, ATC accounts for the greatest percentage of Air Traffic Services³ (ATS) on a global basis, as it is exclusively responsible for the safety of the landing and departing aircraft, and has to execute the actual sequence [28]. Furthermore, the purpose of ATC in the airport TMA is to prevent collisions between aircraft, and between aircraft and obstructions, and to expedite and maintain an orderly flow of air traffic. All these functions will be explained in the sub-sections below.

2.1.1. Air traffic control system structure

The ATC system is characterized by a high number of variables and constraints; numerous sub-systems and strong interactions; limitations and complexity of models used to forecast aircraft movements and traffic evolution; as well as presence of several human operators in the system (e.g., pilots, air traffic controllers), each having certain independence on taking decisions [29]. We talk, here, about a complex system based on the division of the airspace in geographical areas of variable size known as “sectors”. A sector as pointed out by Ball et al. [25] is a volume of airspace for which a single ATC team of one or two individuals has responsibility. The principal constraint on the number of aircraft that can safely occupy a sector simultaneously is the air traffic controller workload. For more details concerning the capacity of a sector and the difficulty found in their computation, we refer the reader to Ball et al. [25].

Such sector unit represents the basic component in the hierarchical structure of the ATC. These sectors are managed by teams of air traffic controllers (a planning controller, an approach controller, and a radar controller). The flight responsibility passes from one sector team to the next [30]. Each one of them is responsible for organizing the traffic according to air routes and aircraft weight levels, and by limiting the freedom of each aircraft while facilitating the respect of aircraft separations, as well as allowing the commercial aviation to remain the safest means of transportation. In every sector, it is clear that the air traffic controller burden depends largely on the traffic density. Obviously, there is a limit to the number of aircraft that air traffic controllers can manage, monitor, and control in a sector.

The jurisdiction of ATC in the system is divided into these parts: oceanic & en-route ATC which is responsible for aircraft operations in the airspace, and airport TMA or Terminal Control Area (TCA) which is responsible for air traffic within the boundary area of 50 NM of an airport [9]. It must be emphasized that the TCA and approach sectors surrounding an airport, are perhaps the most difficult areas for ATC [26]. In our study, we have focused on the present proposal.

2.1.1.1. Terminal maneuvering area. The critical bottleneck in the nation's ATC network is the capacity of the radius region up to 40 NM around an airport (i.e., the terminal area) [31]. Terminal area is a transitional region between en-route and airfield [32]. It is a cylindrical region; wave defined with a radius of approximately 50 NM centered on major airports, and the altitude limit may be of the order of 10 000 ft (ft), as depicted in Fig. 1. The TMA forms the critical interface between the network of airways and the airport (surface), limited by the range of the radar and the cover of communication. At the boundary of the TMA, there are various way-points or fixes, labeled as entry (or meter) gates, through which traffic enters the center [16]. There are also exit points by which aircraft transition out of the center.

The management of the flow of aircraft at terminal areas is certainly one of the most critical aspects of overall ATM [14]. It is explained by the fact that this rendezvous reference of arriving and departing flights is a dynamic and uncertain environment, with constant updates on aircraft state being obtained from surveillance system and airline report [33]. Thus, continuous monitoring by human controllers is almost required. Once

³ ATS: generic term meaning variously, flight information service, alerting service, air traffic advisory service, air traffic control service [27].

more, this area is composed of resource for various types: air segment(s),⁴ common glide path,⁵ runway(s)⁶ and holding circle(s).⁷ It is widely recognized that the runway system, which is a highly constrained resource at major airports [8,36], is the critical bottleneck in a nation's ATC networks [37,38].

2.1.1.2. Runway system. The runway system is the main element that combines airside and ground side of the ATM system [39]. It directly influences the overall airport capacity [5]. Of critical importance, the capacity of airport runway system constitutes a crucial element. The declaration of this process requires a tradeoff between the entities of airport, the various airlines requesting usage of the runway at the airport, and the local provider (i.e., ATC). These parts are designed to achieve at the same time the use of infrastructure and the quality of service, in order to have a functioning being perceived locally acceptable for a given set of constraints. As stated by Bazargan et al. [40], the ACI and IATA guidelines for airport/demand management in 1996 defines the runway system capacity as the hourly rate of aircraft operations (i.e., expected number of landings and take-offs in one hour), which may be reasonably expected to be accommodated by a single or a combination of runways under given local conditions. This is often referred to as the maximum throughput (i.e., airport physical capacity).

The runway system declared capacity depends on the basis of many factors, such as the structure of the airport and surrounding airspace (i.e., TMA); the number and the configuration of runways (e.g., identical, parallel, independent); the runway occupancy; the typology of traffic; the ATC applicable rules; the performances of aircraft using the airport and their individual characteristics; the peak traffic; the meteorological conditions; the environmental considerations; the air traffic controllers workload; and the rules of spacing. All this indeed confirms the importance and the sensitivity of that resource in the air transportation system. Many major airports in the world, such as *Atlanta*, *Dallas Fort Worth*, *Denver*, *New York La Guardia*, *Shanghai Pudong*, and *London Heathrow*, have multiple runways. That was because it is impossible to accommodate all the traffic on a single runway. However, most airports where the traffic load is important have only one runway that is able to flow the predictable traffic, such as *Doha International airport* and the most airports in *China*. In the airport possessing a multi-runway configuration, there should be an air traffic controller responsible for the movement of each runway. This implies that an airport with multiple runways may be somewhat more convoluted than an airport with an only single runway.

The runways host and clear the aircraft's according to the wind direction and strength. Their operating configurations are two: the interdependent (segregated) and the heterogeneous (mixed) modes. For example, *Mexico City International*, *London Heathrow*, and *Dubai International airports* operate two interdependent runways; i.e., the operations on one runway restrict the operations on the other(s). In other words, one runway is dedicated exclusively for approaches and landings, and the other is dedicated exclusively for take-offs. On the other side, *Frankfurt airport* (the largest *German airport*) operates runways that are heterogeneous, i.e., not all operations can be performed on all runways [41]. The

former mode, on the one hand, is considered as the most advantageous [42,43], on the basis that landings have greater priority than take-offs due, among other things, to the higher average of fuel burning and safety measures [36]. This is also evidenced by the fact that this allocation strategy minimizes interactions between runways and reduces air traffic controllers' workload. But, on the other hand, even if it increases the degree of air traffic controller burden [1], the heterogeneous mode was observed as more efficient than the segregated mode, since alternating landings and take-offs on the runway is effective in reducing or eliminating delays due to Wake Vortex (WV) constraints [44]. Moreover, it enhances the capacity to handle airborne and airport traffic [1]. Finally, mention is made of the fact that aircraft operation type (i.e., landing or take-off) of dominant functionality at an airport plays a key role in the runway configuration choice. For example, the employment of the interdependent mode at *Heathrow London airport* is related to the fact that the number of arrival and departure flight is almost the same [36]. Another concrete example is that of *Beijing Capital International Airport* (BCIA), which has three runways operating normally heterogeneously: a runway used for departure and arrival, a runway used for departure, and the last one is mainly used for arrival. In the case of a dense traffic, the three runways could all be operated for landing [8].

2.1.2. Air traffic control authorities

The TMA of an airport is concerned with the complex aircraft operations close to large airports. These operations are usually considered under the umbrella of ATC. The latter consists of two control authorities: the Terminal Radar Approach CONtrol (TRACON) and the Airport Traffic Control Tower (ATCT) [9]. During the landing phase, the aircraft passes from the competence of en-route control center what is known as Air Route Traffic Control Center (ARTCC) that handle the traffic flow outside the TMA [22], to the terminal airspace control center (i.e., TRACON), then to the ATCT. The latter is the reference point of the airport; it is placed so as to be able to visually evaluate the aircraft on taxiways and runway(s). From the ATCT, and at within about 5 NM of the airport and 3000 ft above ground level from the airport [30], the air traffic controller scans the control panels and radar screens, and communicates to the pilot's information about the air traffic, available runway, and environmental conditions. Then, he guides aircraft in the airspace that he controls, and adjusts the flow of arriving aircraft according to the weather conditions, the choice of runway(s) to be used, and the activation of lighting the ground. It might be worthwhile to add that the TRACON is the set of installations that includes radar operations from which the air traffic controller supervises aircraft approaching and departing up to 40 NM and 10,000 ft from the airport [45].

2.1.3. Current practices for landings in the approach paths and runway(s)

2.1.3.1. Air traffic control decisions. The traffic sample is described by the flight plan data of one day at an airport. Each timetable gives all the information needed to reproduce the evolution of aircraft, including the aircraft type, the gate(s) and the runway(s) in-use, the landing time for arrival, etc [46]. Once the flight plan is available at the ATCT, every air traffic controller has to develop a plan to simultaneously support the guiding of a given number of aircraft in arrival (e.g., about 10 aircraft) over a period (e.g., about 10 min), from the time of their entrance into the TMA to their landing on the runway it is assigned to. There is a need to specify that this strategic activity is conducted several months in advance of the day of air operations. Furthermore, it will be necessary to update the plan according to real-time traffic information, that is done at the tactical level.

⁴ Air Segment(s): approach path, typically subdivided into three main segments (initial, intermediate, and final approach segments), defined on the basis of the three airspace point, the initial, intermediate, and final approach fixes [34].

⁵ Common glide path: final air segment before a single or multiple runways before landing [35].

⁶ Runway: defined rectangular area on a land aerodrome prepared for the landing and take-off of aircraft [27].

⁷ Holding circle: air segment dedicated to accumulating aircraft delays during the flight [35].

During the transition from the en-route to the terminal airspace, aircraft arriving from all the directions, descending through the surrounding airspace to enter from different Entry Fix (EF) points, the Extended Terminal Maneuvering Area (ETMA) of the destination airport. Upon entering within the air traffic controllers' radar range, these approach flights appear on the controller's radar screen, and their information including flight number, flight type, altitude, and speed are available at the ATCT [47]. Based on these information's, the air traffic controller has to make decisions so as to expedite the orderly, efficient, and safe of aircraft progress. Such determinations are made in a short time horizon, of about 45 min [48]. Typically, as stated by BenCheikh et al. [49], the ATC will be in charge over the radar horizon, of determining approach paths, runway allocation, and landing times of several aircraft, on the TMA resources which may include air segment(s) and runway(s). This effectively means that the air traffic controller has to provide the necessary pre-tactical instructions to the pilots, so as to keep aircraft moving smoothly along predefined routes, from the ETMA of the airport toward the prescribed Runway Threshold⁸ (RT), following a Standard Descent Profile (SDP). To do this, and in order to achieve reasonable decisions, the air traffic controller is supported by a set of Collaborative Decisions Making (CDM) paradigm on the operational level. Within the framework of controller's support procedures implemented at airports, we would promote the Final Approach Spacing Tool (FAST), which was originally conceived to provide suite advisory information for the TRACON sector controllers. With this beneficial tool, the air traffic controllers may easily be guaranteed balanced runways and sequenced arrival aircraft within 40 NM of and less than 10,000 ft altitude above the airport [50]. For a more detailed explanation of the FAST main features, see the paper of Desai and Prakash [16].

From decent to land, the flight phases could be divided into four segments: arrival descent, initial approach segment, intermediate approach segment, and final approach segment. In traditional method, arrival aircraft normally follow different Standard Terminal Arrival Routes⁹ (STARs) to Initial Approach Fix¹⁰ (IAF). Then, they are vectored by controllers to reach Final Approach Fix¹¹ (FAF). Finally, they sequentially land on the runway in-use [13]. Typically, the decisions encountered several times by the air traffic controller can be broadly divided into routing decisions, timing decisions, and runway assignment. *Routing decisions*, where for each aircraft, an origin–destination route has to be chosen regarding air segment(s) and runway(s), given the traffic regulation constraints. It is a spatial decision, as appointed by Kjenstad et al. [23]. Remark that appeared aircraft were given different routes to approach the runway for landing, and each one of them followed exactly one route [51]. Once a route has been assigned to each of the aircraft's (*Routing Decisions*), the ATC problem in the TCA (with pre-defined routes) is reduced to the Arrival Scheduling Problem (ASP) [52], where the aircraft passing timing has to be determined in each air segment, runway, and (eventually) holding circle (*Timing Decisions*) [53,54]. This means giving each of the aircraft's a scheduling time sometimes known as the broadcast time, which is defined in terms of the time that the aircraft could reach at its cruising speed the desired resource point (e.g., FAF of the air segment, RT of the runway). Finally, an “*allocated runway*” is accredited to each aircraft. The *Runway*

Assignment occurs when the air traffic controller checks the runway or the set of runways, which at some point are considered by the ATCT as the suitable ones to the aircraft type that has to land at the airport. This concern is a tactical decision depending on a number of factors, such as the airport configuration and the arriving aircraft direction (i.e., arrival feeder gate) [22]. The aircraft can be eligible for allocation to more than one runway for landing, and it must be clear that resource after a certain runway occupancy time¹² before the next aircraft is allowed to land. Notably, if there is more than one runway, the importance of runway assignment is clearly evidenced by a balance of the capacity if one runway gets overloaded while another is still able to accommodate more aircraft [18]. For further details about controller's preferences and the factors affected the process of runway allocation, refer to Isaacson et al. [55]. Elsewhere, Liang et al. [13] and Zhang et al. [21] shelled airline's preferences in the subject of runway assignment.

2.1.3.2. Scheduling/sequencing behavior of aircraft in arrival toward the runway(s). An aircraft at least 45 min far away from the destination airport [56], planned to fly from an EF to a prescribed RT along one of a pre-structured path (route), i.e., a sequence of air segments, following a SDP [57], have to be first scheduled, and then sequenced on the Instrumental Landing System¹³ path. The scheduling process is a typical task of the ATC [26], while the aim is to compute the scheduled arrival time which allows the aircraft to join the scheduling airspace point (e.g., aircraft's Scheduled Time of Arrival (STA) to fly to the metering point, the FAF of the final approach segment; aircraft's Scheduled Landing Time¹⁴ (SLT) to fly to the metering point, the RT of the runway), subject to maintain operational and safety constraints [30,60,61]. The scheduling process starts once the aircraft reaches the TMA, about 30–40 min before its planned landing time, and is finalized in advance of reaching the final approach path, about 20–30 min before landing [36]. So, once the aircraft appears at the radar screen, the arrival manager checks whether the aircraft's remaining time to fly to the prescribed reference (i.e., IAF, FAF, and RT) has become less than a specified minimum time, referred to as the “*freeze time interval*”. In case the aircraft's flying time to reach for example the FAF is already less than the *freeze time interval* by the time it enters the TRACON, the aircraft is assigned a STA immediately upon entering the TRACON [6]. Then, once the stream of arriving aircraft is ordered by scheduled arrival times, the air traffic controllers struck the arrival sequences to the meter fixes. After that, aircraft reach the converging point and enter the frozen area in which landing sequence is not changed unless exceptional cases.

The scheduling and sequencing decisions are made in a dynamic environment where the set of aircraft that are going to land or take-off, and the set of aircraft that have already landed or taken-off, are constantly changing during the time [62]. Thus, scheduling decisions have to be taken on the basis of partial data [24]. As far as, due to the uncertainty of en-route operation, the activities of the arrivals sequencing and scheduling remain a tactical decision for the air traffic controllers. In fact, the primary responsibility in this respect rests with the air traffic service provider, although CDM concepts are being applied to these settings, leading to some air carrier participation in scheduling and sequencing decisions [25]. In the early stages of

⁸ RT: located at a point other than the physical beginning or end of the runway.

⁹ STAR: tool which help the arrival aircraft to approach the airport, precisely, from the en-route phase to an IAF.

¹⁰ IAF : starting point of the initial approach leading to the FAF.

¹¹ FAF: located between 4 and 12 NM from the RT (usually in the alignment of the runway).

¹² Runway Occupancy time: amount of time that each aircraft occupies the runway.

¹³ Instrumental Landing System : radio path that the pilots lock and follow in order to perform precision landings [58].

¹⁴ SLT: calculated by trajectory synchronizer equipment after entering the aircraft into the radar range [59]. It is assigned to the approaching aircraft before it reaches the final approach path, approximately 20 min before landing [41].

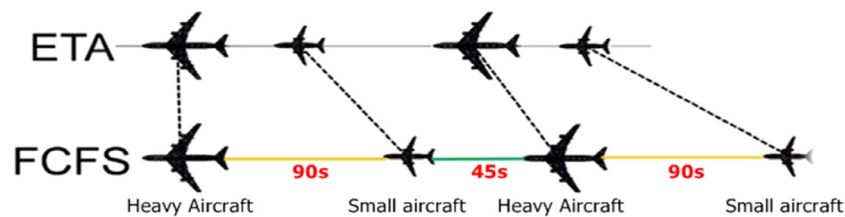


Fig. 2. Aircraft sequenced according to FCFS rule.

Decision Support Tool (DST), we mention the Arrival MANager (AMAN). This system has found their way into operational use in Europe so as to assist the arrival controller's daily work to handle the tasks of aircraft sequencing, scheduling, and metering into terminal areas surrounding busy airports. AMAN provides important information and suggestions on the aircraft sequencing to the air traffic controllers [37], and, calculates the optimized approach landing sequence and certain operational timestamps, particularly during challenging conditions such as bad weather conditions or capacity restrictions [63]. This latter tool is set up, for example upstream (in the approach sectors) at *Hamburg airport*, for accurate sequencing of the arrival flow [23]. For further details concerning AMAN, we refer, on the one hand, the reader to Samà et al. [64] who cited significant benefits of this sophisticated scheduling software's. On the other hand, an important methodology has settled by Rodríguez-Sanz et al. [63] who firstly assessing and predicting the *Adolfo Suarez Madrid-Barajas airport's* level of arrival congestion and delay, under specific operational circumstances, and at a tactical level. Then, the success of this system relies on the fact that the robust model was embedded into AMAN systems. The *Traffic Management Advisor* implemented at large ARTCC in the *United States*, is another support procedure for sequencing and scheduling arrivals [6]. This DST computes for each of the aircraft's the sequence and the STA to the outer meter arc,¹⁵ meter fix, FAF, and RT, in order to meet the sequencing and scheduling constraints. However, with these systems, the air traffic controllers usually have to manually adjust the landing sequences produced automatically; since the existing systems do not usually fully take into account the microscopic details of the aircraft movement required to land in the correct order [65].

Up to now, all major airports all around the world such as *Doha International airport* [38], *Hartsfield-Jackson Atlanta International airport* [66], excepting those were ATC advisory tools are not implemented; employ a policy based on FCFS rule to make flight scheduling on the TMA resources. This means, that once aircraft enter the ARTCC, they are scheduled in the order in which they are predicted to land by using a nominal path flight plan, preferred descent speeds, and altitude profiles [67]. This means even, as shown in Fig. 2, that aircraft are scheduled to the meter fixes based on their Estimated Time of Arrivals (ETAs) by simply applying the required minimum separation between them.

Two distinct FCFS schemes can emerge for the ALP: First Come First Served-System Entrance (FCFS-SE) and First Come First Served-Runway (FCFS-RW). The former sets priorities according to the order at which aircraft enter the TMA system. This event takes place at a certain distance from the runway, of the order of 50 NM. The latter sets priorities according to the projected time at which the aircraft are expected to arrive at the RT [68].

¹⁵ Outer meter arc: approximately 60 NM outside the associated meter fix of the arrival aircraft.

2.1.3.3. Separation criteria for approaching and landing aircraft. The separation requirements are one of the main ATC safety devices in any sector and during all phases of the flight, upon which the air traffic controller has a great responsibility by managing them in tactical level [69]. The separation is described in three dimensions: vertical, lateral, and longitudinal separations. Often the lateral and longitudinal separation standards are combined into only one called horizontal separation standard. In fact, as said by Dear and Sherif [70], the air traffic controller is supposed to keep each aircraft in their jurisdiction separated by at least 1000 ft vertically and 3 NM horizontally at least around the airport, and 5 NM horizontally elsewhere. The air traffic controllers are also responsible for staggering that a minimum separation between the arrival/arrival of one aircraft and the arrival/arrival of another aircraft is maintained. This means creating a correctly separated flow of aircraft toward the runway [71]. More specifically, the minimum required separation governs the minimum permissible distance interval between aircraft lined up in sequence on the approach in order to land on the runway. That therefore means that the successive arrivals moving on the same route must maintain a minimum required separation distance throughout final approach. Add to the latter separation requirements known as "approach radar separation", which is a longitudinal spacing of 5 NM (3 NM in congested area) and a vertical spacing of 1000 ft [37,58,72], we could mention another most applied notion to consider in approach airspace; the "wake turbulence minimum separation" which will be, for its importance, explained in details in the second part. Then, once the aircraft touches down at the RT, and as the runway can be occupied by at most one aircraft at a time, a last minimum separation distance must be ensured from the leave of an aircraft to the entrance of the next one at a runway.

Primarily, the name of the kind of separation, "wake turbulence minimum separation", is related to the fact that an aircraft operating on an air segment faces the risk of instability if interacting with the WV of an aircraft landing or take-off before it. The aircraft WV as a generating lifts consequence is a potential safety hazard for the following aircraft. These Wake Vortices (WVs) have a chaotic evolution, and can cause in imminent turbulence near the aircraft which follows it. Indeed, a number of aircraft accidents are believed to be caused by this phenomenon. The intensity of an inducted WV caused by an aircraft in flight is generally proportional to its weight and the lighter of the following aircraft. For instance, Mullins [73] presents an example of a *Boeing 747* that generates a great deal of air turbulence, and an aircraft flying too close behind could lose its aerodynamic stability. In other cases, Fahle et al. [28] presented an example of a light aircraft (the *Avro RJ 15*). Therefore, to maintain an aircraft's aerodynamic stability and for safety reasons, a relevant minimum longitudinal separation distances (in NM) is enforced between successive aircraft crossing the same FAF meter of the final approach segment and assigned to the same runway. If several runways are available for landing, similar separation depending upon on the relative runways position, are mainly applied for pairs of aircraft moving on different routes to land on different runways [64]. A heavy aircraft usually generates more severe

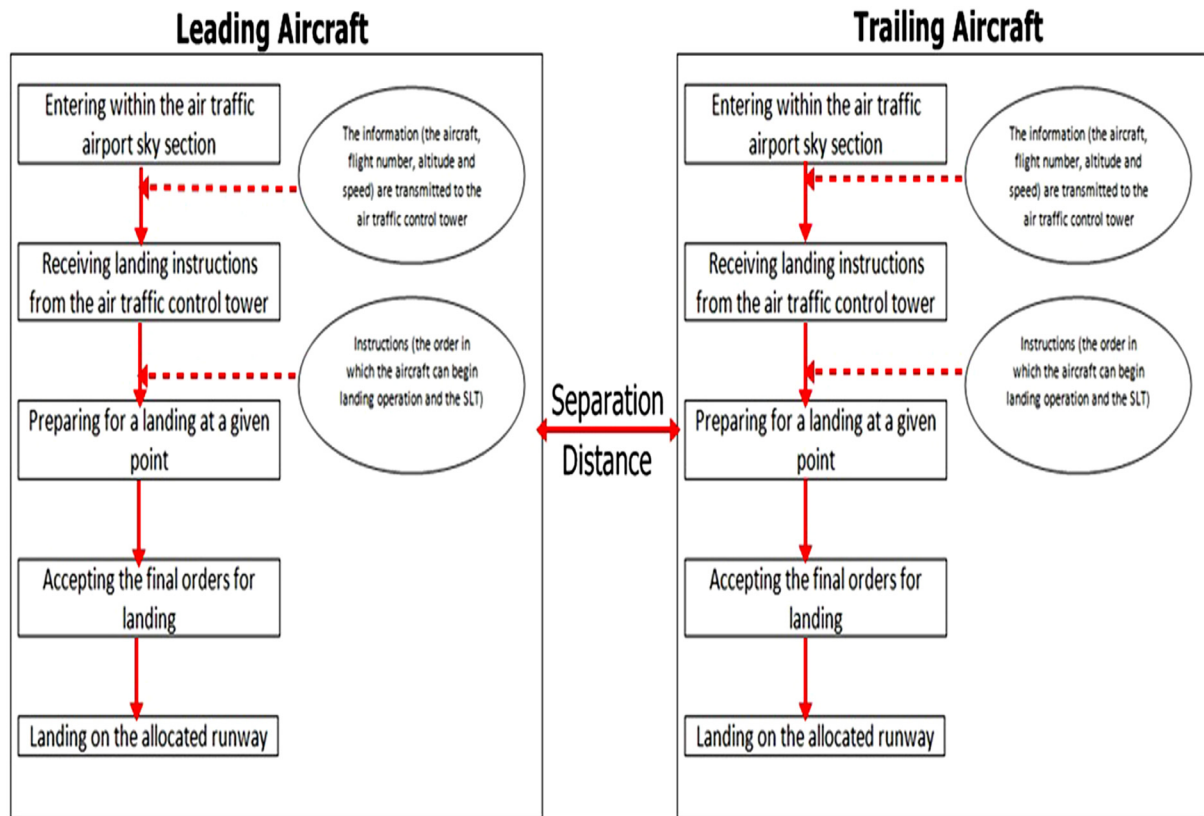


Fig. 3. Chart of landing between a leading and a trailing aircraft.

Table 1

Shows a Matrix of Class depending on the Minimum Separation Distances in the case in which a Runway is used only for Arrivals in Instrument Flight Rules¹⁶ (IFR) Conditions.

Trailing	Leading		
	Heavy	Large	Small
Heavy	4 NM	5 NM	6 NM
Large	3 NM	3 NM	5 NM
Small	3 NM	3 NM	3 NM

wake turbulence than a small one [74]. Consequently, a small aircraft flying behind a large aircraft needs more minimum separation distance than the reverse order (and vice versa). The Fig. 3 is a schematic of the necessary safety separation distance to be respected during landing to guarantee that the trailing aircraft does not suffer from the WVs induced by the leading aircraft. These separations are based on the preceding aircraft categories.

Separation requirements are mandatory and recognized by international aviation regulations [10], such as the FAA and the International Civil Aviation Organization (ICAO). The separation requirements mandated by ICAO, are reported in Table 1. These requirements occur during all the approach phases for each air segment of the TMA. They are determined by several factors, such as the weight class to which the aircraft pair belongs; the types of operations involved: the arrival followed by another arrival, the arrival followed by a departure; the relative orders and the landing velocities; the length of the final approach; the occupation and the configuration of the runway.

Regarding the classification of aircraft, the FAA and other civil aviation authorities around the world classify aircraft into a small

number of classes for terminal area ATC purposes. For example, we found the international classification of aircraft into three types presented by ICAO, which is based on the Maximum Take-Off Weight (MTOW): *Heavy*: aircraft capable of MTOW of more than 136 000 kilograms (kg), regardless if they are operating at this weight during a particular phase of flight; *Large*: aircraft with MTOW between 136 000 and 7000 kg; and *Small*: aircraft with MTOW of 7000 kg or less. The Civil Aviation Authority (CAA) in United Kingdom has increased the original ICAO three wake turbulence separation groups to five categories: *Heavy*, *Upper-Medium*, *Lower-Medium*, *Small*, and *Light* [75]. On the other side, the FAA defines four classes, based also on MTOW: *Heavy*, *Large*, the *Boeing* (a class by itself), and *Small* [75]. Recently, notably with the emergence of the very large aircraft, the ICAO defines four categories of wake turbulence, namely *Light*, *Medium*, *Heavy*, and *Super* [58]. The latter category puts into a distinct wake the aircraft capable of MTOW of 560 000 kg [76], such as the *Airbus A380*, and the *Airbus A388*.

The separation between two aircraft is considerably influenced by the air traffic controllers' experiences and the pilots' reactions, and may not make full use of the system's capacity [77]. This is due to the fact that required separation distances expressed by the ICAO can change, for example, in the case of unexpected departure delays and weather changes during operations. Moreover, in the conditions of low visibilities (whatever is the weight class of aircraft); the minimum separation distance becomes larger [71]. On the other hand, in a particular airport, noise abatement procedures may require a different separation requirement between aircraft, if they land on the same runway [47]. Furthermore, legislation makes it possible for air traffic controllers to reduce the minimum separation (e.g., during normal weather conditions) if they have an aircraft in sight [72]. A closer look at these particular evidences nevertheless allows us to deduce that the separation requirement is a variable quantity, which

¹⁶ IFR: set of regulations that dictate how aircraft are to be operated when the pilot is unable to navigate under Visual Flight Rules (VFR).

cannot be limited to the minimum separation requirements set by ICAO.

As noted above, the difference in aircraft types and performances is not the only interruption affecting the aircraft ability to achieve the proper separation distance from another aircraft. Furthermore, several interruptions may exist, including the route chosen for consecutive aircraft in each TCA resource [54], temporary change of flight plans, weather, wind conditions, pilot response, and Flight Management System¹⁷ (FMS) [66]. Yet, the air traffic controller should ensure, for tactical considerations, that landing rates will always be close to the ceiling of system capacity. Alternatively, he has to monitor aircraft speed at the approach phase by making the necessary changes to prevent the distance between the aircraft falls under the minimum value. The air traffic controller can also order a corrective maneuver (e.g., go-around [39], horizontal maneuvers or altitude changes [6]) in the case of loss of separation.

2.1.3.4. Air traffic control practice. Once the arrival aircraft crosses the outer meter arc of the ETMA, it could traverse the air segment following a SDP, and proceed toward the runway, i.e., it could have touched down and reached the gate without further delay, given that the necessary separation distance from its predecessor has elapsed. However, unforeseeable events or sudden changes could occur in air sector, such as gate unavailability, airspace saturation, bad visibility, weather conditions, or the occupation of the air segment or the runway by another aircraft...etc. The aircraft cannot initiate its approaching/landing procedures in a similar situation, and, it is forced by the air traffic controller to “wait” in the sky. Unfortunately, this waiting process is complex. On the one hand, the aircraft cannot put up acceleration and deceleration in a large range. Most obviously, the speed is heavily constrained by the aircraft type, the altitude, the weather, etc, and, the landing speed of any aircraft type is generally different from the landing speed of another aircraft [57]. That has been tied to the strict restriction in the approach area, and aircraft's own capability (e.g., the fuel per aircraft is limited [59]). On the other hand, the waiting process results in a nonlinear amplification of airborne delays, those needs to be absorbed in the air. Therefore, in order to ensure the management of the inbound traffic immense ballet, there are specific undesired traffic control procedures in flight, that have to be requested by the air traffic controller, to make an aircraft wait before landing. Typically, the air traffic controllers control incoming traffic, and when it is necessary and timely to order a maneuver, they issue specific instructions and commands to pilots. Then, the pilots, after an unavoidable delay, execute the instructed maneuvers [78]. It is a complicated task for the reason that the air traffic controllers must determine and transmit control actions in a short time frame.

The well-known current practices of trajectory controlling for making an aircraft wait to approach/land are speed adjustment, airborne holding, and vectoring [62], which are explained below. There are other airborne-spacing solutions, such as heading change or flight level change [8], which are not defined in this part since they are not frequently proposed in a congested situation.

- Vector Flight Spacing (VFS): the vector is made in a manner that the aircraft follows the shortest route between two points so that it makes a turn with a small rotation (i.e., the route extension by using radar instructions in terms of headings [72]);

¹⁷ FMS: an on-board multi-purpose navigation, performance, and aircraft operations computer designed to provide virtual data and operational harmony between closed and open elements associated with flight from pre-engine start and take-off, to landing and engine shut-down.

- Holding Pattern (HP): the HP takes place in the TCA, in what is called the waiting areas or “stack” as shown in Fig. 4. Aircraft are instructed to join a holding stack, localized in correspondence with the IAF [57]. They are stacked at different altitude levels, where they pass from the cruise flight to the approach phase, waiting for the instructions of the ground controlled approach¹⁸ as to when each of them will start its landing. Once stranded in a holding circle, the aircraft must fly at a fixed speed. After performing a half circle of a loop, it can reverse direction in order to continue the landing procedure. For each of the aircraft's, there are a maximum number of allowed half circles as prescribed by the air traffic controller [79]. Once an aircraft at the lower level is cleared to leave the holding stack [30], it proceeds along the prescribed standard trajectory, and the air traffic controller would make sure that in normal conditions, this aircraft will not be in conflict with others. Once the aircraft is close to landing, the air traffic controller calls the lowest aircraft in the HP, and directs it to the runway axis for landing. Bianco et al. [57] and Samà et al. [52,80] assume that holding circle resources capacities are nonbinding, and Samà et al. [65] undertake that there are no aircraft sequencing decisions in the holding circle, which reveals that the aircraft can enter/exit in different orders in that resource; and
- Acceleration and deceleration of the aircraft by the fact that air traffic controller, on the basis of a discredited value [48], requires the pilot to modify aircraft speed by slowing down or speeding up.

The most used maneuvering actions are, respectively, the VFS and the HP. Indeed, in China, a large amount of radar vectoring is frequently used by approach controller to manage both arrivals and departures. Elsewhere, several HP may be defined in the same TRACON system [81], and an aircraft may be forced to fly unnecessary HP several times [30]. Air traffic controllers always hate executing this latter procedure because; a HP usually induces larger delays than VFS, and tries to reduce the total of their number [82]. It is worthwhile to mention that the air traffic controllers prefer to reduce cruise speed rather than spending the excess time flying a HP, because as Mulfinger and Sadosky [83] said, the absorption of delay by speed control has solely the potential to reduce air traffic controller workload by avoiding path changes.

We dissected the decisions made by the air traffic controller, but there are some maneuvers that could be taken by the pilot in case of finding a problem not discovered by the former responsible. On the other hand, the pilot can decide not to land as the air traffic controller consent, for example, if he meets runway incursion¹⁹/excursion.²⁰ In such a case, he can decide itself to go-around. The go-around is made when the aircraft leaves the approach path; it starts again in the height, and is placed again in the axis of the runway. This undesired flight maneuvers is rarely recommended by the ATCT. Finally, let us get back to the DST concern, particularly their usefulness regarding the control aspect. Surprisingly, we would like to draw the readers' attention to the fact that such systems (e.g., AMAN) fails to capture for the aircraft appropriate methods, such as vectoring, path stretching, speed changes, or holding to meet its time or position in the sequence [13].

¹⁸ A ground controlled approach is a type of service provided by air traffic controllers whereby they guide aircraft to a safe landing, based on primary radar images.

¹⁹ Runway incursion: any occurrence at an aerodrome involving the incorrect presence of an aircraft, vehicle or person' on the protected area of a surface designated for the landing and takeoff of aircraft [84].

²⁰ Runway excursion: a veer off or overrun from the runway surface. These surface events occur while an aircraft is taking-off or landing, and involve many factors ranging from unstable approaches to the condition of the runway.

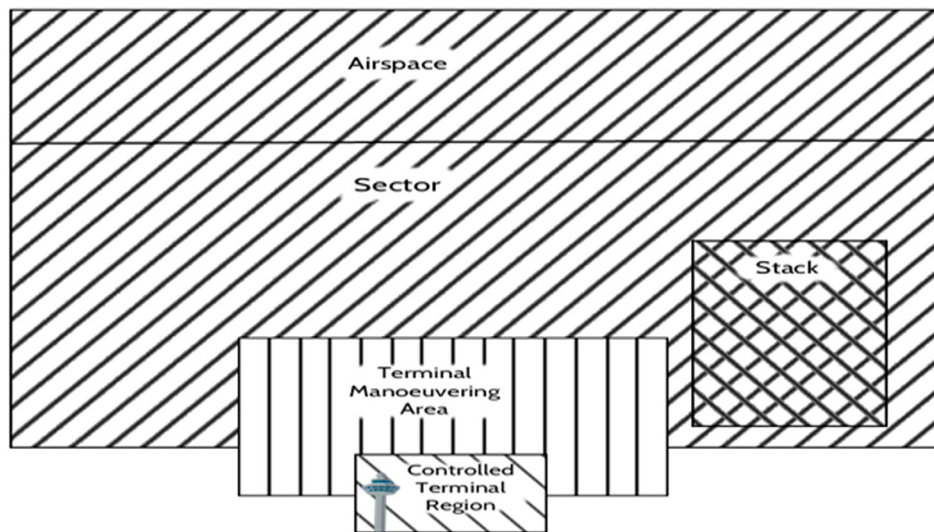


Fig. 4. Structure in and around TMA.

2.2. Air traffic flow management control practice

Even in the context of control practice, we note that once an aircraft has started its final approach, changes in the landing sequence cannot be made without demanding a large workload from the air traffic controllers [37,58,72]. On this basis, and included with the growing of the air transportation system complexity and the tight coupling of its various elements, it falls within the intervention of the ATFM. This view is dictated by the fact that the ATFM contributes to the safety, efficiency, cost effectiveness, and environmental sustainability of an ATM system [85], and, moreover, part of the ATFM enhances the efficiency of airport operations by dynamically optimizing the allocation of airport resources for arriving and departing aircraft [86].

As regards the cases for which the ATFM intervention is shown in the landing procedure, we can quote a conflict between two aircraft. According to ICAO regulations, two aircraft are considered to be in conflict if their horizontal separation is less than the minimum aircraft separation and their vertical separation is less than 1000 ft [8]. Furthermore, a conflict between two aircraft could be looking to the situation of loss of minimum safe separation between them [26], precisely the longitudinal diagonal one [80], when they crossed the same air segment or runway [87]. It should be underscored the potential route conflict, given that this type of conflicts can occur when the aircraft requests the TCA resources in its route.

When any potential conflict is detected, it must be solved in real-time by adjusting the off-line plan (i.e., time table). The ATFM, depending on the conflict, is asked to intervene by taking the suitable control action. For example, when some potential conflicting routes are detected, the ATFM could take suitable re-scheduling actions including re-routing some aircraft to other resource in the same airport or in other airports [87]. Currently, many ATFM tools are operated at airports, such as the Air Traffic Command Center (ATCC) which coordinates the flow management in the *United States*. It aggregates all the information collected by the 22 regional centers to re-schedule and re-route flights when a hazardous event (e.g., bad weather) occurs so as to minimize the delay costs caused by congestion [78].

2.3. Landing operation identified hurdles and alternative solutions

2.3.1. Identified hurdles

The management of aircraft landings in the TMA differs from one period to another. The previous management philosophy

occurs under normal traffic and when the aircraft are operating in good conditions. Recognizing that even in normal traffic period, the air transportation system is affected by high uncertainty, due to several effects (e.g., wind, imprecise knowledge of aircraft parameters, FMS settings). However, it is considered that the current situation, marked by the appearance of peak traffic periods, is a bit more complicated. This complexity is compounded by, on the one hand, the emergence of congested airports. It is particularly noticeable that, a congested airport is frequently defined as the airport where the number of flight that can be scheduled per hour (or other unit of time) is exceeding the “declared capacity” of the airport (i.e., AAR). Indeed, several of the busiest airports in the world have reached their maximum capacity, thereby making air traffic congestion in the airport surface, as stated by Clare and Richards [88], to become a major constraint on the available capacity of the air transportation system. In terms of its effects economically, the air traffic congestion reduces the turnaround efficiency. Remember that although it has a major environmental impact in terms of air pollutants and noise emissions, all of which negatively impact the local region and the quality of life of near-airport communities. Furthermore, congestion causes concerns for air traffic controller workload and increases runway incursion. On the other hand, it should be noted as well that, at such periods, excessive flight delays are often observed. These delays that are created mainly by imbalances between demand and capacity induce a lot of impacts. These include for example, the unnecessary fuel consumption, the gas emission and the time inefficiency [89], the passenger discomfort, and the disruption of the airlines schedules. Therefore, the low of flight efficiency should be recognized in the TMA today.

We wish to signal that, at present, the real-time management for landing a team of aircraft relies on the judgments of the air traffic controllers [88,90]. These skilled employees, according to their knowledge and experience, make the appropriate decisions in this regard and take all necessary actions to continue the successful ways of aircraft. The air traffic controllers’ task is delicate whether for normal or peak periods. In all cases, they have a very short period of time to determine the aircraft landing times and positions in the landing sequence, and to issue also the appropriate control actions necessary to obtain a smooth progress [91]. Moreover, they also have to deal with the complicity and variability of various anomalies, such as aircraft accident, airport closure, and bad meteorological conditions. We

are thus observing that the air traffic controllers play a key role in the current centralized air transportation system.

In view of these latter complications, we differentiate that, the viable accomplishment of landing operation is not simple. Nowadays, the situation becomes more and more difficult, for several reasons. One of the most important is the fact that the demand for runway occupancy reaches or even exceeds the capabilities of the whole system [29]. Let us take the concrete example of the air transportation system in the *United States*, which is operating for a long period close to its capacity limits in the most major airports [33]. Another aspect that must be taken into account is the air traffic load which does intend to rise in the near future, and this increase is not being matched by an appropriate expansion in capacity and more effective management of demand [92]. Added to these latter aspects, looking strategies to keep pace with the increasing demands for efficient control, scheduling, and landing of larger aircraft number, will make further the landing operation management a mission requiring effective and urgent interventions from the different head.

2.3.2. Alternative solutions

Several tactics have been proposed as regards the enhancement of the landing operation efficiency near TMA. On the one hand, injecting additional capacity in the TMA, which is seen as strategic measure, and, on the other hand, the continuously use of the existing resources in better ways, were the most distinguished. We note also the scheduling interventions and the Ground Holding (GH) as techniques allocated at pre-tactical level.

2.3.2.1. Improving terminal maneuvering area capacity. As we have mentioned above, the TCA is the bottleneck of the entire ATM system, specifically; the capacity of runway(s). Runway complexes of major airports are very scarce and expensive resources, and their capacity is gravely limited by separation requirements between aircraft [93]. Consequently, even small enhancement pertaining to runway throughput can have not only a significant impact on such a system wide delays [33], but also the efficiency and the effectiveness of the landing operation process. In fact, the transportation solutions trend to expand the airport's terminal capacity by two options. The first one is about building airports, and the second one is about constructing extra runways at existing facilities [92]. For the first option, we note recently the opening of the new *Istanbul Airport*, which was planned as the largest airport in the world serving 150 million annual passenger's capacity. We may cite, by way of example for the second option, the case of building 5 new parallel runways at *Beijing Daxing International airport* [8], and the new parallel runway which has been constructing at *Brisbane airport* in *Australia* [36].

The construction of new airports or new runways was usually considered not to be realistic option due to several limitations, such as the environment, policies, aerospace and parasite limits, noise disturbance caused by aircraft landing as well as take-off, and the geographical constraints [94,95]. In addition, the capacity expansion at airports decisions is rooted in strategic that require considerable investments and long construction lead times [38]. Since new airport/runway(s) expansion options take a while to implement and may not be a cost effective solution, aviation authorities are seeking methods to utilize the currently available capacity of the resource(s) at an airport as efficiently as possible [89].

2.3.2.2. Improving terminal maneuvering area capacity utilization. Until recently, it was believed that during the busiest time, assessing the best way to use existing airport resources such as runway(s) can help to avoid the congestion and airborne delays, as well as increase the airport capacity [5]. One promising approach

to get for this state is one that aims to improve the performance of managing the bottleneck operations related to arrivals, departures, runways, and taxiways [16,96]. To overcome this challenge, a higher level of semi/fully automation is needed in management of aircraft near TMA. It could be achieved, for example, through increasing the efficiency in handling the scheduling strategies and the optimization of Arrival Scheduling and Sequencing (ASS) techniques at an airport [41,62,97,98]. At present, the FCFS is the practical scheduling rule that is commonly used by air traffic controllers' at all major airports all over the world, as outlined above. This traditional rule was observed as frequently reliable and robust in today's operations for managing the arrival traffic, driven by its simplicity of implementation while satisfying the ALP constraints, and also as it minimizes the standard deviation of time delays [99]. Moreover, FCFS seems to be a fair procedure to all airlines, as the preferences of these latter stakeholders in the scheduling aircraft process are not executed [100]. In fact, it generates fair schedules when delays must be absorbed [101]. Furthermore, it requires minimal intervention from air traffic controllers since there is no required re-sequencing [102]. However, its disadvantages are quite obvious for producing excessive delays during peak traffic periods [103], since it does not take advantage of separation standard asymmetries (see, e.g., Zhan et al. [94] for an instructive example between a *Boeing 727* and a *Boeing 747*). In addition, this principle generates a negative effect on runway throughput, aircraft tardiness, and passenger delay [30]. Andreeva-Mori et al. [100] took these disadvantages to mention that the FCFS rule provides rarely the best sequence. Therefore, an optimized approach that incorporates this broader rule becomes more appreciated. In fact, it is observed that the optimized FCFS for the problem of sequencing the simultaneous arrival and departure on a single runway with assigning priorities to aircraft landings, based on practical size instances observed at *Doha International airport*, improves upon the FCFS rule by reducing the makespan by 4.3% [38]. This contributes to the fact that a large margin improvement of the terminal area capacity utilization is possible with new technologies based on optimization techniques. Hence, the provision of efficient scheduling/sequencing methods with the ability of powerful tool for optimizing the air traffic controllers' flight scheduling and guidance procedures becomes one of the main transportation solutions. These quick and intelligent tools are capable to cope with absent large increases in capacity given that, on the one hand, they improve the TMA resources capacity during aircraft operations and the resilience at busy airports, and on the other hand they leverage the possible workload involving in re-scheduling effort [1]. We refer the readers to Samà et al. [64] for additional explanations concerning the subject of the advanced scheduling systems and how they need to behave.

2.3.2.3. Scheduling interventions (re-scheduling). Re-scheduling of flights is needed in numerous situations, heavy traffic in air segments, Holding Patterns (HPs), runway schedules, and airport surface operations [104]. Regarding the runway schedules, it is imperative that the scheduling interventions are a feasible option for mitigating delays in the short and medium terms, given any particular state of aviation's infrastructure (airports and ATM) [86], which consequently improved capacity utilization. The policy of shifting the relative position of aircraft according to FCFS rule is commonly adopted for a personalized intervention on the scheduling of landings. This means that each aircraft must land within a pre-specified number of positions of its order in the FCFS sequence [48]. For example, an aircraft positioned at the beginning of the sequence can be shifted to the end. Such procedure in fact avoids the aircraft from being excessively delayed and deals with uncertainty, inaccuracy, and non-determinism. It is to be made clear, that deviations from FCFS are allowed if the aircraft

currently being scheduled can be assigned a landing slot between previously scheduled aircraft without requiring these aircraft to be re-scheduled [6]. However, the TMA is an extremely dynamic environment [91], and air traffic controllers have no possibility to implement an effective landing sequence that allows large deviations from FCFS sequence for two reasons [68]:

- (i) The system affords the air traffic controllers only limited flexibility in re-ordering aircraft; and
- (ii) The large deviations from a nominal or a “priority” schedule may be unacceptable to airlines from a fairness standpoint.

The problems of how best to, re-order, permute, or swap position of arriving aircraft have received greater attention in the literature. In this context, the Constrained Position Shifting (CPS) method is addressed to minimize aircraft landing delays by dynamically controlling aircraft shifts, i.e., changing the approaching aircraft sequence in the TMA of an airport. Several researchers in both the *United States* and *Europe* have used CPS as a model of fairness, and have worked toward developing fast technical solutions for scheduling within CPS framework (see, e.g., Balakrishnan and Chandran [33], Ma et al. [48], Dear [68], Dear and Sherif [70]). The main role of this method in the ALP is that it prohibits sequences which are intolerably biased against some aircraft categories. This is particularly in a “dynamic” environment, where the landing of some aircraft may be continually postponed due to the arrival of aircraft categories which receive higher priority. Samà et al. [65] observe that such algorithms can be a fairer comparison against real-world behavior, since air traffic controllers usually modify the FCFS solution by switching groups of two or more aircraft around, in order to improve bottleneck resources throughput. It is not a coincidence, because since CPS method had been incorporated into the proposed approaches, the results came up with a fast and flexible solution for planning. This is due to the multiple advantages supplied by this method (see, e.g., Dear [68] who quotes CPS advantages). There are, besides this, excellent perspectives of this methodology for an effective TMA system regarding the future to the extent of its flexibility for the provision to the pilot when entering the system, the possibility to specify the aircraft speed landing and the desired route of control.

2.3.2.4. Ground delays program. Most flight delays in the busiest airports are created by imbalances between demand and capacity [86]. Measures are required in order to prevent or convert especially airborne delays caused by shortage of landing capacity at the destination airport. In that context, Bertsimas and Frankovich [3] have been among who argued that forcing aircraft to be held at their origin airport is one option to consider when there are arrival capacity shortage throughout the air traffic flow network. This means that certain aircraft are delayed before they depart from their origin airports through the anticipatory assignment of GH delays. Indeed, the so-called Ground Delays Program (GDP) procedure comes into effect when there is inclement weathers either en-route or at a flight's destination airport. The GH delays policy is therefore appearing to be a promising solution for reducing severe congestion and delay costs. The reader could look into [105] and [106] for more clear description of this scope. There is a need to advocate that GDP is a part of the ATFM mission, given the ATFM procedures efforts to transfer airborne delays to ground delays. Moreover, optimally allocating airport capacity to all users during the fall in short-term rates of airport capacity, and limiting the foreseen negative impacts (e.g., severe flight cancellation, flight delays), are included among the ATFM purposes [9].

2.3.2.5. The support of air traffic controller. With regard to workload issues, the stakeholders should be alert to the complexity of the air traffic controllers' job. This orientation will sought to assist air traffic controllers, and in increasing their own ability to manage all operations, especially in complex traffic situations. In fact, governmental agencies are already looking for solutions that guarantee better conditions to these professionals [2]. The subjects of air traffic controller workload and the maximization of the effectiveness of these individuals to play their role as traffic managers will undoubtedly lead to concerted efforts toward providing to each one of them, “*the right amount of data, in the right form, and at the right time*”. Answering these expectations will lay a firm foundation for a reliable, accurate, and feasible ATC automation. This means that the requirement for performing Communication Navigation and Surveillance (CNS) techniques should be emerging. In addition, there would be great benefit in terms of CDM in the ATC environment for safety and efficiency [89], given that these tools provide in advance the information for all parties (airport, airlines, and ATC) [22]. It must be emphasized that the development of good and intelligent DST to use these information is recommended.

The development of new advanced computer technologies will be helpful to air traffic controllers for managing the arrival flow during peak hours. It also reduces airspace congestion in low-capacity situations [9]. This meant that automation will not only be limited to the gathering and storage of data and the graphical representation of current aircraft positions and speeds [53], but also, and in particular, to the provision of guidance to air traffic controllers in carrying out their “*intelligent*” tasks. Automation, in fact, does not have take-over the role of the human decision-makers. It has the potential to significantly reduce the air traffic controllers' burden in order to be able to argue on its optimal decisions. Nevertheless, this option needs a long period of times and large investment to make it operational, and the efficiency is often low [97].

3. Aircraft landing problem from a theoretical point of view

3.1. The aircraft landing problem complexity

To recapitulate the previous parts, it must be remembered firstly that the commercial aviation operations are supported by what are probably the most complex transportation system and possibly the most complex man-made system in the world [25]. Second, the landing operation is not an easy task to go about in civil aviation, as it might be seen. It requires not only immediate attention and considerable effort from air traffic controllers, but also powerful decision support tools in order to guarantee a number of successful landings for one time. Arguably, the complexity of the operation in question does not in practice only, but also in theoretically terms. It is indeed generally acknowledged that the aircraft landing operation in an airport platform, comprising one or several runways, consists in ensuring the landing of a set of aircraft, which can cope with congestion, delays, and potential circumstances. To address these tasks, it is necessary to ensure that the amount of time is set at a level that is reasonable, and, which will allow to handle quickly all the possibilities that might come forward and to deal with those eventualities. It is clearer that those alternatives are not compatible, given for example the real-time control requirements, the capabilities of computers today, and also, and in particular, the difficulty of defining all the preliminary combinations involved between aircraft. To highlight this last distinction, we note that even an instance with only 10 aircraft; we will have to make 3,628,000 comparisons and, 1,307,674,368,000 comparisons with 15 aircraft [29].

The determination of optimal sequences for aircraft landings is theoretically conceivable, firstly by examining all the possible sequences, and then by selecting the most favorable ones. However, as stated by Kjenstad et al. [23], finding good global solutions manually is too hard and very time consuming. This complexity is partly due the required computational effort, which is a factorial function depending on the number of aircraft and runway(s), and even beyond the potential of the fastest computers currently used in the operational environment of the ATC system. Hence, considering these shortcomings and limitations; the aircraft landing operation has been reminded as a Nondeterministic Polynomial-time (NP)-hard problem from the scientific literature [94,107–109]. In effect, the ALP was played a prominent part in the literature with a lot of research studies carried out on modeling and developing exhaustive algorithms to propose solutions for this air operation. In the following sections, after presenting how the ALP has been fixed, we are going to highlight the constraints and objective functions related to this issue that will lead to submit the distinctions. Then, some ALP modeling approaches are briefly outlining in this section.

3.2. How the aircraft landing problem has been dealt?

Successful management of traffic flows emerging the TMA entails, routing, scheduling, sequencing, and assignment of the homogeneous set of aircraft to a single or multiple runways including when it may be necessary the control action and the resolution of potential conflicts. We note, however, upon analyzing the large number of recognized research about ALP, that the problem had been dealt in several ways. Particularly, we can see that the latter issues were either handled in isolation or jointly. On the one hand, Ma et al. [48] could be taken as illustrations when they have solved the sub-problem of routing (i.e., merging problem). Their methodology in addressing this issue consist of a route network graph, while acknowledging that each aircraft follows exactly one route corresponding to its entering point. The basis of the routing of aircraft is expressed differently in [110]. The difference between the last two works lies in the metering points. In [110], the arriving aircraft has to be routed from its landing runway to its stand or hangar, while in [48], the arrival aircraft is allowed to fly from the EF to the RT of a single runway. On the other hand, the management of landings on the runway(s), which is commonly referred in the literature as the Runway Scheduling/Sequencing Problem (RSP) (see. e.g., Chandran and Balakrishnan [91], Kopolke et al. [111]), Aircraft Landing Scheduling (ALS) (see. e.g., Cămară et al. [2], Ernst et al. [61], Mokhtarimousavi et al. [59]), or ASP (see. e.g., Xu [112]), or whether Arrival Sequencing Problem (ASEP) (see. e.g., Kwasiborska [76], Andreeva-Mori et al. [100], Psaraftis [103]) has been treated on several occasions. Even optimizing ASS is presented, when researchers tackle at the same time the arrival scheduling and sequencing missions (see. e.g., Hong et al. [17], Salehipour [19], Eun et al. [107], Hu and DiPaolo [113]). It is also interesting to note that a large majority of publications addresses the case of mixed operations, which involves the allocation of both landings and take-offs to the runway resource (see. e.g., survey papers of Bennell et al. [30,45]). Wagar et al. [114] have already explored for this latter field, on the one hand, the scheduling of arrival landings and crossings and, on the other hand, departures, for the four runways of *Charlotte Douglas International airport*, ranked as 8th in the *United States* and 23rd in the world. Yet it should be pointed out that the ALP and the Aircraft Take-off Problem (ATP) are often considered as separate problems, and the former has received greater attention than the latter. It all boils down to the fact that arrival operations are somewhat more important and critical than departure ones [7,16].

For the past years, we have observed that the research work took another way in its progression with the launch of looking at a particular aspect of the ALP. First of all, more efforts made on integrated optimization models for terminal airspace operations. This means the partial or full integration of flight concerns (e.g., routing, scheduling, sequencing), whether landing or take-off operations, is for not only the runway(s) resource, but although included another resources of the TMA, such as the air segment(s), the taxiway(s), and the holding circle(s). In this field, the autonomous system of Nikoleris et al. [6] designed for a mix of conventional aircraft and UAV, performs at the same time the ASP and the resolution of conflict. For our part, we submitted a contribution in [115], when assessing a solution for the more general scheduling, sequencing, and assignment, including the control action, of aircraft landings on a TMA containing air segment and runway resources. Elsewhere, and also as part of reducing efforts of air traffic controllers in the ATC sector, we have of course seen great focus on adopting a more comprehensive and analytical view of the issues that have arisen during the landing operation in real-time, such as re-routing, re-sequencing, and re-scheduling procedures. In this regard, we mention Samà et al. [35] who made a significant contribution on the simultaneous re-routing of both landings and take-offs on a TCA which can incorporate several air segments and runways, while scheduling and sequencing the aircraft on those airport resources. Finally, we notice that the researchers become more aware when their approaches are devoted for a specific airport under study, with its own configuration, equipment and layout, all for the sole purpose of meeting the development trend of the air transportation system. We would, in particular, like to acknowledge the works of Hong et al. [17] and Liang et al. [13]. The first group investigated three sequencing and scheduling algorithms for handling inbound traffic through a merging arrival technique, the Point Merge System²¹ (PMS). For more description of the PMS technique, we refer the readers to Errico and Di Vito [116] who present a detailed overview of this concept. The second group presented a system that integrated simultaneously the merging including the control, and the resolution of conflicts that could occur to the arrival operations, from the EF to the RT of the runway for which they are assigned and sequenced. The model was designed to the parallel runways of the actually most important, largest and busiest international aviation hub in *China* and the second busiest airport in the world, *BCIA*, where around 1700 flights from 94 airline companies operating in that airport.

The analysis of the majority of the prior papers on ALP supports that all the relevant issues cited above have looked at the case, whether and where only a single runway is used or multiple runways operate simultaneously for landing operation. Furthermore, there is who, for further precision, checks the efficiency and the validity of their approach through data extracted from real-world traffic case, in order to increase the level of realism in search of solutions. Farbadi et al. [4], for example, presented a solution for the arrival and departure assignment and scheduling over a single and parallel independent runway, while contrasting the case of *Doha International airport* single runway with the two runways newly constructed at *Hamad International airport*. On its part, Ma et al. [48], and, in order to address the ASEP, the computational experiments were assessed for the single runway of *Paris Charles De-Gaulle airport*. This difference was also generalized in [117], when the experiments are performed by simulating a real scenario of aircraft landings extracted from the major airport in *Tunisia*, *Tunis Carthage airport*.

²¹ PMS: new STAR design for sequencing arrival flows, which have been developed by the EUROCONTROL Experimental Center EUROCONTROL 2006. PMS is implemented in different international airports such as *Oslo*, *Dublin*, *Seoul*, *Paris*, *London City*, *Incheon*, and *Jeju* [17].

3.3. Aircraft landing problem dominant constraints in the air traffic control system

We rely on the explication presented by Baccouche [118] to summarize the practical management of aircraft landings, in order to highlight the ALP constraints. The latter mentions that the air traffic controller gives a landing order to every aircraft so that it has the time to fly in safety conditions from its current position to the runway. However, he has to make sure that an aircraft lands at the appropriate position, and while respecting the safe separation with all other aircraft. The first condition involves that for every aircraft, the landing time is constrained to be within predefined landing time window, whereas the second condition means that required separation in terms of time must elapse between landings. These two real-world constraints were handled as indispensable constraints for the ALP, as illustrated below. In connection with this clarification, the CPS rule also was, as we have seen above, presented as a crucial rule that each aircraft must obey in the sequencing process. It is further recalled that there are several constraints which need to be fulfilled for the ALP. The details are also presented below.

3.3.1. Landing time window constraint

Aircraft emerges at a given appearance time,²² which is when it reports to the ATCT for landing [119]. This is, indeed, where the air traffic controller should determine the scheduling time, enabling for that aircraft to fly from the EF to the landing runway. As we are taking the landing procedure, we will be dealing with the SLT which is defined for each aircraft to arrive at exactly the RT reference point. The SLT represents the time that enables the aircraft to land while flying straight to the runway, at its most economical and preferred speed commonly referred to as the cruise speed. Meanwhile, the SLT has to be as defined by BenCheikh et al. [120] as close as possible to the Preferred Landing Time (PLT) of the aircraft, also known as the Target Landing Time (TLT). The TLT may correspond to the aircraft's flight plan, the airlines timetable (i.e., the time at which an airline advertised a flight as being scheduled to land in its timetable), or the time used by the airport in their plans for assigning a gate to the aircraft or for unloading the baggage [24]. The SLT is bounded by an Earliest Landing Time (ELT) and a Latest Landing Time (LLT). The ELT represents the time when the aircraft could land while flying directly to the runway at its fastest speed (which is not the most economical speed for the aircraft) with no maneuver (e.g., vectoring or speed adjustments). It is calculated by considering both, the allowed speed and the delay [48]. For the LLT, it represents the time when the aircraft could land while flying at its most fuel efficient airspeed, while also holding for the maximum allowable time [121]. The latter term, which is determined from fuel limitation and maximum allowed delay that a flight can incur [122], differs from one aircraft to another [123]. It is important to underline that the former parameter is used more for calculating passenger and airline on-time performance, and the latter parameter is used for calculating additional fuel consumption and measuring ATM efficiency [21].

The earliest/latest landing times are considered in many studies. This is obviously because an aircraft cannot put up acceleration and deceleration in large range due to the restriction at the terminal area and aircraft fuel capacity [124]. Furthermore, it is proved that assigning each aircraft an ELT during the flight permits not only to absorb delay in a more fuel-efficient manner (e.g., speed reduction), but also helps in reducing the variability of

the arrival traffic [14]. For the ELT value, the researchers either assume that an aircraft's earliest time is the TLT [30,83,93], or limit the possible earliness as one minute [91,112,125], or whether five minutes [13] less than the TLT. Small values are committed for this parameter, because it is often uneconomical worthwhile to move the landing time forward by more than a minute [67] due to fuel consumption considerations [126]. For the LLT value, Chandran and Balakrishnan [91], Lee and Balakrishnan [125] and Balakrishnan and Chandran [33] set the aircraft's latest possible arrival time at 60 min after the TLT, implying that an aircraft would not put on hold for more than an hour. In the same context, a number of papers that have appeared in the literature (see, e.g., Beasley et al. [127], Xu [112]) aimed to reduce the ALP complexity, assume the LLT value (which does not correspond to the reality) to be sufficiently large to be of no consequence to the matters at issue. The time between an aircraft' earliest landing time and aircraft' latest landing time is the landing time window. The latter depends on the remaining distance to be covered, the aircraft's maximum and minimum velocity, and the remaining fuel [128,129]. Hence, the SLT for an aircraft must lie within time windows [130], and, the aircraft needs only one landing time [131]. Furthermore, Wu et al. [90] recommend that the SLT of each aircraft shall not be earlier than the ELT, or it will affects the landing of the follow-up aircraft and increases the burden of ATC system. As a result, it is clear why the landing time window is treated as a hard constraint in the ALP processing. Ultimately, as a practical recommendation, particularly where the size of time windows is concerned, it might preferable to require a short time window (e.g., 30 min). This is due to the fact that longer time windows can hardly be considered because of perturbations occurring the day of operations [110].

3.3.2. Minimum separation time constraint

As we have already shown in Section 2.1.3.3, to maintain an aircraft's aerodynamic stability and for safety reasons, a minimum separation distance (in NM) is defined. The latter term is based on the aircraft categories that must be respected during approach/landing to guarantee that the trailing aircraft do not suffer from the WVs induced by the leading aircraft. We would stress that the separation in distance can be translated into a minimum separation time [52] for many reasons, such as the limitation of the variability in the speed of aircraft within the TCA, the convenience of time-based system for detecting conflict and metering flows [13], and also the fact that each aircraft moves at standard speeds in the TCA [64]. Therefore, the separation distances are dealt with by converting them into separation times, using the assumption of fixed landing speed which depends on the aircraft categories [75]. A representative one is shown in Fig. 5. These separation times were derived from official separation distances issued by FAA [33]. It is noted that in order to convert a 'distance' into a 'time', the ground speed is used rather than the air speed [14]. This conversion is established at given airspace points, such as the RT, the outside marker, and the beginning of a free flight path. Among other factors, the separation times determine the maximum number of aircraft which can safely use a runway in a certain time period.

Two kinds of separation requirements can be modeled for the ALP, complete and successive separation [93]. The successive separation method means that the separation is established between the front and back aircraft (e.g., Psaraftis [103] and Kapolke et al. [111] present researchers who have opted for this method). Whereas the complete separation method means that the separation is established between the landing of an aircraft and the landing of all successive aircraft in the sequence (e.g., Beasley et al. [127], Beasley et al. [43], Moser [132], and Diallo et al. [121] present researchers who have opted for this

²² The appearance time in Heathrow airport is between 13 and 20 min before unconstrained landing time [24].

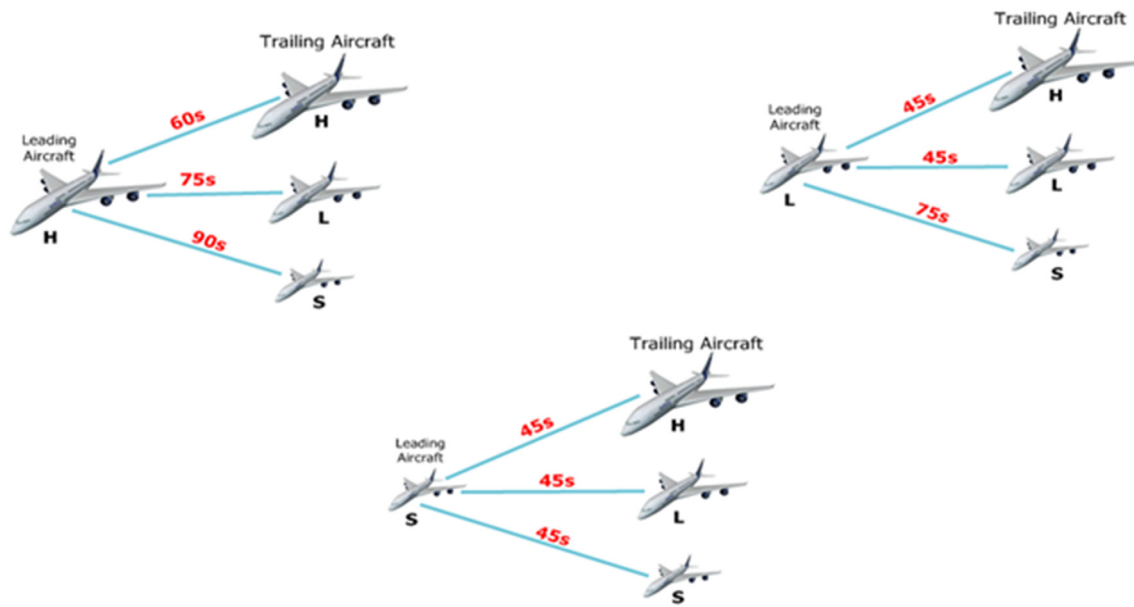


Fig. 5. Separation time at the RT meter (in s) between a leading aircraft (heavy, large, or small) and a trailing aircraft (heavy, large or small).

method). As stated by Dear [68], the former is sufficient to ensure a complete separation, for the reason that in the real-world environment the interference of aircraft which are not scheduled consecutively is a rare event. Nevertheless, in the opinion of Moser and Hendtlass [119] the separation time for each aircraft needs to be met not only with the immediate preceding aircraft, but with all its predecessors.

3.3.3. Constrained position shifting constraint

Any deviation before or after the TLT would cause customer's dissatisfaction, flight security decline, delays for other aircraft, and penalty costs. Adjustments to the Target Landing Times (TLTs) are possible, even if a limited re-ordering in the landing sequence can be made to allow the occurrence of unexpected schedule disturbing events or accumulated errors in time during the descent. In order to cope with such a scenario, the most important constraint that may be included so-called CPS concept. In particular, two CPS parameters are introduced:

- The well-known Maximum Position Shifting (MPS) stipulates that, in the initial landing sequence, the number of forward or backward position to each aircraft shifts with respect to its position in the FCFS sequence by a number of positions not greater than a fixed threshold. The maximum number of allowed positions to be shifted is denoted by k ($k = 2, 3$ for the most part of landing runway(s) system), and the resultant environment is referred to as a scenario of k -CPS, as shown in Fig. 6. Fig. 6, taken from the article by Lee and Balakrishnan [125], represents an example of the CPS network for five aircraft and for the value $k = 1$;
- The Relative Position Shifting (RPS) is the parameter that allows air traffic controllers to shift the aircraft from the position in the previous re-arranged sequence, by a number of positions not greater than a fixed threshold, depending on its present position. This parameter limits the pilots' and controllers' workloads. A detailed example for a sequence of 30 aircraft, with the RPS values associated equal respectively to 0, 3, and 5, is given in [34].

Furini et al. [58] were confirmed that, small values for the range of position that an aircraft can hold in the schedule, can efficiently solve the sequencing problem of mixed operations (i.e., arrival

and departure) on a single runway. Indeed, Dear and Sherif [133], Ball et al. [25], and Liang et al. [13] showed that, by setting MPS to a small number, such as 2 or 3, one can obtain most (e.g., 60%–80%) of the potential benefits offered by unconstrained optimal sequences, and at the same time ensures reasonable fairness in accessing runways. In all cases, Wu et al. [90] present fundamental explications on both the MPS and the RPS values.

When the resolution method will be based on the CPS constraint, it should be noted that MPS and RPS parameters must be met simultaneously in the sequencing process. This link is justified by the fact that on the one hand, the MPS brings benefits to the safety of aircraft since it can facilitate the position swapping in real-time. On the other hand, the RPS at the same time improves the flight safety and reduces the air traffic controller workload [90].

3.3.4. Aircraft landing problem additional constraints

Landing phase counts among the most dangerous phase in the flight process. Therefore, aircraft are able to get much more strict constraints during it. Specific constraints can exist for the ALP by observing the real functioning necessities. Indeed, researches are expressed a variety of operational constraints which can be imposed in the ALP modeling. These kinds of constraints that are discrepant from the landing time window, the minimum separation times, and the CPS rule, should be satisfied as much as possible. The most typical of these constraints are the following:

- *Precedence/priority constraints*: With respect to aircraft landings, certain precedence relationships related to the landing scheduling/sequencing tasks may be required. Precedence constraints are pair-wise requirements on aircraft that stipulate when one aircraft must land before another. For example, "fast" aircraft do not have to overtake "slow" aircraft as they come into land; and aircraft already in a standard trajectory segment have to land in the order they are already in. Such constraints are important in current ATC automation systems. Their sources are the airlines themselves that have precedence constraints arising from their strategies on banking [91]. On the one hand, Andreussi et al. [134], when optimizing landing and take-off sequences to especially the TMA of Rome, have introduced precedence constraints. In fact, they have given the priority to the aircraft arrived in a critical situation of kerosene against an aircraft taken-off,

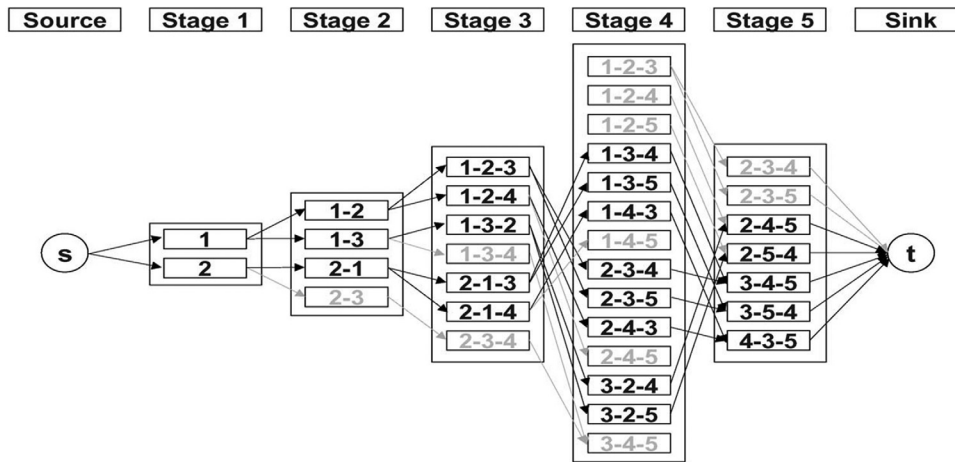


Fig. 6. An example of the CPS network for five aircrafts and for the value $k = 1$.
Source: Lee and Balakrishnan [125].

and to the aircraft which made a maximum delay. Of it share, Hong et al. [17] introduce priority consideration for the ASS missions; by assigning priorities for landing to aircraft in arrival that reaches the TMA at the earliest time. Aircraft priorities are also taken into account in the approaches proposed by Ghoniem et al. [135] and Ghoniem and Farhadi [136], destined to solve combined arrival and departure sequencing problems over multiple parallel runways. They have given predominately the priority to *heavier* aircraft, as opposed to *larger* and *smaller* one. On the other hand, Riahi et al. [36], Samà et al. [65], Ghoniem and Farhadi [136], and Hancerliogullari et al. [137] who're proposing approaches for landing and take-off sequencing problems, were prioritizing arrivals over departures. This consideration comes from the case in practice [4], since in general a landing aircraft in the air has more risk than a departing aircraft which can wait at ground level with fewer risks.

-Overtake constraint: An overtake means one aircraft passes alongside, over, or under another aircraft sharing the same trajectory segment. In the TMA, the extra workload by overtakes could be too intensive to be accepted by air traffic controllers. Therefore, it is undesirable in many cases [107]. Indeed, the detailed model in [54], designed for re-ordering and re-routing landing and take-off operations, precise that the overtaking is not allowed precisely within the air segment resource of the TMA. This would imply that the aircraft ordering must be the same at the entrance and at the exit in/from air segment [64]. In this context, for the ASEP based on multi-approach routes case, Zhang et al. [77] forbade that one aircraft overtakes another throughout the same arrival route. In other side, for a better sequence of aircraft from the border of the TCA to a dedicated landing runway, Yu et al. [99] introduced precedence constraint for successive aircraft entering the TMA at the same entry way point, to not allow an overtaking between them.

-Maximum and minimum allowable airborne times: Based on operational and technical considerations, such as limited fuel, air-speed, etc, each aircraft has minimum and maximum allowable airborne times, which have to be treated as hard constraints [22, 59, 138]. In our detailed mathematical model intended to solve the ALP for a single and multiple runway [115], we are modeled the constraint of maximum allowable airborne time, when the time fixed for the aircraft class to approach the air segment shall not exceed a parameter that defines the maximum approach delay time. Tian et al. [32], for their part, required at the arrival fix

reference that the air holding time of each arrival aircraft cannot exceed a maximum air holding time.

-Capacity constraint: In the landing and take-off scheduling process, D'Ariano et al. [10] and Samà et al. [52] grant multiple capacities for air segment resources in the TCA. In [4], we also see the examination of the notion of that runway capacity in light of some factors like the runway physical configuration and operating mode (segregated vs. mixed). For our part, we have opted to work on the AAR in [117], when a constraint states that the demand in terms of number of aircraft to land must not exceed the declared capacity.

-Assignment constraint: Within this constraint, Capri and Ignaccolo [139] and Ben Messaoud et al. [115] have introduced the primordial constraint for the runway assignment task whose the principle is restricted to the extent that at a time, the presence of one aircraft on the runway imposes no other aircraft can use it. From their parts, Kapolke et al. [111] propose for the ALS on a single runway, a constraint which deals with the assignment of several aircraft to the same time window of a given size.

3.4. Objective functions

The air transportation is a highly complex and interdependent system as there is many stakeholders. The national authorities responsible for the safety, the organizations of regulations of aviation, and the representatives of the air transportation industry in particular airlines and airports, are respectively the crucial stakeholders about this system. Every one of them aims to achieve a set of interests which meet their needs. If, for instance, we were the airport managers, we would see that their challenge at the practical level of the runway system management lies at simultaneously achieving the safety, the efficiency, and the equity. While in general airports care about passengers' throughput, which is the primary source of their revenue, and also hope to reduce airfield congestion [21]. On the other hand, if we put ourselves in the position of airlines, which are the major stakeholders in the air transportation system [33]; we notice that goals would be to lower the cost induced by aircraft (e.g., minimizing fuel consumption), since fuel efficiency in operations is a major factor impacting the profitability of the airline industry [24]. The Air Navigation Service Provider (ANSP) shall ensure the safety and efficiency of aircraft on its part. Elsewhere, governments and the public are paying more attention to the environmental impacts of airline operations due to noise and emission issues [66]. As noted below, the goals of various stakeholders are different and

often contradictory, as well as their achievements should be given in a reasonable period of time. Moreover, the air transportation is a system including technical, human, and natural factors characterized by a high degree of uncertainty. As Dear [68] emphasizes, in the face of the current acute system, the overall system performance is naturally sensitive to match the chosen objectives.

The choice of an appropriate objective function is a difficult issue, since it does not depend only on the adopted point of view, but also the current state of the airport. For example, during peak periods, it would be preferable to focus on efforts on improving the rate of capacity (e.g., the maximization of runway throughput); whereas in less congested periods, the emphasis on the criterion relating to the total cost that could be implemented is desired (e.g., the reduction in fuel usage). From this, we can conclude that the basic objective of the ALP in air traffic automation is a balance between traffic demand and airport capacity over one or multiple runways, while minimizing the total of landing aircraft delays or the respective cost. In this context, see the work of Samà et al. [65] which examine the trade-off between different classical performance indicators for the scheduling literature of take-off and landing aircraft at an airport with air segments and independent runways. They distinguish that an objective function of the cases studied, is not generally recognized. Since there is no well recognized objective function, its choice is of great discussion in the literature. This, on the one hand, is due to the fact that it relies on the perspectives and interests of various players, including airport officials, passengers, and airlines, and, on the other hand, those functions, that are depend on a large number of parameters, are extremely irregular. By far, and for perfectly legitimate reasons, several objective functions related to the management of runway(s) that host landings, have been considered. In the operation research literature, optimization goals can be divided into single and multi-objective goals. It is to be made clear, that multi-objective implies when considering two or more objectives simultaneously. The quality of the ALP, addressed whether separately or coordinated with the ATP, can be classified in the following objective functions of competing.

3.4.1. Single optimization goal for the aircraft landing problem

-Throughput objective: It is similarly clear that, among the important aspects that could be highlighted in relation to the resolution of ALP is the use of existing critical airport resources (i.e., runway(s)). In turn, the maximization in processing capacity of the runway system is pursued as a goal commonly shared by the service providers; in other words, it means the maximization of the number of aircraft operations that can be accommodated in the runway per unit of time. This consideration translates into the maximization of airport capacity use (i.e., throughput), which can be viewed as minimizing the completion time or the makespan [57], or also minimizing the LLT aircraft in a sequence [103]. For their part, Bianco et al. [34,57] sustain the minimization of the maximum completion time for the right landing and departure sequences on a given and multiple runway. The minimization of the maximum completion time involve the maximization of the throughput, as previously describes in [103]. Balakrishnan and Chandran [33] also adopted this latter objective function, and discussed its practical importance for arrivals or departures RSP at major *United State airports* case, the *Dallas Fort Worth airport*. In view of this, see the Eq. (1)

$$\text{Min } W_{\pi} = X_{\pi_{\text{end}}}, \quad (1)$$

where W is the *makespan*, which is the landing time of the last aircraft in the scheduled sequence π (denoted by π_{end}); π is the final landing sequence which indicates the order of all

the aircraft; and X_{π_i} is the landing time of each aircraft π_i . We resolutely inform that Eq. (1) is the objective function employed by Xu [112]. Moreover, it is important note that minimizing the LLT (or maximizing “throughput”) should not necessarily be the desired objective for optimal sequences, as declared by Ball et al. [25].

-Delay objective: Landing delays was an intrinsic objective for many researchers and practitioners, since airborne delays had been one of the reasons for poor performance in the entire ATC system. It should be noted that the delay of an aircraft is partly due to the delay seen in its entrance the TMA, and partly to the additional delay caused by the resolution of potential aircraft conflicts in the latter airspace. Researchers in streams choose in the processing of ALP to minimize the first part (see. e.g., *Minimizing the total of deviation from the TLTs* [131]; *Average delay minimization* [33]; *Minimizing the total airborne delay* [94]). Yet the second part (i.e., *Maximizing delay minimization*) was welcomed in several models (see. e.g., Lee and Balakrishnan [125] model intended for the ALS on a single runway case; D'Ariano et al. [10] model intended for landing and take-off sequencing on a set of air segment and runway; and Samà et al. [80] model intended for the routing and scheduling of both landing and take-off operations on holding circles, air segments, and runways). On the one hand, Eq. (2)

$$\text{Min } A_{\text{delay}} = \text{STA}_i - \text{ETA}_i - \Delta t \quad f_i \in F_{AA} \quad (2)$$

extracted from [97], represents an illustration of how *minimizing the total delay* is. Note that A_{delay} is the total delay time of the whole flight sequence; STA_i is the actual time of flight f_i arriving at the airport; Δt is a certain range to express the fact that the actual time later than the estimated time within Δt is not regarded as delay; f_i is an approach flight; and $F_{AA} = \{f_{A1}, f_{A2}, \dots, f_{Au}\}$ represents the collection of approach flights. On the other hand, Eq. (3)

$$\text{Min Max } \sum_a^A \sum_w^W \sum_s^S (AET_{w(a)}^{EF} - MET_{w(a)}^{EF}) + (SLT_{w(a)}^{FAF} - TAT_{w(a)}^{FAF}) \quad (3)$$

represents an illustration of how *maximizing delay minimization* is. The objective consists in the minimization of the maximum consecutive delay over all aircraft at the two meter fixes, the EF of the TMA and the FAF of the air segment. Regarding Eq. (3) which is extracted from [115], A is the set of approach aircraft waiting to be instructed; W is the set of aircraft classes; S is the set of air segment; $AET_{w(a)}^{EF}$ is the actual entrance time which presents the arrival time of the aircraft class at the EF meter; $MET_{w(a)}^{EF}$ is the minimum entrance time of the aircraft class to reach the EF meter; $SLT_{w(a)}^{FAF}$ is the scheduling landing time that allows the aircraft class to approach efficiently the air segment at the FAF meter; and $TAT_{w(a)}^{FAF}$ is the target approach time of the aircraft class to reach the FAF meter of the air segment.

There is a need to specify at the level of the second part, we found more details and contributions when modeling the objective function. For example, the goal in [Samà et al. [53,87]] was to effectively reduce for the combined re-ordering and re-routing aircraft, the propagation delay at both the entrance in the TCA and the runways meters. They proved that such indicator forms an important component in the context of ATC real-time. This is particularly useful because it may simultaneously reduce aircraft travel time and their energy consumption. In addition to [Samà et al. [53,87]], Samà et al. [80] carried out more specific work using the following notation “exit delay” for the aircraft delay factor. The exit delay of each aircraft which is determined at the RT meter was composed by the unavoidable delay (i.e., the consequence of a possible late entrance in the TCA which cannot

be recovered by aircraft re-scheduling), plus the consecutive delay (i.e., the additional delays caused by the resolution of potential aircraft conflicts in the TCA). The total of these two components was the subject of many papers (see, e.g., D'Ariano et al. [10,54,140]). Consideration is given in [65] to the development of a new goal viewed as the minimization of maximum consecutive delays for a landing and take-off scheduling solution. The maximum consecutive delay minimization is inspired on the makespan minimization. In fact, it has the advantage to compute more compact schedule within a TCA with interdependent runways and air segments, compared to the minimization of the total consecutive delay.

Let me draw your attention to the *Punctuality Objective*, in relation to the *Delay Objective*. The punctuality is used as an indicator of fairness of the ALP, which was little investigated or may be indirectly studied. In this respect, Kopolke et al. [111] and Samà et al. [64] distinguished that the objective of maximization of punctuality is equivalent to the minimization of the deviation to TLTs, which are planned in a mid-term horizon and published in the flight schedule (see, e.g., Pinol and Beasley [47] and Beasley et al. [75]).

-Operational costs objective: An additional penalty has to be paid if an aircraft operation (i.e. landing or take-off) is further delayed with respect to an initial solution. For example, a penalization should be imposed when an aircraft landing before its TLT [130] and this penalty will grow with the difference. As is clear, the former penalty is proportional to the amount of deviation [74]. It depends on the respective operation class; i.e., operation type and aircraft class, and on the length of delay [41]. As a consequence, contributions to minimize costs incurred from delays due to early and in late landing were numerous (see, e.g., Câmara et al. [2], Salehipour [19], Vadlamani and Hosseini [138], Awasthi et al. [141]). Bencheikh et al. [142], for example, set up a goal for minimizing the total deviation costs from TLT, including the costs of both earliness and lateness landing, in order to propose a solution for the ASS procedures. It is important to note that the cost could be a linear, piecewise linear or non linear function [21]. In fact, Eq. (4)

$$\text{Min} \sum_{i=1}^n (\vec{t}_i \times \vec{c}_i + \overleftarrow{t}_i + \overleftarrow{c}_i) \quad (4)$$

described this objective function type. The objective detailed by Xie et al. [143] consists more thoroughly in the minimization of the total cost of landing deviation from predefined target time for a given set of landing aircraft. Note that $\vec{c}_i$ is the cost of late landing of aircraft i , $i \in [1, 2, \dots, n]$; $\overleftarrow{c}_i$ is the cost of early landing of aircraft i , $i \in [1, 2, \dots, n]$; $\vec{t}_i = \max(0, t_i - T_i)$ is the delay of landing aircraft i where t_i is the SLT of aircraft and T_i is the TLT of aircraft i ; and $\overleftarrow{t}_i = \max(0, T_i - t_i)$ is the earliness of landing aircraft i (landing before the target time) where t_i is the SLT of aircraft i and T_i is the TLT of aircraft i .

In addition to the cost of flight delay, there may be other costs to consider such as fuel consumption. For this purpose, some researchers (see, e.g., Pinol and Beasley [47], Ernst et al. [61], Kopolke et al. [111]) focused on the minimization of fuel costs of all aircraft ought to be landed, by meeting their most economic TLTs at their preferred speed. We see another stream of existing literature where the objective function as the cost function for the ALP is usually comprised of the total average airborne delay (see, e.g., Balakrishnan and Chandran [33], Eun et al. [107]). An increasingly number of recent studies has produced goals related to the cost that has a direct relationship with the most important stakeholder of the air transportation system, the "airlines". For example, Mokhtarimousavi et al. [59] found a way to minimize some extra costs that may be imposed on the airline for the ASS,

such as costs yielded by additional stay of aircraft at the airport, apron, and parking. Moreover, Rosenthal and Eisenstein [144] proposed another cost factor is that of passenger delays concern. The reasons of this choice are that we know that when some unexpected disruptions occur, passengers are inconvenienced, and various costs are generated to their carriers, such costs that an airline incurs to put them on later flights, or to put them up in a hotel overnight. Consequently, the carriers wish to minimize the number of delayed passengers, in order to save money, and also improve (or at least not harm) their customer satisfaction ratings. This is a point on which Rosenthal and Eisenstein [144] are focused, in order to re-schedule a sequence of arrival on a single runway. The Soomer and Franx [71] case which has been carried out by Eq. (5)

$$Z = \text{Min} \sum_{i=1}^N C_i \quad (5)$$

is taken here as an example in the sense of an illustration. This equation is expressed as the total scaled costs where each airline j with N_j arriving flights provides a convex piecewise linear cost function for its flight, and C_i is the decision variable cost for landing flight i .

We have also observed other objectives in which the researchers tackled one particular aspect of the ALP.

-Workload leveling objective: With regard to this item, Beasley et al. [75] dealt with the ALS on a single runway so that the controllers' workload is balanced. In addition, Zhou and Jiang [97] composed multi-objective optimization model for scheduling approach and departure flights at multiple runways. Note that the balance of the controllers' burden is among the goal to be achieved in the model. The Eq. (6)

$$\text{Minimize } Z_{\max} - Z_{\min} \quad (6)$$

illustrating the problem of balancing workload is depicted from [127]. It should be noted that $Z_{\min}$ represents the workload on the most lightly loaded runway, and $Z_{\max}$ represents the workload on the most heavily loaded runway.

-Environmental objective: The major environmental issues of descent and landing operations in the TMA are the noise footprint on the ground, the fuel used in this sector, and carbon emissions of the descending aircraft [116]. We note that, notwithstanding the stronger need for environmentally friendly aviation, there is a very limited literature of operational models that directly consider the negative environmental impact of air traffic, which is a source of interest for the government. The few studies available on this subject include on the one hand [100] that tried to minimize the total fuel burn in order to resolve the ASEP on a single runway. On the other hand, Mokhtarimousavi et al. [59] succeed in minimizing the carbon dioxide pollution by the aiming to minimize fuel consumption cost in the ALS process. In other cases, Rosenthal and Eisenstein [144] marked that reduced overall fuel burn costs will be translated into lower carbon dioxide emissions. Finally, it is imperative to inform the reader of the accurate model of Sölveling et al. [145]; one model has been proposed to provide insights regarding the value of environmental optimization for scheduling the arrivals and departures on two runways. Especially, they sought to capture the savings that could be completed through environmentally schedules, which could significantly produce benefits for both airlines and society. In this classification one should not neglect the efforts dedicated by Tian et al. [32]. These researchers had paid increasing attention to the emissions and noise levels in terminal areas. In fact, they put forward a complete framework that, at the same time, aims at

reducing the delays of arrivals incurred at the arrival fixes, the emissions of the pollutant, and noise level as shown in Eq. (7)

$$\text{Min } EN = \min \left\{ \sum_{m \in M} [\text{Normal}(E_m)] \right\}. \quad (7)$$

We would stress that EN denotes the sum of normalized emissions of each pollutant; E_m is emission of pollutant m ; and M is the set of pollutants which are respectively NO_x , CO , and HC . It must be emphasized that Tian et al. [32] have concentrated on a case study of one of the busiest multi-airport terminal areas in China, the Shanghai terminal area. Finally, with aircraft have large noise levels over the background noise compared to other transportation means, the noise restrictions are considered in [146]. They provide a linear bi-objective model for the ALP on multiple runways that minimizes not only delays but also noise impact and total fuel consumption.

-Control objective: For the management of terminal approach and landing phase of flights, Artiouchine et al. [82] opted for a goal based on minimizing the number of possibilities which the aircraft may use the HP procedure. We will profile the objective function of Ben Messaoud et al. [115] for this particular classification. Eq. (8)

$$\text{Min} \sum_a \sum_w X_{w(a)}^H \times (AET_{w(a)}^{EF} + (TM_{w(a)} \times V_{w(a)})) \quad (8)$$

is expressed as minimizing the maneuver time that the aircraft class performs, because as we all know, the aircraft class is forced in certain circumstances by the air traffic controller to perform a maneuver. When it comes to this issue, it is clear that $X_{w(a)}^H$ is a binary variable to indicate if the aircraft class $w(a)$ performed a holding procedure, and $AET_{w(a)}^{EF}$ represents the actual entrance time of the aircraft class at the EF meter. Since then, the air traffic controller is limited to the number of holding procedures performed by aircraft class, the number of holding procedure that the aircraft class should perform without exceeding the maximum number of holding procedures permitted by the ATC is a decision variable. This operation would therefore result in a further delay undergone by the aircraft class $w(a)$. It is clear that this latter term will be equal to the number of holding decided ($V_{w(a)}$) multiplied by the time of maneuver ($TM_{w(a)}$) defined for a single holding procedure.

3.4.2. Multiple optimization goal for the aircraft landing problem

Tang et al. [147] pertinently designed the arriving of aircraft from different air routes at an airport as a two objective optimization (Eqs. (9) and (10)). Specifically, they considered simultaneously minimizing the total delay undergone by a number of aircraft to land

$$TLT = \sum_{i=1}^N STA_i, \quad (9)$$

and, the total cost caused by early/late landings of the aircraft

$$TC = \sum_{i=1}^N (STA_i - ETA_i) \times \delta_i. \quad (10)$$

In a quest to better understand the objectives, we quote that

$$\delta_i = \begin{cases} D_i, & STA_i \geq ETA_i \\ A_i, & STA_i < ETA_i \end{cases}; \quad STA_i \text{ is the STA for aircraft } i \ (i = 1 \dots N);$$

ETA_i is the expected time of arrival for aircraft $i \ (i = 1 \dots N)$; D_i is the cost per unit of time for landing after the ETA_i for aircraft i ; and A_i is the cost per unit of time for landing before the ETA_i for aircraft i . Correspondingly Guo et al. [148] presented

a multi-objective to solve the ALS problem. They introduced a general landing cost function to encompass runway throughput maximization and average delay minimization. They mainly focus on two objectives to be minimized: total delay which means minimizing the sum of delay time for every aircraft i if it is postponed

$$f_1 = \sum_{i=1}^N d_i \quad (11)$$

and total cost which means minimizing the sum of additional cost caused by every landing which is at its preferred time, whatever earlier or later

$$f_2 = \sum_{i=1}^N c_i. \quad (12)$$

$$\text{Note here that } d_i = \begin{cases} 0 & \text{if } t_i \leq T_i \\ t_i - T_i & \text{if } t_i > T_i \end{cases};$$

$c_i = \begin{cases} g_i \times (T_i - t_i) & t_i \leq T_i \\ h_i \times (t_i - T_i) & t_i > T_i \end{cases}$; g_i is the penalty cost (≥ 0) per unit of time for landing before T_i for aircraft i ; h_i is the penalty cost (≥ 0) per unit of time for landing after T_i for aircraft i ; t_i is the landing time for aircraft i ; and T_i is the TLT for aircraft i .

The most recent Zhang et al. [21] work provides *Min-Sum expression* and *Min-Max expression* which are taken as a multi-objective while modeling the ALP. For *Min-Sum expression*, Zhang et al. [21] choose the total dwell time²³ in the terminal area

$$\sum F_j \quad (13)$$

and the total flight delays

$$\sum T_j \quad (14)$$

in which the former is completion time-related and the latter due date-related. For *Min-Max expression*, they select only the maximum dwell time

$$\text{Max } F_j. \quad (15)$$

In the list of concrete achievements by the proposed approach, the efficiency completed can simultaneously improve punctual performance, enhance runway utilization, reduce air traffic controller workload, and maintain equity among airlines.

3.5. Distinctions of operational constraints and objective functions

For a definitive distinction of the operational constraints used in the literature for the ALP, we note first of all that the integrated models, proposed by Bianco et al. [34,57] and designed for the scheduling and sequencing of both landings and take-offs on multiple runways, represented in a uniform framework several operational constraints (e.g., longitudinal and diagonal separations time in specific airspace regions), and different runway configurations (e.g., crossing, parallel, with or without dependent approaches). Second and despite its fundamental significance, it is clear that few of the articles dissect the capacity constraint. Finally, to sum up, none of those researchers have considered the atmosphere condition and especially the unavailability of runway until Mahmoudian et al. [123]. Runways unavailability and climate limitations are considered to be constraints in the latter, which made the offered solution for the ALS to be more realistic.

²³ The dwell time is a surrogate of air traffic controller workload, and minimizing this criterion reduces air traffic controller workload [21].

We note some important distinctions according to the most objective functions applied for the ALP resolution, which are extracted sometimes from the related works. First as regards the objective functions definition, as we have seen above, the air transportation system is associated with multiple actors (airport, airline, ATC, governorate). Each of them has its own criteria, which might create a great deal of discussions. Therefore, the negotiation and the regulation between these multiple actors, who have particular needs and different points of view, are required. We also notice few proposed approaches able to simultaneously model all the three components (safety, efficiency, and equity), and the lack of a generally recognized performance indicator to optimize, which would require putting special attention to the objective function definition.

The second distinction was deduced by Rosenthal and Eisenstein [144]. These authors point out that numerous papers have studied flight delay and recovery (e.g., Lieder and Stollatz [41], Ernst et al. [61], Beasley et al. [127]), but show that none of them considers including transfer payments among the carriers, or mentions the inherent problems with private information from different carriers.

The third distinction is regarding the most effective goals that may be appointed for a smooth ALP resolution. On the one hand, Lee and Balakrishnan [125] have dissuaded the different trade-offs that might have existed in the TCA, including the throughput, the average delay, the delay costs, and the operating costs. These scholars numerically show that delay minimization is a favorable objective of the ALS for the single runway case as it usually entails a high runway throughput, while throughput maximization may result in excessive delays. Those distinctions were confirmed by a lot of researchers. The first one related to the minimization of delay is justified by the common preference for these objective functions so far. Indeed, the result in [64], which addressed the integration operations of ground, take-off, and landing aircraft, suggests that the objective functions related to the delay minimization should be preferred over other performance indicators. Furthermore, Riahi et al. [36] use the total weighted tardiness of a schedule as objective, to allocate landing and take-off operations to multiple runways. They warned that this goal not only allows reducing delays, but also maximizing runway usage, thus reducing congestion at airports. Faye [109] proved on its part that a sequence giving the maximum runway throughput might generate tardiness or earliness that penalized airlines and passengers, which confirmed the second distinction presented by Lee and Balakrishnan [125]. On the other hand, Eun et al. [107], based on the point of view of users, declared that the minimization of the total costs which include the cost of waiting for obtaining service, is the more appropriate goal to achieve efficient sequences for the ASS processes. Ghoniem and Farhadi [136] observe in essence with the same subject that minimizing the total deviation from target times of take-off or landing operation has the benefit of conforming to the prescribed landing and departure times, which in turn can help to guarantee smooth and efficient airport operations, and would appeal to airlines and passengers alike. Finally, Ji et al. [129] who consider the dynamic case of the ASS problem, has paid attention to this challenge. They have in fact proven the makespan objective function to be as effective to underpin the safety at the TMA, compared to two investigated objective functions, the total airborne delay and the extra costs, respectively.

The fourth distinction by Samà et al. [65] revealed the existence of several of performance indices, reflecting the interests of different actors involved in ATM, such as the aircraft punctuality, the utilization level of airport resources, the cost incurred by different airline companies in terms of delays, the broken flight connections and energy consumptions, and so on. These indices

should all be taken into account in the development phase of schedule that can be measured in terms of aircraft arrival times at the EF and the RT meters.

The fifth distinction is relative to the mathematical shape of the objective functions. In the literature, scholars used either linear or non linear function. For example, Pinol and Beasley [47] and BenCheikh et al. [149] presented a mathematical formulation with a linear and non linear objective functions for the ASS on the runway. In some situations, the linear objective is perfectly reasonable (e.g., if the suggested model is designed to land aircraft as fast as possible; or if a multi-runway model is intended to balance workload between runways). But according to Beasley et al. [75], the non linear objective would in many cases be necessary. It is in fact essential that we look at how a significant number of factors could be easily integrated and combined into a non linear objective function. Obviously, this will bring many candidate results to the decision maker such as landing sequences with desirable characteristics.

The sixth distinction is that, there are scholars who choose to resolve the ALP with an objective function, and they who are aiming to propose multi-objective functions in order to properly reflect the preferences of several players. Mokhtarimousavi et al. [59] studying the ALS on a single runway, conclude that on one side the important step for running the model consists in the adequate selection of multi-criteria objective function since it can potentially satisfy the interests of all parties or provide acceptable compromise between them. Samà et al. [65], who are examining the landing and take-off scheduling with interdependent runways and air segments of the TCA, view that on the other side the optimization models with a single objective function are more suitable than multi-objective approaches. Under these circumstances, it is understood that most of the existing methods solve the ALP via the optimization of one predefined measure, and the study on this problem from multi-objective aspect is still few [148]. This is especially so given that the multi-objective ALP is time-consuming and a robust method is needed to solve it. However as we have described in details, in practice, multiple measures of performance, or objectives, are often required to be taken into account simultaneously when determining the optimal solution to this real-world problem. Therefore, the ALP can only be a multi-objective optimization problem with lots of criteria.

The last distinction is concerned about the lack of personalized models that brings attention to the load of staff on the ground. It is worth mentioning that after aircraft landings on the airport; the ground staff workload is becoming very large. None of the previous approaches has elicited the impact of landing sequence(s) on staff workload on the ground before the work of Boysen and Flidner [128]. The latter introduced three novel objectives of full support to the subject of staff on the ground for the ALS issue. These objectives are aimed at leveling the ground staff workload by evenly spreading over the planning horizon, the number of landed passengers, the number of landings per airline, and the number of landed passengers by airline. Mathematical models along with difficult results were developed, and precise heuristic solution procedures are also presented.

3.6. Modeling techniques

The modeling which consists in minimizing or maximizing an objective function under a set of constraints is a crucial step in solving the ALP. You can find in the literature, a multitude of models designed for the ALP modeling. This is demonstrated by the fact that many factors, including different decision variables, objectives, and constraints are needed to be taken into account in the problem formulation. This sub-section seeks to identify the most modeling techniques represented in the existing literature.

3.6.1. Mixed integer programming

Most researchers, such as Heidt et al. [39], Murça and Muller [62], Samà et al. [65], and Beasley et al. [127], deal with Mixed Integer Programming (MIP) formulations for the modeling of the ALP, or a particular aspect of the landing process, or even landing and take-off operations. This is mostly based on the assumptions that the Mixed Integer Linear Program (MILP) prohibits real-time applications as the computation time is very sensitive to changes of the dependent variables [126], presents the advantage of greater flexibility in modeling, and allows easily to introduce the desired extensions (e.g., the permissible landing time window, the minimum separation time) [124]. In addition, this technique has the possibility to pact with the complexities of the practical scheduling/sequencing aircraft landings within the ATC environment: control, separation times, latest times, runway allocation, and objective function. At present, according to Sabar and Kendall [108], MIP formulations have been able to find high-quality solutions for the ALP, and even obtain optimal solutions for small instances. However, the performance of that approach deteriorates for largest instances. The commercial MILP solver namely IBM-CPLEX is used to formulate and solve the MIP formulation of the ALP (see, e.g., Beasley et al. [43] who treated the ALS on a single and multiple runway; Diallo et al. [121] who treated the ALS for the single runway case; Ben Messaoud and Ghedira [20] who treated the ASS on a single and multiple runway).

3.6.2. Queuing theory

It was normal to envisage the ALP as a particular system of queues. In this context, Bauerle et al. [150] examined a model for aircraft landing procedure at an airport with up to two runways based on queuing theory concepts. The problem was modeled (with dependent services times) wherein aircraft are represented as customers of various types and runways as servers. Each customer's service time (i.e., aircraft) corresponds to the time of separation between the aircraft and its successor. Different queue models can be used to represent the ALP, according to the number of available runways and the way of functioning of every runway (i.e., segregated or mixed).

3.6.3. Relationships of the aircraft landing problem with other optimization problems

The ALP toward an airport platform including a single or multiple runways bears similarities with other well-known job sequencing problems in manufacturing. In the same way, it was seen that the aircraft operations, whether only arrivals, or only departures, or mixed mode, are known to be an application of the Traveling Salesman Problem (TSP) [3] and the Vehicle Routing Problem (VRP) [135].

3.6.3.1. Job shop scheduling. The Job Shop Scheduling (JSS) modeling approach considers the possibility of the airspace interactions between aircraft in order to compute a better quality of aircraft scheduling solutions in terms of delay management and air traffic flow coordination in the TCA [80]. Academic researchers viewed therefore the aircraft operations as an extension of the JSS (see, e.g., Ahmed and Alam [5], Zhang et al. [21], Samà et al. [52], Bianco et al. [79], Briskorn and Stolletz [93], D'Ariano et al. [140], Psaraftis [151]). Regarding the JSS approach specifics, the runways correspond to the machines and the aircraft corresponds to the jobs/tasks. The TLT associated with each aircraft is the ready time of the job. The SLT assigned to the aircraft corresponds to the start time of job. Service times depend on the order of arrival aircraft, while the completion time over the runway corresponds to the time when the aircraft go off the runway. The JSS approach consists in organizing all of the jobs and assigning them to the available machines (i.e., runway(s)); so that the total cost of

treatment is minimized, and with the respect of the dates as soon as possible and latest of execution (i.e., it is about the ELT and the LLT of each aircraft), the regulations times relating to machines which are dependent on the nature of the operations (i.e., it is about the minimum separation time between landing aircraft), and the order of the operation in the jobs. Moreover, the runway occupied time in ALP could be considered as the processing time in the machine scheduling problem [21].

From the set of observations, each scholar employs the JSS as assessment modeling according to their need. For example, the authors in [57] present a JSS approach dividing the approach path into discrete segments and representing each of them by a machine in order to sequence landing and take-off aircraft on all the prefixed routes. This representation of the TMA as a multiple resource problem, allows them to represent the evolution in time of aircraft positions in the latter airspace. Another alternative was presented in [80] where a detailed JSS formulation is proposed for the crossing of landing and departing procedures on a possible set of paths in the TCA, taking into account safety rules and different performance indicators. Holding circles, air segments, and runways are the resources on which [80] depend upon in the model. It is important to note that the JSS model proposed by Samà et al. [52] is the most detailed and comprehensive model offered in this context, in which each operation denotes the traversal of a resource (e.g., air/ground segment, runway) by a job (i.e., take-off or landing operation).

Girish [74] has a different opinion with regard to this issue. He discriminates the important difference between the ALP and the single machine scheduling problem. He explains that in the ALP, the separation constraint must be satisfied not only between immediately successive aircraft in the sequence, but also between all pairs of aircraft. However, in single machine scheduling, the separation requirement between completion times of non-successive jobs does not exist.

3.6.3.2. Traveling salesman problem. For better accuracy, the ALS on a single runway can be abstracted to an open TSP with ready time, where the runway is regarded as the node and the aircraft as the traveler [112]. In fact, the intercity distances represent the separations between aircraft, and the time windows to visit each city are the landing time widows. Concerning the ALP on multiple runways, it looks like a multiple TSP. Bianco et al. [152] make the first attempt to model the ASS issues as a permutation problem similar to the TSP. Then, Ghoniem et al. [38] make contributions and modified the work of Bianco et al. [79] in order to take advantage of the structure of an asymmetric TSP with time windows for sequencing the landing and take-off operations on a single runway based on static information. Ghoniem et al. [38] affords the construction of certain partial convex hull restrictions to further tighten their representation by using the Reformulation Linearization Technique (RLT). Liu et al. [98] propose from their parts a formulation based on Time-Dependent Traveling Salesman Problem (TDTSP) to solve the stochastic ALP. For the sake of conciseness, the TDTSP is a version of the classical TSP, where the costs for an arc between two successive nodes depend on the time period on which the first node is visited.

3.6.3.3. Vehicle routing problem. Briskorn and Stolletz [93] specifically mention the resemblances between the single and multiple runway ASS and the VRP. Indeed, they consider the set of customers partitioned into subsets correspond to the set of aircraft partitioned into classes. They assume that distances between customers only depend on the subset the customers are contained in. Then, the distance between two customers corresponds to the separation time between two aircraft. Each customer has a service interval in which he has to be served and a TLT in this interval as to when he prefers to be served. Lower and upper bounds of the

interval correspond respectively to the ELT and the LLT. Finally, runways correspond to vehicles can dispatch. The objective of this work was to serve each customer in his service interval such that total penalty for deviation of service time from TLT is minimized.

4. Aircraft landing problem classifications

It is possible to mention that an ALP unified description does not exist across the literature. In most cases, the description depends on the researcher's choice. For example, Sabar and Kendall [108] choose to meet the expectation of the ATC, whereby the goal is to assign for every aircraft reaching the TMA, a landing time and a runway with a minimum total cost deviation from the TLTs, and while respecting the time windows constraint and the separation time between two aircraft landing on the same runway. Samà et al. [52,80] have linked farther the ATC and the ATFM functions in such a way that potential conflicts between the departure and landing flights are efficiently avoided when traversed a set of paths, through, respectively, the TCA resources including holding circles, air segments, and runways. In light of these, it is possible to identify four classifications of the resolution to the ALP: type of information, scope of model, framework of model, and type of resolution technique.

4.1. Type of information

The first classification of the ALP is based on the type of information. There are basically two types, namely:

- *Static Air Traffic Models*

Static, or off-line, or even deterministic model; it is the ALP version where some information's are known in advance. The set of aircraft that are going to land and must be scheduled and sequenced, the number of runway, and the position and the speed of all aircraft, are often been initiated as known information in the different research works when dealing with static case (see. e.g., the reviews in [2,17,93,141] and [143]). It is possible to insert the static approach into a dynamic system that solves a particular issue of aircraft operations with static information (see. e.g., Samà et al. [153]).

- *Dynamic Air Traffic Models*

Dynamic, or on-line, or even stochastic model; it is the case where decisions about the landing times for aircraft should be made as, time passes, and, the situation changes (e.g., aircraft that landed, new aircraft that arrives in the airport radar range). The ALP with dynamic information (see. e.g., Beasley et al. [43], Sölveling and Clarke [102], Ji et al. [129], BenCheikh et al. [120], Hu and Chen [154]) requires the computation of an aircraft schedule every time of reach of a new incoming aircraft.

4.2. Scope of model

The second classification of the ALP is the scope of model used in the resolution, which can be classified into basic or detailed models. The former includes only the runway(s) resource in the TMA, while the latter schedules aircraft on other related TMA resources.

- *Macroscopic (Basic) Air Traffic Model*

The choice of a simplified model to build an optimal solution for the ALP by several researchers such as Lieder et al. [15] is motivated by the fact that the runway(s) is often the bottleneck of the TMA.

- *Microscopic (Detailed) Air Traffic Control Model*

Recently, it was deducted that the complex system for runway is not always the main system bottleneck in the TMA, but that the latter may also occur on other relevant resources. It may happen, for example, in the fixes [57], in the air segments [53],

and at the stand area [110]. In this context, the detailed model to solve the aircraft operations has been the subject of several studies. A detailed model of the landing and departing procedures traversed a set of possible paths in the TCA resources (holding circles, air segments, runways), emerges from [80]. In addition, Ben Messaoud et al. [115] present a compact MILP for aircraft from their entering the TMA to their reaching the terminal point, the RT.

4.3. Framework of the model

The third classification of the ALP is based on the framework of the model. Note that there are who is looking to solve the problem as a whole (*centralized approach*), and those who split up the problem to be addressed into smaller and computationally tractable problems (*decentralized approach*) (see e.g., Furini et al. [58], Mahmoudian et al. [123], Ji et al. [124], Rezaei et al. [131], Samà et al. [153]). Rolling Planning Horizon (RPH) [41], Receding Horizon Control (RHC) [94], and sliding windows [97] are common approaches for dividing the problem under study. Furini et al. [58], for example, present a Rolling Horizon Approach (RHA) which partitions a sequence of aircraft operates as an arrival or departure into chunks, and solve the aircraft sequencing problem individually for each of these chunks. In a further example, Ma et al. [48] introduce a Time Decomposition Approach (TDA). It consists in portioning the whole of the time interval under consideration into several overlapping time windows, and in solving the aircraft merging and sequencing problems in each such sub-window individually. We note for this classification a comparison between centralized and rolling horizon approaches in [87] that studying take-off and landing operations scheduling and re-routing aspects.

4.4. ALP resolution techniques

The fourth classification is based on the algorithmic approaches. In fact, the literature contains a variety of approaches that have been examined for the ALP. They can be classified into two categories: the exact approaches (e.g. Linear Programming (LP) [135], Dynamic Programming (DP) [15]), and heuristic approaches (e.g. Genetic Algorithm (GA) [139]). Each of these approaches has its own advantages within a certain application field. Wu et al. [90] give a suitable statement with regard to these two approaches, and declared on one side that exact methods can quickly find an optimal solution, allowing it to be applied into the ATM system. On the other side, heuristic methods have made an immediate effect on it good way at solving the Combinatorial Optimization Problem (COP), and can yield perfectly satisfactory solution within a reasonable time. In this section, we will present the approaches that have been used to solve the ALP on a single and multiple runways: exact approaches in Section 4.4.1, heuristic approaches in Section 4.4.2, simulation method in Section 4.4.3, and software systems in Section 4.4.4.

4.4.1. Exact approaches

■ Dynamic Programming

The DP is an efficient and flexible method seen the benefits of producing quickly different optimal solutions. As Psaraftis [151] said, other features that make the DP approach so attractive are the relative ease for purposes of computer coding, the capability of a specialized constraints examination, as far as the effective "re-optimization" ability. We added that the DP meets the TMA requirements, mainly scheduling and re-scheduling when new events or updates of data are obtained. In both contexts, the DP exhibits fast computation times, as confirmed by Lieder and Stolletz [41].

Psaraftis [151] reported the use of DP-based approach for the static version of the ALS over a single and two runways as early as 1978. Two objective functions were envisaged in this work: the throughput by landing the last aircraft in the sequence as soon as possible and the delay by minimizing the sum of the waiting to land times for all passengers in the system. As an extension to its previous work, the ASEP on a single runway is solved by analogy to the TSP in [103], where the cost of the graph represents the landing cost and the route represents the aircraft landing sequence. The aircraft are classified by their classes in this work, and there is no constraint for the permissible landing time so that the flights are available to land immediately. Moreover, he assumes, first of all, the separation requirements to be class dependent, and, secondly, penalties to be non-decreasing in landing time. The DP algorithm created in this work incorporates the CPS strategy. Knowing that the incorporation of the CPS principle has not driven the increase in the DP algorithm computation time in the order that computational effort and implementation occurs only on instances in which the number of aircraft is small.

The work of Psaraftis [103] is generalized by Venkatakrishnan et al. [31]. They modified it in a heuristic manner to take account of time windows constraint to improve the dynamic case of ASEP on a single runway. For the separation times between landings, they built a statistical model based on real data observed at *Logan Boston airport*. Venkatakrishnan et al. [31] developed two DP approaches subject to minimize the cost of aircraft delay in the TMA. In both approaches, aircraft are frozen in the landing sequence once they are ready to land. The difference between these two common approaches is that one is leaving the aircraft time window unchanged, while the other is gradually reducing the size of the time window as an aircraft approaches rising terrain. The computational results were presented for six data sets involving up to 92 aircraft and 2 runways.

As mentioned above in Section 3.3.2, a special case of the TDTSP with time windows, was proposed in [152] to model the steps needed to solve the ALP. Afterward, Bianco et al. [152] presented an efficient DP algorithm, whose objective is to minimize the sum of all distances traveled from the origin to all the other cities to solve the problem denoted as “TSP with the cumulative costs”. Computational results were presented for the solution of problems ranging from 15 to 60 cities. The results demonstrate that problems up to 35 cities can be solved exactly and problems up to 60 cities can be solved within 3% from optimality.

Chandran and Balakrishnan [91] motivated by Dear's CPS strategy, developed an algorithm for generating the ALS on a single runway. The two objectives of increasing runway throughput and increasing reliability, which are apparently conflicting aims, are considered in the present work. They included various operational considerations, such as time window restrictions, minimum separation requirements between successive aircraft, and precedence constraints. Moreover, Chandran and Balakrishnan [91] modeled the distribution of the error, which presents the difference between the actual landing time and the SLT, as a triangular distribution. The contribution of this work is the use of uncertainty to the ELT. Hence, the DP algorithm proposed in this paper has the ability to accommodate several sources of uncertainty, and is efficient for a real-time application. Lee and Balakrishnan [125] extend the latter measures in the previous framework to solve the static version of the ALS on a single runway. A DP algorithm was designed as a method for the determination of the minimum landing cost, while giving the option to provide the delay cost of each aircraft. Simultaneously, the proposed algorithm explores both the CPS and the Time Advance (TA) strategies. The latter is used to decrease the average delay incurred by aircraft, which is an approach that clearly allows the speed up and/or the assignment of a shorter route to the

aircraft in order to arrive earlier than its TLT. It has proven that such strategy is an efficient method to reduce the average delay time without changing the order of the aircraft [67]. Lee and Balakrishnan [125] simulated a fictitious instance of 30 aircraft extracted from the *Dallas Fort Worth airport* in the *United States*. However, Hu et al. [155] stated that the approach proposed by Lee and Balakrishnan [125] does not adapt well because of the limitation of the optimization algorithm applied. Balakrishnan and Chandran [33] made, in an advanced work, a model for the scheduling of whether landing or take-off operation on a single and multiple runway. In this paper, maximizing runway throughput is the core problem in respect of accounting for various operational considerations (i.e., time window restrictions and precedence constraints). In this work, some degrees for fairness are recorded where Balakrishnan and Chandran [33] limit the number of overtaking maneuvers. One main contribution of this paper was to design the first class of scalable DP algorithms for the time-indexed RSP within the CPS framework which is based on two steps. First, the scheduling problem is casted on a directed acyclic graph in which every feasible CPS sequence is represented by a path in the network, and was then solved using a DP approach. The results demonstrated the benefit of using the CPS for the case of greater arrival rate. Balakrishnan and Chandran [33] and Lee and Balakrishnan [125] shall conclude from their propositions that the problems of enhancing throughput, decreasing delay, and ensuring fairness can be effectively modeled and solved in polynomial time (i.e., linearly on the number of aircraft), while accounting for most operational constraints. In addition, there is a greater benefit to re-schedule when the fleet mix is more homogeneous.

The new MILP formulation for sequencing both arrivals and departures proposed by Furini et al. [37] contribute to give a better utilization of airport capacity. Both single and multiple runway settings are considered in this formulation. The objective function aims to minimize the total cost caused by the delay in a sequence. The length of the horizon of the RHA used is defined by two time windows for a limited length of time; the first one is dedicated to freeze the sequence, while the second represents the optimization horizon using a fixed number of aircraft. The approach proposed with this problem over instances based on *Milano Linate airport* outperforms the FCFS procedure. In a subsequent work and as the approach provided by Balakrishnan and Chandran [33] is of significance, Furini et al. [156] generalize their approach that based on discrete time to build an approach in continuous time. It has been shown that the DP method have been conducted in the latter works. However, in [33], the DP algorithm is presented over a so-called CPS network, whereas the DP algorithm in [156] is applied directly to the sequencing of arrivals and departures. Furini et al. [156] focuses exclusively on the static case in order to minimize the sum of weighted deviations from the scheduled arrival times of the aircraft under a minimum separation time between consecutive aircraft. They have investigated several methods by using upper bounds and completion lower bounds in order to reduce the state space of the DP algorithm. Numerical experiments and a comparison with the MILP approach of Beasley et al. [127], based on real-data sets of *Milano Linate airport* (a large sequence of 173 aircraft), are presented. Computational experience though indicated that, from a computation time point of view, the DP algorithm could markedly outperformed the MILP formulation with a comparatively short computation times. The following challenge aroused in the suitability of the proposed DP in the exploration of large instances, and that raised its increasing use in the real-world cases.

Since the DP approach was an attractive method, it was introduced to provide a new optimization algorithm for the field of

aircraft scheduling in [15] and [41]. Lieder et al. [15] resolve dynamically the ALP on multiple independent runways making via use of the aircraft classes and with general assumptions (e.g., positive target times, limited time windows). The method developed in this paper does not consider stochastic events, e.g., changes in the TLTs of approaching aircraft or weather conditions. They assume short planning horizon of approximately 45–60 min, in which it is possible to calculate the TLTs with a high precision and a reliable weather forecast. The numerical experiments show that problems with up to 100 aircraft can be optimally solved within seconds. Lieder and Stolletz [41] generalize the DP approach proposed by Lieder et al. [15] for realistic RSP of landing and take-off operations at *London Heathrow* and *Frankfurt airports*. They have considered complete and diagonal separations, and, interdependent and heterogeneous runways, with the knowledge that *London Heathrow airport* has two interdependent runways and *Frankfurt airport* has four heterogeneous runways. In addition, they propose a RPH heuristic for solving large problem instances. In fact, the RPH divides the large RSP instance into a number of smaller but connected instances that are then iteratively solved. The analyses in the present work demonstrate how the DP approach compared with the FCFS scheduling discipline led to gain additional runway throughput, while still reducing delays. The analyses of the RPH heuristic performance with respect to total delay cost and computation time are also presented. Despite the ability of the DP approach to support realistic problem instances solving within short computation times for both runway settings, Lieder and Stolletz [41] encourage the use of a heuristic approach if the computation times are too long for practical use.

After their surveys for the RSP (i.e., Bennell et al. [30,45]), they devised practical applications for both the static and dynamic fashion of the ASS on a single runway in [24]. A multi-objective formulation is investigated in this paper which takes account of runway throughput earliness and lateness, cost of fuel arising from aircraft maneuvers, as well as additional flight time incurred to achieve the landing schedule. These objectives are bounded by the constraints linked to the ALP: the time windows for each aircraft's landing time, the minimum separation times between consecutive landings, and the limits of deviations (MPS). We note the development of several resolution methods for static and dynamic cases in addition to the DP approach. For example, the Iterated Descent (ID) and the Simulated Annealing (SA) algorithms are proposed for a resolution of the static core. The dynamic problem is tackled periodically up-dating the previous schedule, with an ID or DP solution approach and using a RHA. We also note that in the dynamic case, the frequency of update time and the length of the time window, when aircraft are available for scheduling, are investigated. The results of the proposed algorithms using random data and real data from *London Heathrow airport* show that DP, ID, and SA approaches improve upon FCFS rule across all objective function elements. However, the discussion in this work indicates that the ID in the static case, particularly because of its more modest and predictable computational requirements, is preferred to DP and SA algorithms.

Recently, Faye [109] proposes a mixture of exact and heuristic approaches for the static case of ALP on a single runway. The goal in this work is the provision of quick solutions in terms of computation time for the two issues; the ASEP and the assignment problem. The objective was the minimization of the total cost of landing deviation from predefined TLTs, subject to the ALP constraints: the time windows, the RPS rule, and the safety separation constraints between two successive aircraft landing on the same runway. An exact DP algorithm is made-up to compute the optimal landing times for a proper sequence of aircraft. Then, in order to test the latter polynomial time algorithm performance comparatively than LP method, the two methods are embedded in

an iterative algorithm, based on SA framework. The SA algorithm starts from a given sequence of aircraft and seeks for a sequence of minimal cost. The computational tests, which performed on real data sets of *Leopold Sedar Senghor airport of Dakar* involving small and large instances (10 to 20 aircraft and 100 to 500 aircraft), show many contributions. First, the algorithm provides important gains in speed for the single case of the ASS tasks. Second, the heuristic based on the proposed algorithm is much faster than the one using LP. Finally, Faye [109] pointed out that the proposed model is generic enough to be applied for arrivals and departures.

■ Branch & Bound

There were several research works that employed the Branch & Bound (B&B) to solve ASS problems in an exact way. This method is based on an implicit and intelligent enumeration of the set of feasible solutions. Basic ingredients of the B&B algorithm are two steps as its name suggests: the separation referred to *branching*, and the evaluation. The former consists in decomposing the set of the solution into several subsets which in their turn are divided according to an iterative process. This issue can be visualized as an enumeration tree. The nodes of the tree correspond to the sub-assemblies, and the leaves correspond to the practicable solutions. The evaluation occurs by avoiding the useless exploration of some nodes in order to accelerate the search for the optimal solution. Its strategy consists in calculating, at every level of the tree, the value of every node, in order to decide which subset of the node will be developed. Preliminary research of B&B algorithm to the dynamic nature of sequencing of airplane inside a TMA with one runway dedicate for landings and take-offs is due to the [157] work. The authors mentioned that the initial solution for the B&B algorithm could be obtained by a heuristic approach. The problem under study is compared with the JSS view with constant separation times, but no detailed computational results were given.

Brinton [60] proposed a special technique called Implicit Enumeration (IE) for scheduling landing aircraft and allocating them a runway in an airport with multiple runways. Originally developed by the research center NASA, this algorithm has been implemented in the TMA tool of the Center TRACON Automation System (CTAS). Brinton [60] adopted the principle used in [68] indicating that the optimal results can be obtained by restricting the number of aircraft, which in turn leads to a decrease in the computational requirements. Therefore, it developed a B&B algorithm with binary variables based on enumerating all possible aircraft sequences. Branches in the tree were discarded when the cost of a partially constructed sequence exceeded the best-known feasible solution. The aircraft in the sequence with the minimum of cost will have the priority to be allocated to the assigned runway. For computational reasons, Brinton [60] suggested applying his enumeration algorithm in a heuristic manner via a moving window approach. The dynamic case of the ALS issue is handled in this paper via freezing the position of an aircraft in the landing sequence and associating costs due to changes in the SLT, but no detailed computational results were presented. The proposed technique in this work has been criticized by Isaacson et al. [55]. They declared that the proposed IE algorithm did not adequately address the issue of representing workload as a series of terms in a cost function, what was in fact found unacceptable by air traffic controllers.

Ernst et al. [61] are concerned with the static version of the scheduling problem on a single runway, featuring landings, take-offs, or indeed, a mixture of take-offs and landings. They adopted a JSS approach for modeling the problem. The objective consists in assigning a penalty to an aircraft arriving or departing after the desired arrival or departure time. A key element of the proposed approach is the presentation of a specialized *simplex* algorithm

which very rapidly calculates the aircraft's times by constraining time windows. Hence, they develop a space search for the problem based on heuristics that includes the *simplex* algorithm and a heuristic-based constructive to generate a proper sequence. The heuristic algorithm and the *simplex* method are used to obtain upper and lower bounds for the B&B algorithm which look for the space of disturbance. The approach proposed by Ernst et al. [61] which could be extended to multiple runway case, was tested on instances involving up to 50 aircraft and 4 runways.

An interesting work was presented by Beasley et al. [127] who are committed to solving the static ALS for the cases of both single and multiple runway system. A mixed integer zero-one approach is used to consider the cost of a solution adjustment procedure if an aircraft is further delayed. The MIP formulation of Beasley et al. [127] is probably the most cited up to date [15,111]. Thus, several authors (see e.g., Farhadi et al. [4], Lieder and Stolletz [41], Furini et al. [58], Briskorn and Stolletz [93], Prakash et al. [96], BenCheikh et al. [149]) have proposed generalizations, reformulations, or Valid Inequalities (VI) to tighten the *Beasley* formulation. For example, Ghoniem and Farhadi [136] follows the classical formulation of Beasley et al. [127] to model the sequencing of both arrivals and departures on multiple runways, but eliminates auxiliary binary variables that were used by the latter to indicate whether a pair of aircraft are assigned to the same runway or not. Beasley et al. [127] introduced restrictions on the number of landing in a given time period (over a period of half an hour). The B&B algorithm based on the relaxation of the binary variables in addition to some constraints are proposed as a resolution method. They also suggested an effective heuristic algorithm (whatever is the number of runway) and a LP based on tree search. Computational results for both the heuristic and the exact algorithm are presented with a number of problems designed to test involving up to 50 aircraft and 4 runways. Although their algorithm demonstrates its efficiency in terms of both solution quality and computational time, Yu et al. [99] observed that the computational cost for this work is quite high. In contrast, Ghoniem et al. [38] declare that the continuous relaxation is relatively weak in *Beasley* model. Finally, Guépet et al. [110] show that the MIP of Beasley et al. [127] already offers poor performances because of weak linear relaxation due to the big M constraints.

The scholars in [107] present an efficient static optimization for the dynamic case of the ASS subject to MPS constraints. It is the first work to be concerned with enhancing the range of control actions, using flight delay options in order to increase the effectiveness of the model for generating advisories to air traffic controllers. By taking into account airline preferences, Eun et al. [107] had already tried to limit the aircraft delays and the deviations from the TLT. The airborne delays are modeled in weighted manner so that the air traffic controllers allowed airborne aircraft to wait using only a discrete set of delay values. The involvement of these optimization variables in the proposed algorithm is justified by the limitations of voice communication incurred between air traffic controller and pilots. Eun et al. [107] were formulated the ASS problems as LP with Linear Matrix InequalityS (LMIS) that allows the computation of the lower bound of cost for the best first search in the B&B algorithm, and proposed Lagrangian Dual Decomposition (LDD) for computational efficiency. The solution approach is analyzed and validated using real flight data from the *Gimpo airport* in *South Korea*. The simulation results show that the proposed algorithm using LDD approach has a significantly better computational performance than other scheduling algorithms, particularly for congested airspace situations. According to Wu et al. [90], it is important to underline that the physical meaning of the weighting co-efficient presented by Eun et al. [107] is not clear enough.

Recently, interesting studies have been further dedicated for the simultaneous problems of scheduling and routing landings and take-offs. The aim was the management and the control of aircraft at the TCA level, composed of interdependent runways and air segments. In their working group (D'Ariano et al. [54, 140], Samà et al. [52,87]); we note greater development in the proposed models. Therefore, we are trying to capture the most important considerations for all of them. Practically, first, these researchers (i.e., Samà et al. [52,53,80,87,153,158] and D'Ariano et al. [140]) generalize the approach of Bianco et al. [34] which submits further relevant aspects of the TCA. Waiting time before landing, traveling in feasible time windows, hosting multiple aircraft simultaneously in air segments, and blocking single capacity of the runways, are explicitly modeled by Bianco et al. [34]. Then, the scheduling aspect was modeled with higher levels of detail based on the alternative graph model introduced by Mascis and Pacciarelli [159] that is able to model aircraft movements. It is important to stress that all these works deal with the objective of makespan minimization, and the experiments were conducted whether for *Rome Fiumicino airport*, the world busiest airport in the world by passenger traffic and the main airport in *Italy*, or for *Milan Malpensa*, the second busiest airport in *Italy*.

First, D'Ariano et al. [10] made a basic model for the ATC sequencing of take-offs and landings in a TCA where aircraft routes are decided at a preliminary step. You will find a detailed illustration of the formulation for each specific TCA resource in this work. From a methodological point of view, two versions of the B&B algorithm are exhibited, a version with holding circles and another one without the inclusion of this resource. Then, the two versions are designed to model the sequencing task of arrivals and departures according to safety regulations, the required separation in time among aircraft at air segments and runways. Four heuristics are also proposed as ways to problem solve. Those were exactly the First In First Out (FIFO) rule, the Job Greedy Heuristic (JGH), and the Arc Greedy Heuristic (AGH) with two evaluation criteria, the Avoid Most Critical Completion Time (AMCC) and Avoid Most Similar Pair (AMSP), respectively. These methods are indeed used to obtain the initial upper bound of the B&B method. The B&B algorithm experienced of its share the single machine Jackson Preemptive Schedule (JPS) for computing a lower bound on each air segment and on each runway. The results of different approaches on practical size instances of the *Rome Fiumicino airport* are presented in terms of the computation time, the number of feasible schedules, and the number of optimal solution. The B&B algorithm performed for practically the scheduling heuristics and often reached optimal solutions in short computation time. D'Ariano et al. [54] developed an iterative approach for solving the combined aircraft re-scheduling and re-routing with flexible routes in their subsequent work. The approach is evaluated in terms of propagation delay minimization. The re-scheduling problem is modeled as a generalized JSS problem and formulated via alternative graphs that are able to enrich the [34] model. To solve the aircraft re-scheduling sub problem, D'Ariano et al. [54] used the truncated B&B algorithm proposed by D'Ariano et al. [10]. For the aircraft re-routing sub problem, they have been proposed a Tabu Search (TS) algorithm. They describe and test re-scheduling and re-routing algorithms for Aircraft Conflict Detection and Resolution (ACDR) on practical size instances of the *Rome Fiumicino airport* and for small time horizons (the horizon of traffic of 15 min). The computational results demonstrate the better performance of the TS algorithm in reducing aircraft delays for small disturbances.

Work continued on investigation the potential of optimal aircraft operations scheduling for the detailed ATFM in [53]. It was based on a RHA. At the first stage, they divide the problem into two sub problems: a re-routing problem in which a route among

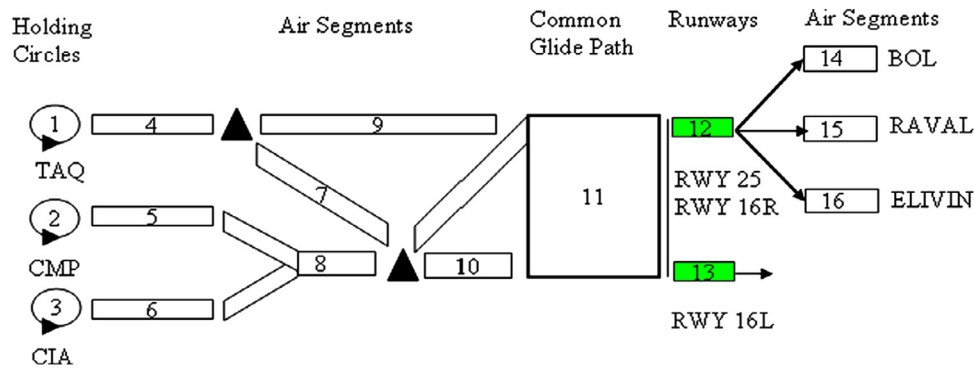


Fig. 7. Schematic view of Roma Fiumicino TCA.
Source: Samà et al. [153].

a set of re-routing possibilities is associated to each aircraft, and re-scheduling in which a scheduled time is assigned to each aircraft operation. Then, the problem is modeled via an alternative graph formulation, and solved by aircraft re-scheduling and re-routing algorithms used previously in [54], namely the truncated B&B algorithm of D'Ariano et al. [10] and the TS algorithm of D'Ariano et al. [54]. Computational experiments are based on practical size instances from the *Milan Malpensa airport*, and are evaluated in relation to different indices: minimization of computation time, delay, and travel time. There are similarities between the two approaches presented in [53] and [54]. The significant difference is in the comparison between the various approaches found in [53]. The latter compare the truncated B&B algorithms for aircraft re-scheduling with fixed routes, the TS scheme for combined aircraft re-scheduling and re-routing, and the FIFO rule. The analyses demonstrate that the B&B finds the best possible solutions in terms of delay minimization, while B&B and TS approach better minimize travel times.

In the year 2013, Samà et al. [153] solved the problem of the assignment of the passing time to each aircraft at all resources of its path (holding circles, air segments, common glide paths, and runways), as shown in Fig. 7, derived from the present work.

In this paper, a detailed model for the two static and dynamic cases and for both landing and departure procedures is presented in order to manage busy traffic situations with a large number of delayed aircraft. The authors presented an alternative graph formulation, and an overlapping chunking rule specifically introduced for the dynamic case where real-time information is provided by ATC system. The RHA method which divides the problem in multiple steps is used for the latter purpose. A B&B algorithm and heuristics are addressed in [153] in order to resolving the different cases. The suggested algorithms are evaluated on practical size instances derived from real scheduling data regarding traffic flows in *Rome Fiumicino* and *Milan Malpensa* terminal control areas. Experimental results proved that the scheduling algorithms based on static information are able to compute a feasible schedule in a dynamic environment in which information on the aircraft landing is updated during the traffic prediction. Besides, compared to FCFS rule, the B&B algorithm considerably minimizes aircraft delays and travel times, but at a greater of the cost of computation time.

During 2014, work continued on studying the simultaneous real-time scheduling and re-routing problems by Samà et al. [87]. They also propose a decomposition of the overall problem into two sub-problems, as in the case in [53]. In particular, routing decisions will be taken regarding the runway resources, since it is proved that this type of routing is more promising compared to re-routing aircraft on air segment(s) previously employed by D'Ariano et al. [54]. Samà et al. [87] use the alternative graph to

obtain a reliable MILP formulation concerning the ATFM issue. The performance of simultaneously addressing the scheduling and the routing algorithms was investigated on traffic situations of the *Milan Malpensa airport* in terms of computation time, number of route changes, aircraft delay, and travel time. Samà et al. [87] also use the TS algorithm of D'Ariano et al. [54] for solving the problem related to aircraft re-routing. Computational results demonstrate the effectiveness of the optimization procedures compared to FIFO rule. Indeed, the maximum consecutive delays are reduced up to 99% when using the optimization-based solvers. In [87], it is important to note a specific combination related to solvers (FIFO, B&B for scheduling aircraft with default routes, B&B combined with TS for scheduling and re-routing aircraft, MILP for scheduling and routing aircraft) and solution approaches (centralized or RHA). Finally, when pressed on the analyses previously presented by Samà et al. [53], we note that the picture has changed here. Indeed, the new MILP model addressed by Samà et al. [87] whose IBM LOG CPLEX MIP 12.0 is the solver, presented better results in terms of delay and travel time minimization compared to the B&B with TS algorithms.

In [158], three models are addressed for the scheduling of landing and take-off operations without and with re-routing process. The difference between the models lays in the measurement of potential aircraft delays at different meter fixes of the TCA in disrupted traffic situations. The first model measured aircraft delays at the RT. The second model measured the delays of arriving aircraft both at the RT and at the EF of the TCA, while the delay of departing aircraft is measured at the runway only. The third model is an extension of the second one with the addition of *deadline constraints*²⁴ for landing aircraft at the EF of the TCA. Various optimization-based approaches are implemented to solve the models. First, the truncated B&B of D'Ariano et al. [10] is adopted to solve the aircraft scheduling and re-routing to near optimality. In particular, a binary branching scheme is implemented in order to take one scheduling decision at a time. Second, a MILP model is investigated for simultaneous aircraft scheduling and routing. Finally, the TS algorithm of D'Ariano et al. [54], combined with the B&B algorithm, are illustrated for the aircraft scheduling and re-routing. It is interesting to recognize that centralized and RHA control paradigms are implemented and compared for the three models. The optimization methods are compared to the FIFO rule, presumably been used as a surrogate for the behavior of air traffic controllers. In [158], we can discover some original contributions related to the computational results which could be useful for researchers. These are truly significant considering that the approaches are tested for practical instance

²⁴ Technique to speed up the solution algorithms, developed in the areas of scheduling theory and constraint programming.

of the *Milan Malpensa airport*, and a detailed comparison of the various solution approaches is reported in terms of computation time, number of optimal solutions, number of constraint violations, and aircraft delay minimization. The objective function which is expressed in terms of maximum consecutive delay minimization exclusively at the entrance of the TCA and at the runway(s) is the first contribution. Samà et al. [158] indicate that there is a trade-off between minimizing delays at the EF meter and delays at the RT of the runway. The minimization of the former delays is necessary for optimizing the coordination with the management of neighboring areas, while the minimization of the latter delays is necessary for minimizing passenger and freight delays, and ground operations delays. The second contribution is about the RHA framework. Samà et al. [158] stress that it is a robust approach, since a feasible schedule is always obtained when using the proposed optimization-based approaches for all instances tested. Finally, details should be supplied about results achieved. As this matter, we note results with/without pre-processing, results for the entry delays, and results for the entry delays plus a disrupted runway. For the delay instances, we have retained that the computational results show the high potential of the optimization-based approaches compared to FIFO rule which was not able to compute a feasible schedule, especially for disruption instances. For example, the B&B algorithm combined with the TS algorithm better minimized the objective function of each model variant compared to the B&B and the FIFO rule, but it was less competitive than the MILP approach. Overall, the MILP was the best approach for the 30 min, 60 min, and 180 min delay instances, either in the centralized or RHA framework. Samà et al. [158] reported on the current need of developing exact methods and good lower bounds to the instances cited.

There is a big similarity between [87] and [158] works. The latter tends to confirm this point of view by saying that their mathematical structure on the ATFM problem and the ATC problem studied in [87] are similar, even if the scope moves from planning to re-plan in real-time. For this reason, we are going to deconstruct just the contributions found in [158]. The first contribution is consistent with the fact that the proposed approaches include the management of severely traffic disturbances (e.g., multiple delayed landing/departure aircraft and a temporarily disrupted runway). The second contribution concerns the extension of some previous algorithms. In the first part, the algorithms in [D'Ariano et al. [10,54]] are extended by means of a procedure to speed-up the reaching of a feasible schedule (i.e., *deadline constraints*), and then the extended version of Samà et al. [153] is described to deal with routing flexibility.

For their parts, Samà et al. [80] built a microscopic formulation to the ATFM issue at the TCA, in such a way that the resulting schedule in the path chosen is conflict free. They include in the mathematical model, in particular, relevant TCA aspects, such as holding circles, intervals for aircraft speeds, multiple capacities of air segments, and blocking constraints at the runways. Two specific objectives functions are analyzed, respectively, the minimization of the maximum consecutive delays and the travel time spent by the aircraft in the TCA with their combination in order to find a reasonable balance.

Building on work arising from [54], D'Ariano et al. [140] introduce alternative formulations for real-time routing, scheduling, and sequencing of landing and take-off operations with fixed routes. They consider the problem of minimizing the delay propagation with respect to the off-line timetable. D'Ariano et al. [140] applied a JSS to model the subjects covered. Then, exact methods (the B&B algorithm), scheduling rules, and heuristic are implemented using extensive computational experiments carried out for the *Rome Fiumicino airport*. Test experiments show that the support provided by these optimization models and algorithms

has a significant potential for improving the performance of the TMA as well as reducing the air traffic controllers workload. They also highlight that the B&B algorithm is able to solve instances of practical size within a time limit compatible with real-time application, and it outperforms the heuristics.

These detailed papers in combination with performing results require a mandatory comparison paper. Therefore, in the year 2017, Samà et al. [65] do not focus on the investigation of, new algorithmic approaches, solutions, or special mathematical properties for the different topics addressed previously, but rather on the comparison of (near) optimal solutions in terms of several performance indicators. Criticizing the RHA proposed in [153] by saying that the rolling horizon approaches or other problem decomposition approaches cannot compute the optimal aircraft operations schedules solution for large-scale problems, lies in their basic distinction.

Most recently, Samà et al. [64] present a single optimization model which incorporates taxiway operation into the microscopic MILP model presented previously for managing the scheduling of the bottleneck operations related to landings and take-offs, so that the stress on the airport resources is larger. The overall scheduling problem which consists of scheduling aircraft from the border of the TCA to the gate, and vice versa, is formulated as a JSS model with additional constraints, such as inter-aircraft separation requirement and interdependent runways. Several indicators are to be optimized. They correspond respectively to the maximization of runway throughput, the minimization of aircraft approach time, the air traffic controllers' workload approach time of aircraft, the aircraft taxi-time, and the arrival/departure delay. In [64], regarding landing operation, we note the formulation and the analysis of two air scheduling policies, *wait-on-route* (i.e., HP) and *free the route*. In particular, Samà et al. [64] proposed and evaluated exact and heuristic methods for the ground and air operations, building on the former papers. Regarding the exact method, the B&B algorithm analyzed previously by D'Ariano et al. [140] is used to minimize the delay by exploiting a lower bound, based on the solution of a preemptive single JSS. Samà et al. [64] have used the same lower bound and implications discussed previously in [140]. The computational experiences based on fictitious instances from the main international airport in *Netherlands*, the *Amsterdam Schiphol airport*, yields very promising results in terms of solution quality and computation time. Samà et al. [64] have proven to be successful in further enhancing the coordination of 70 aircraft partitioned into four different size categories (heavy, medium, small and light) between taxiway scheduling and airborne scheduling. This strong performance was built on solutions produced in terms of delay and travel time. Optimal or near optimal solutions are found within low computational burden which ensure the schedule feasibility in real-operations. Larger buffer times for the other operations taking place at the gate e.g. refueling, cleaning, baggage handling, are therefore guaranteed. This raised the level of realism of the proposed approaches.

The most recent research work of Rodríguez-Díaz et al. [146] proposes a solution for the routing and ALS on multiple runways that minimizes noise impact, total fuel consumption, and delays. Most of all, they define how to assess the noise impact. Then, a stronger formulation under WV separation and CPS restrictions (e.g., $k = 2, 5$) is made. Among the novelties introduced to respect separation restrictions, we observe that the real-time of arrival of flight plan is determined by its departure time plus the flight time of the selected route, and the delay that can have. In addition, the flight is assigned to one route and there are not considering introducing any change in the STARs (i.e., fixing arrival routes). Particularly, the filled flight plan route ends at the point where a Standard Terminal Arrival Route (STAR) begins. These ties correspond with real-world where the STAR, approach route, and

landing runway are usually assigned to the pilot during the en-route phase of the flight. The main results come out from a set of instances consisting of a small sequence of flights (5 aircraft) approaching an airport with various landing runways under high and intermediate congestion. The linear model, solved by the LINGO optimization software, succeeds in achieving the most suitable landing runway and the landing times that minimize noise. More specifically, the research over the populations under study shows potential improvements of up to 4.5% reduction of total fuel consumption (without increasing noise levels) only by modifying the sequence of arrivals, and up to 43% of reduction in noise impact (without extra fuel consumption) by comparing with real data from *Madrid Barajas airport* (the biggest airport in Spain with four runways).

4.4.2. Approximate methods

On one side, the most RSP models are inherently hard to solve from a theoretical perspective [125]. On the other side, the ALP is an NP-hard optimization problem, making it more and more difficult to solve as the size of the problem increases [108]. To deal with such situation, most practical implementation resort to utilize simple heuristic method (4.4.2.1), or meta-heuristic approaches (4.4.2.2). These methods have been successfully applied by several authors (see. e.g., Bianco et al. [79], Beasley et al. [43]). This goes back to the fact that the heuristic is deemed as a better representation of what the air traffic controllers are doing [160]. Moreover, the major advantage of heuristics is the possibility to deliver a feasible solution at any time.

4.4.2.1. Simple heuristic method. The specific heuristic solution is a production method adapted to a particular problem. It is based on a specific “priority rule” with an in-depth route or in width (e.g., FCFS and CPS disciplines, explained previously). Dear and Sherif [70] is among the most distinguished works in this field. The static and dynamic cases of the ASP on a single runway are among the issues discussed. They presented the CPS concept to solve the sub-problem of scheduling of aircraft. For the assignment task, they left the decision to the air traffic controllers. These authors have considered maximizing the throughput and minimizing the delay as objectives, and, minimizing the separation time between landing aircraft as constraint. The CPS was examined via fast-time simulations involving 500 aircraft. The obtained results indicate the effectiveness of this method.

A particular case has been presented by Beasley et al. [75], where a modern heuristic technique called “Population Heuristic”, is provided for scheduling aircraft waiting to land on a single runway during a severe congestion, at *London Heathrow airport* which has two runways. This airport has its own peculiarities and characteristics in terms of runway operation (segregated mode) and the categories of aircraft. The operational data used for the validation of simulation results indicates a reduction in expectations of the order of 2% to 5% in the volume of traffic, compared to current operational practices. However, the problem formulation is static, and nothing is said about its implementation in a highly dynamic environment.

Bianco et al. [34] take into consideration approach path and exit routes, and aim to modelize the TMA inbound and outbound traffic flows as no-wait job-shop problem for an airport with prefixed routes and two interdependent runways. The paths of the aircraft in the TMA are decomposed into machines which represent runways, and the processing times take into account the separation distances between aircraft. Conflicts are resolved, while scheduling and sequencing aircraft operations on the different resources. Bianco et al. [34] address several objectives, such as average delay minimization, minimization of maximum delay, and throughput maximization. The problem is solved by the CPS algorithm which is a fast dynamic local search heuristic. The

computational experience in [34] is based on an implementation of the proposed model and algorithm to two realistic data from the *Milan Malpensa* and *Rome Fiumicino airport* terminal areas. The performance analysis is done by considering three different scenarios: a real scenario with no terminal condition, a congested scenario, and a high congested one.

As we have seen above, Bauerle et al. [150] examined the queuing process of aircraft arriving at an airport, and its implication for the airport capacity. A number of heuristic routing strategies, such as *Coin Flipping*, *Type Splitting*, *Round Robin*, and variants of their *Join-The-Least-Load* rule, have been addressed to solve the ALP on two runways. These strategies are numerically analyzed and compared with respect to the average delay they cause. It turns out that some modifications to *Join-The-Least-Load* could give the best results.

Soomer and Franx [71] and Soomer and Koole [161] recommended new objective functions for considering the ASS issues at a runway, in order to introduce fairness aspects in the ALP and by reconciling airlines companies' needs and point of view. On the one hand, Soomer and Franx [71] assign a landing time to each flight, taking into account the cost incurred by airline companies. On the other hand, Soomer and Koole [161] considered an additional objective which represents the airlines' preferences. The writers in [71] studied the static case of the ASS on a single runway by respecting the rules of separation, and knowing that the proposed model can be used in a dynamic setting by recalculating the schedule every time the circumstances change. Soomer and Franx [71] used the *Beasley et al. [127]* MILP and solved their model with a knowledge based on a specific local search heuristic. In this paper, the model was tested with moderate and heavy surface traffic scenarios in a simulated environment. Indeed, it has been proven that this heuristic is able to solve a number of large instances with over 100 aircraft in few minutes. In addition, the results indicate a substantial decrease regarding the costs for airlines, and an average improvement of 30% in cumulative delay over FCFS practice. Such contributions are still theoretical so far. It is reasonable because Serhan et al. [162] have insisted that airlines are not willing to share their sensitive business data, including their operational cost, the number of passenger on each flight, and the number of transfer passengers. In addition, it is difficult to select metrics that guarantee equity among competing airlines.

4.4.2.2. Meta heuristic algorithms. It was proved that applying meta-heuristics could help improve operator's decisions for landing queue management and consequently decrease solutions costs [2]. Therefore, numerous advanced meta-heuristic algorithms have been proposed in search for good quality solutions for the ALP. Meta-heuristic methods can be classified in two families: methods based on the improvement of a single solution, and methods based on the improvement of a population of solutions. This approach attempts to combine an evolutionary algorithm (or perhaps some other heuristic search algorithm) with an algorithm that belongs to the family of exact methods. The most widely used meta-heuristics are the following: Genetic Local Search²⁵ (GLS); Scatter Search (SS) Pinol and Beasley [47]; TS D'Ariano et al. [54]; Ant Colony Optimization (ACO) Wang et al. [163]; SA (Awasthi et al. [140], Salhipour et al. [164], Rodriguez-Diaz et al. [146]); Iterated Local Search (ILS) Sabar and Kendall [108]; Variable Neighborhood Descent (VND) Salhipour et al. [56], Variable Neighborhood Search (VNS) Salhipour et al. [56], Flower Pollination Algorithm (FPA) Zhou et al. [130], and Bat Algorithm Xie et al. [143]. Several of the proposed algorithms have also been

²⁵ GLS: an extension of the GA achieved by incorporating local search into a GA framework [95].

hybridized in order to combine interesting properties and to take the best from each of them.

■ Genetic Algorithm

The GA is a global search meta-heuristic which aims to evolve a population of solutions through a process of selection, crossover, and mutation during a number of iterations [49]. The successes in applying Genetic Algorithms (GAs) for the resolution of the ASS missions have advocated a strong potential of evolutionary algorithms in dealing with this type of NP-hard problems [94]. Abela et al. [42] have chosen in this field the GA combined with the B&B algorithm to solve the static case of the ALS on multiple segregated runways. The GA provides an upper bound of the optimal solution for the problem. This value is used during the research of the B&B algorithm based on a relaxation of an exact formulation of the problem. The MIP model developed for a predefined scheduling horizon solves the sub-problem of scheduling aircraft landing times whilst keeping the weighted deviations of all aircraft minimized. Yet the assignment of aircraft on the runways was not taken into account. Computational results for the scheduling case with several scenarios from 10 to 25 aircraft of two types to 1 to 4 runways were tested.

Cheng et al. [165] represent one of the first applications of the GA. They provide four different GA schemes to put emphasis not only on the ASS, but also on the assignment of incoming aircraft to a number of runways. The GA parameters are: the coding of the solutions (by two chromosome representations for sequence and runway assignment, by one chromosome representation for sequencing and runway selection priority, by one chromosome with chromosome selection based on fitness, and by a chromosome representation defined as mathematical operations and functions), the selection (by the roulette, by the rank, and by the tournament), the two-point crossover and the mutation by inversion. The findings of this work exemplify the importance of the selection of the chromosomal representation for a genetic search problem. Computation results were provided for the four approaches using instance including 12 aircraft and 3 runways.

On their part, Ciesielski and Scerri [166] designed two different GA adaptations to stem the ALS dynamically in two runways case. The proposed approach consists of finding the best solution within 3 min, updating the situation after each appearance/landing of an aircraft in order to find the best solution. There was no link between the previous set of landing time decisions, and the new one. In this coding, the non-feasible individuals are not eliminated but penalized. For the experiments, two data sets from *Sydney airport* incorporating the use of its both runways were generated: one contains 28 aircraft arriving in a 37 min period and the other contains 29 aircraft arriving in a 38 min period.

A specific GA is building by Capri and Ignaccolo [139] for the ASEP on a runway by the introduction of departures. The GA works with a fixed population of chromosomes that are encoded as integer vectors, and the reproduction is made at random. In this paper, the model definitively proved that it is very effective and fast to produce optimal solutions, following the simulation of a sequence of 44 aircraft.

Hu and Chen [154] set the minimization of the sum of the difference between the predicted and the allocated landing times of each aircraft as objective to resolve the dynamic version of the ASS on a runway. On the other side, these writers did not use the GA in an isolated way, but rather integrated into the RHC concept, which increased the real-time implementations of the reported GA with largely potential. The results of the simulation of a sequence consisting of 30 aircraft show that the fairness of this method to solve the ASS tasks in a dynamic environment is respected. Hu and Paolo [113] continued work to design binary-representation-based GA for issues of ASS. They consider two

objective functions: the minimization of both the total airborne delay of all aircraft and the maximum length of all arrival queues. The first objective function emphasizes the operating cost of airlines, while the second focuses more on the efficiency in the use of airport capacity. They also explained the GA applications to cope with ASS problems on multi-runway system. They presented a GA that uses the relationship between aircraft for making chromosomes. This activity makes designing of a uniform crossover operator possible, which was difficult to run for those GAs that have been designed directly for aircraft arrangements in entry queue. The results were presented in a simulated body compound of 12 aircraft arriving to 3 runways using the Monte Carlo (MC) simulator.

Pinol and Beasley [47] investigated an interesting formulation of the static version of the ASS for interdependent and heterogeneous runways. This formulation which considers extensions of the previous model given in [127] was performed by a MIP $\{0, 1\}$ with 0 (P^2) variable and 0 (P^2) constraints, where P is the number of aircraft. The latter formulation considers extensions of the previous model given in [127]. Time windows and separation distances between two successive aircraft are assumed to be runway dependent. These new criteria involve respectively linear and quadratic constraints. Then, two population-based heuristic solution approaches with special features which are in essence of GAs are presented. This includes the SS and the Bionic Algorithm (BA). Both algorithms use a local search technique to optimize the new individual's time line. The benchmark solver used a local search not discussed in details. The results of the simulated body of 500 aircraft and 5 runways showed good solutions that can be produced cost-effectively and relatively quickly. We note that SS solves the problem faster than BA in the different cases studied. This common discrepancy is in parental selection scheme. In each iteration of the simulation the specific feature of SS algorithm is that a unique set of parent is built and only one child is generated and improved. However, in the BA, several sets of parents are constructed from the current population. Thus in every iteration many children are generated and must be improved. Furthermore, the SS exceeds the BA with respect to the quality of the solution when the goal is linear, and, BA exceeds the SS when the goal is non linear. In relation to this latter work, Liu [98] have ascertained that in real life situations the runway dependent version holds more than the interdependent version employed in [47]. Then, he depicted a solution procedure based on the integration of the *threshold accepting mechanism* into the GLS algorithm for solving the static case of multiple runways ASS with runway dependent attributes. In fact, the proposed GLS algorithm has been able to efficiently find the best arrangement of runway allocations, sequences, and corresponding landing time for arriving aircraft, by minimizing total delay and following the separation requirements based on aircraft types. Liu [98] used the five test instances previously created and investigated in Cheng et al. [165] and Pinol and Beasley [47]. On the one hand, the superiority of the GLS is in terms of finding the objective function values and computational time as compared to GA, BA, and SS algorithms. On the other hand, it is in the position to solve the runway dependent ALP when the number of runways is five which corresponds to the major hubs in real world, or when a runway restriction is imposed.

Kupfer [126] is part of the development of a solution for the ASS to two runways. He contributes to this issue by taking into consideration a specifically runways layout: dependent, very closely, spaced, and parallel. We note, however, that the schedule computation occurs at a specific coupling point. In this respect, aircraft are coupled for parallel approach segments. For this reason, we notice the route information as new input that was not discussed in the previous works. The purpose of the MILP model

was to minimize the makespan, taking into account temporal, pairing, sequencing, separation route, and grouping constraints. Then, Kupfer [126] proposed a set of scheduling methods. A general GA and an improved one where a *Greedy algorithm* was embedded into the GA were among them. These methods were compared with two specific FCFS scenarios. The first one represents a possible schedule under Visual Meteorological Conditions (VMC), and the second one is the schedule that would be applied under Instrument Meteorological Conditions (IMC). The findings of the simulations using a set of 20 aircraft, distributed over 30 min, show that the advanced scheduling methods (e.g., the improved GA) resulted in better schedules compared with the FCFS method. For evaluation to serve its desired scope “*the runway layout*”, it was expected that the establishment of safe and efficient schedules for simultaneous aircraft landings to closely spaced parallel runways is one way of increasing in capacity at hub airports without increasing airport footprint.

Since the Multi-Objective Evolutionary Algorithms (MOEAs) provide the controllers a lot of choices with different level of tradeoff between the total landing time and cost [147], this method have attracted the attention of many researchers in the last 20 years. Yet, the mathematical model in [59] involved a tree-objective for the static ALS problem on a single runway. Then, an elitist multi-objective evolutionary algorithm named Non-Dominated Sorting Genetic Algorithm *II* (NSGA- *II*) was introduced to solve the problem for two instances including 20 and 50 aircraft. In the tests, it was observed that a correct sequence in an appropriate time was achieved by reducing the costs imposed by the system and fuel costs.

In the same year 2015, a multi-objective optimization model is formed by Zou and Jiang [97] as in [59]. The model is dedicated to control the scheduling of aircraft operations, approach, and departure flights, on multiple runways. In this process, several factors are considered, such as the weather, ATC rules, flight safety interval, runway throughput, flight delays cost, and air traffic controller workload. The GA combined with *sliding time window* algorithm is the proposed method for resolving the issue. The simulation results reveal that the improved algorithm gets the better of optimization performance in terms of computational time and work efficiency compared with FCFS algorithm and the traditional GA.

Ahmed and Alam [5] studied the scheduling and sequencing for integrated arrival and departure in a single runway scenario. They develop a target for runway throughput capacity, and proposed an improved GA with a *dynamic window*-based optimization approach to optimize and integrate the sequence of arriving and departing aircraft for a given time window. The evolved optimal arrival and departure sequencing was analyzed for different aircraft configurations, using the *time space diagrams*. The results of simulation indicated that the proposed model compared with FCFS rule succeeded in obtaining optimal sequence that takes lower processing times as well as achieves a higher throughput approaches. The results also demonstrate the strength of evolutionary algorithms in dealing with NP-hard problem from real world.

■ Ant Colony Algorithm

The research methods by the Ant Colony Algorithm (ACA) have a special feature: the search for solutions is made on the basis of all the population and not from a few best individuals (as it was the case in the GA). However, ACO has been proven to be suitable and very promising in solving various COP with dynamic nature because of its flexibility and adaptation to environment changes [120].

The first efforts are due to Randall [167] who proposed an approach based on ACA to solve the RSP at a runway. The objective was to minimize the penalty cost. The basic principle used in this

algorithm was to divide the landing window into several equal intervals. An aircraft lands in one of these time intervals according to the value of the pheromones. These latter are more important when approaching scheduled landing times. This algorithm does not take into account the aircraft assignment on the runway.

The paper of Zhan et al. [94] makes a first attempt to propose a new solution framework for the ASS problem on a runway using an Ant Colony System (ACS). This algorithm is introduced in real-time scenario and is based on the RHC approach in order to optimize the scheduling from the robustness and effectiveness of the queue model on arrival flights. It appears that at present, the latter approach is suitable for the ASS, especially when considering dynamical environment with rapid growing changes [90]. The ASS issues were formulated in the form of permutation problem to ensure an optimal sequence for aircraft landings with a minimum airborne delay. The proposed algorithm is further enhanced by a two-opt exchange heuristic local search. Then, they generate the model to test problems derived from an uncongested situation, a normal situation, and a congested situation, with 30 aircraft as well as with 60 aircraft. The results of the FCFS rule, the GA-based approaches developed in [154], the ACS without embedding the RHC technique, and the proposed improved algorithm are respectively presented and compared. The experimental results indicated that the proposed improved algorithm outperforms the FCFS rule, the ordinary ACS without embedding the RHC technique, and the GAs, in achieving the ASS with high robustness, effectiveness, and efficiency. In [94] and [58], the aircraft are all participants at all times in the sequencing procedure, but in reality the terminal area has a limited capacity. This essential factor was often ignored in these latter references, according to Wu et al. [90].

BenCheikh et al. [142] and BenCheikh et al. [149], count among those who several times have developed the ACA for solving the ALP for a single and interdependent runway. First, the ACA is used by BenCheikh et al. [142] to generate a subset of good solutions which are later considered as initial population for the GA method to solve the ASS on single and multiple runways. Then, BenCheikh et al. [149] tackled the static case of the ASS for single and multiple runway cases. A heuristic was proposed, firstly, for scheduling aircraft landing times on a single runway from an order determined by a priority rule. Secondly, the ACA was improved by incorporating a local search heuristic for the assignment of aircraft landing time for each runway level, to consider the extension to multiple runway case. The ACA was improved to be adapted to the dynamic nature of the problem where two forms of effects are considered: the cancellation of a flight and the closing of a runway. This adaptation is at the level of the graphical representation by the addition or delete of an aircraft or a runway. The algorithm has been tested on instances involving 10 to 50 aircraft and 1 to 5 runways.

The proposals in the most recent paper of BenCheikh et al. [120] are the modification of the ACA version in [149] dedicated to the static case, in order to solve the dynamic version of the ASS and their assignment on multiple runways. The forms of effects of the dynamic aspect considered in this work are the re-scheduling and the freezing time, which differs from those applied in [149]. In this formulation, the objective is to minimize the total penalty cost, and the displacement function which corresponds to the total move of the current solution from the previous one. An improved ACA combining the displacement function, and a local heuristic which used to improve solutions given by the colony, was proposed to resolve the problem. The algorithm has been tested on the benchmarks of Beasley et al. [126] involving up to 50 aircraft and 1 to 4 runways. Finally, the improved ACA is compared with the FCFS rule, the ACA before the improvement by the local search, and for the first time in the literature with the approach proposed in [43].

Some similarities exist between the ALP with carrier aircraft and ALP with civil aircraft. We can consider that the former meets an ALP on a single runway with a limited number of aircraft. The basis for this distinction is the fact that there is only one available runway for landing on the deck. Furthermore, the number of aircraft to return to the deck is usually much smaller than that of aircraft which are allowed to land at a particular civil airport. In the other case, the ALP with carrier aircraft is considered to be a very sensitive issue compared to the ALP with civil aircraft. Landing on an aircraft carrier is complicated and more dangerous in comparison with land-based aircraft, since a particular series of conditions need be taken into account [168]. The consideration of the safety issue, when the carrier aircraft must satisfy certain conditions before they get permission for landing is noticeable in the former case. For example, the aircraft that has the problem of insufficient fuel must wait for the air refueling tanker²⁶ before being permitted to land; the aircraft with severe damage has to land urgently on the land base. In addition, the chance that carrier aircraft fail to land at the first attempt (i.e., Failed to Land Aircraft (FLA)) is much higher than that for civil aircraft (i.e., FCFS rule). Wu et al. [90] in this context establishes a realistic model for the ASEP on the deck for the static and dynamic cases. Firstly, the landing mission, the traffic in the terminal area, and strategies for a FLA case are described to build a *conceptual decision-making model*. Then, a mathematical model based on a mixed integer 0–1 programming formulation is presented under the constraints of SLT interval, and the constraints of RPS and MPS between two neighboring sequencing process whilst taking account FLA cases. In the last step, the appeal for an improved ACO considering the constraints of RPS and MPS as a problem solver is justified by the fact that this method can more easily generate feasible landing sequence than the *parthenogenetic algorithm* and the Particle Swarm Optimization (PSO) approach. The results of the advanced algorithms compared with the Least Fuel First Service (LFFS) principle show that the modeling method for the ASEP and the improved ACO-based-solver has a better performance in optimizing landing sequence in terms of landing time consumption and cost function. More specifically, the dynamic method outperforms the LFFS principle-based method resulting in a higher level of safety and efficiency than the static scheduling method when arranging the landing mission.

Xu [112] investigated a solution that would meet the ALS on a single runway by making use of the aircraft classes. Minimizing the makespan is defined as a goal and the ALS is transferred to an open TSP. The ACA based on the WV modeling method is proposed for resolving the problem. The data in [154] are used to test the efficiency of the proposed algorithm. In fact, the approach was compared against the general ACA, the CPLEX, and the approximation algorithm created in [169]. The numerical results validate that the new method has better performance than these latter methods. In addition, it was deduced that the proposed approach which has been tested on file containing 30 aircraft, finds the optimal solution within 15 s and allowing it to be promising method for the improvement of the scheduling system efficiency from a theoretical standpoint.

■ Hybrid Technique

The pooling principle of several meta-heuristics or exact methods could lead the development of more successful heuristics called *hybrid*. The hybridization of algorithms aims to mix two or more different methods in a harmonious way to retain only the most interesting characteristics of each of these methods. We want to begin by Moser and Hendtlass [119] who solve the dynamic case of the ALP on a single runway by the investigation

of the adaptive capabilities of Extremal Optimization (EO). The main goal of this model is to minimize the penalties for early late landings compared to the TLTs. After the search for the best permutation, the EO solver's task hybridized with a local search heuristic find the optimal permutation of the aircraft in the active window in latter part of the optimization process. Computational results presented for a number of aircraft varying between 10 and 50 proved that the approach was highly competitive.

By analogy with the JSS problem, Artiouchine et al. [82] have modeled the terminal approach and landing phases by assigning landing times to aircraft for a single runway, in order to minimize the use of the HPs. This is achieved by maximizing the minimum time elapsed between two consecutive landings, and by giving more than one time window to each aircraft. All time windows, where the mechanisms as vector spaces and HPs are modeled, have the same size. Artiouchine et al. [82] provide a MIP formulation that allows studying problem complexity and identifying solvable cases in a polynomial form. The authors propose a general hybrid technique called "Branch and Cut (B&C)" to solve the problem with arbitrary time windows.

BenCheikh et al. [142] handled the static case of the ASS on single and multiple runways. In a first part, they proposed a mathematical programming model based on the improvement of the modeling problem initially presented in [127]. Their strategy is to reduce the number of constraints which from their views will reduce the computational time. In this work, the objective function essentially consists of minimizing the total cost of penalties under the constraints of time windows and the minimum separation time between two successive landings on the same runway or on different runways. The second part of this project is the formulation through a graphical representation, the ASS as a JSS with a partial orders and alternative sequences. Then, they proposed hybrid method consisting in appealing the ACO in order to generate a good subset of feasible solutions, considered as initial solution for the GA approach. The hybrid algorithm was tested on instances for a number of aircraft varying between 10 and 20, and a number of runways between 1 and 3. Subsequently, these writers developed a specific hybrid algorithm for the static case of the multiple RSP [49]. The resolution consists of two computational heuristic search techniques, namely a TS algorithm which hybridizes to the GA. The computational results are presented the scheduling of 10 to 50 aircraft, and their assignments to 1 to 4 runways.

Yu et al. [99] examined the ALS on a single runway. The goal of this related work was to minimize all penalties incurred by aircraft, while respecting the landing time windows and the minimum separation time between every pair of aircraft. They developed an approach named Cellular Automata Optimization (CAO) as a method of resolution. This approach solves the ALS in two major steps. The first one is to get a feasible landing sequence through simulating the landing process using the Cellular Automata (CA). A local search strategy is then used to further optimize the landing schedule generated by the CA model. The second step is to determine the exact landing time of each aircraft. The CAO approach was compared with popular approaches of related studies: LP-based tree search, ACO, SS and BA algorithms [47]. All of these methods were evaluated on the same *Benchmark* initially expressed in [127]. In the context of this work, satisfactory solutions are obtained by the CAO approach within 4 s in most cases studied.

In the same context of hybrid techniques, Hu et al. [155] pursued the implementation of the static ALS problem on a single runway. The objective consists in minimizing the total landing time. They suggested the transfer of neural networks "Mutation Hopfield Neural Network (MHNN)" as a method of resolution. The main idea of MHNN is to integrate the Estimation Distribution

²⁶ A process of transferring aviation fuel from one military aircraft (the tanker) to another (the receiver) during flight.

Algorithm (EDA) in the Hopfield Neural Network (HNN). On the one hand, the EDA which is a global search technique is still a hot topic in field of evolutionary computation [124]. On the other hand, the HNN usually has local research capacity and fast and powerful operation. The test sequence consists of 20 aircraft which should land in a lapse of time equal to 25 min. The MHNN was compared to the HNN and its interrupted version, the EDA, the GA, and the ACO, in order to test the performance of the suggested algorithm. The results using the MC simulator show that the MHNN sequence reduces both total landing time and total delay better than the FCFS sequence.

In [170], the ASEP has been considered when minimizing the total cost of the deviation from the TLTs and minimizing the completion time of the landing sequence. The Fuzzy Programming Approach and an estimator were used for solving the scheduling aircraft's landing for a single runway in a static fashion. The choice of the FPA for multiple objectives of the ASEP depends on its effectiveness. In fact, this approach can properly formulate vague information related to the objective functions, and when there is an inherent uncertainty in the relative importance of these goals. A set of experiments for a set of small-sized problems involving from 3 to 9 aircraft and for a set of large-sized problem of 10 to 20 aircraft is offered. In this paper, they conclude that the degree of satisfaction generally decreases with increasing the number of aircraft. However, the degree of satisfaction for a sequence of more than 20 aircraft is less than unity, which makes this approach not particularly useful in real environments.

In previous work, Salihpour et al. [56] design the VND and several local search algorithms to offer an efficient and practical solution approach to the multiple RSP. The objective is to allocate to the available runways a set of aircraft to land in order to minimize the total delay of the aircraft's landing time. All problems are real-life instances and refer to *Dallas Fort Worth airport* based in *Dallas, Texas*, in the *United States*. Two algorithms are then designed as a resolution method in [164]. The first one is the SA hybridized with the VND, and the second one is the SA hybridized with VNS. This choice is motivated by the fact that the SA which is a randomized optimization strategy could find tractable and approximate solutions to complex deterministic COP [26]. These scholars are modeling the ASS on multiple runways by a MIP model. The goal behind this work consists in minimizing the total deviation from TLT under the two principal constraints in the ALP: time window and minimum separation time between two aircraft landing on the same runway or on different runways. The computational result show that the SA can obtain the very high quality of optimal solution for instances up to 100 aircraft, which seems to be the largest instances solved today [109]. It is also capable of finding high results in a short time for the problems with up to 500 aircraft and 5 runways.

Nithyanandam and Mohankumar [11] presented a technique which describes the execution time and penalty cost of each aircraft for the ALS in the static case. Moreover, this work presents a discussion of modeling a number of issues that commonly may occur in reality, such as aircraft selection, precedence restrictions, and runway workload balance. Then, the artificial Neural Network (NN) is utilized for finding the optimized set of timings, and the GA is employed for optimizing point in the runway for aircraft. The proposed hybrid technique has succeeded in producing the optimization in penalty cost, along with speed and distance of aircraft. Nithyanandam and Mohankumar [11] found that their approach made valuable contributions when the timing factor penalty cost is accounted, which eradicates the congestion occurred between the emergency and normal landing.

In addition to the exact B&B approach they offered (see the sub section of *exact methods*), D'Ariano et al. [140] solve the landings and take-offs scheduling and routing problems by testing different *dispatch rules* and *greedy heuristics*. The AGH is a family of

heuristic algorithm which takes decisions based on global information available from the GA. The choice of this hybrid strategy is justified by the need for searching new feasible solutions (using the depth-first strategy) and promising selections (using the best first strategy).

The research stream cited in the previous section does not explore exclusively *exact methods*. In fact, we notice how Samà et al. [52] develop meta-heuristic algorithm for the real-time traffic management problem of scheduling and routing landing and take-off operations at busy TCA. Samà et al. [52] present a MILP in order to minimize the maximum positive deviation from the target landing and take-off times. They propose the TS and the VNS algorithms as resolution methods. The algorithmic framework starts from a good initial solution for the sequencing and scheduling of aircraft landing and take-off with pre-defined routes. This solution was the truncated B&B algorithm presented previously in [10]. Meta-heuristic algorithms are then applied to improve the solution by changing the routing of some aircraft. The neighborhood of a solution is characterized by the number of aircraft to be re-routed. The remarkable quality of the solutions based on meta-heuristic is done through a comparison with the experimental results of the combination of the B&B scheduling algorithm and the TS re-routing algorithm. The computational experiments demonstrate that VNS outperforms the TS in term of computation time. This work will be continued and further developed through Samà et al. [35] who as mentioned previously, treat the re-routing, scheduling, and sequencing of landing and take-offs operations and their assignment to several air segments and runways. The problem is modeled as a generalized JSS via alternative graphs with which the potential conflicts are solved. In this work, we note that the proposed alternative graphs have the ability to model any 4-dimensional route for the aircraft flying in the TCA. While most of the related work done assumes 3-dimensional routes. This paper considers the same objective, presented previously in [10,140], and also the MILP formulation proposed in [87] which was solved by the RHA developed by Samà et al. [153]. This implies that the present work is an improvement of the results previously presented. The contribution we can capture is the new developed hybrid meta-heuristic, based on TS and VNS schemes. The new methods are developed in order to respond to two issues that are not taken into account in [Samà et al. [87,158]]. The first issue concerns the extent at which different solution methods might outperform the TS algorithm and the RHA concept. The second issue is the studying of algorithmic improvements in order to reduce the time spent to achieve good quality solutions. These algorithms using real-world instances of the TCA of *Milan Malpensa airport* [87] outperform the results gathered by the TS proposed in the latter work in terms of computation time with respect to both the MILP solver and the previous versions of AGLIBRARY.

In 2015, Sabar and Kendall [108] called to an ILS heuristic to solve the ASS and their assignment on a single and interdependent runway. They considered minimizing the total landing deviations from the PLT as a goal. Sabar and Kendall [108] made up of first Randomized Greedy (RG) heuristic to generate an initial solution for the ILS. Then, the *Benchmark* instances introduced in [127] is used to demonstrate the effectiveness of the proposed algorithm. The results of the ILS with multiple operators for perturbations were compared with the results reported by the state of the art methodologies (e.g., the meta-heuristic methods proposed in [46], the hybrid algorithm combined SA and VNS in [164]). The analyses prove the high quality solution achieved by the ILS compared to the SS, BA, and SA algorithms, in terms of both solutions quality and computational time. In addition, the results demonstrate the benefit of using multiple perturbation operators as well as time-varying perturbation strength.

Furini et al. [72] continued to propose solutions for the static aircraft sequencing including both arrivals and departures on a single runway after their previous works (i.e., Furini et al. [37, 156]). Minimizing the total cost due to the weighted delay of the aircraft in the sequence during a predefined time window was fixed as a goal. The separation time and the delay of each aircraft that respect the lower and upper bounds are considered as constraints in this paper. Two MILP are reviewed to formalize the problem, one of them is the variant of the problem studied in [127] and the other is the Furini et al. [37] model. New approaches have been proposed in this work, which does not mean that the authors do not reuse techniques previously presented (i.e., Furini et al. [37, 58, 156]). Indeed, they reuse the RHA while expressing some rules for deciding how to partition a given aircraft sequence and their effects on solution quality. In addition, the TS heuristic and an Iterative Rolling Horizon Approach (IRHA) are proposed as resolution methods. The TS finds solutions to the sequencing concern in a short computation time, and the IRHA improve on the FCFS rule for real air traffic instances from *Milan Linate airport* by using different chunking rules. The experiments are based on the simulation of two days, where the former contained 100 scheduled flights (54 arrivals and 46 departures) while the latter contained 121 scheduled flights (68 arrivals and 53 departures). Moreover, the two days simulations were merged by the authors into one in order to obtain larger and more congested instances. The results showed the efficiency of the heuristics against FCFS method.

A more detailed study of the ASS for both single and multiple runway cases are investigated by Girish [74]. The solution is characterized by the development of a hybrid PSO algorithm coupled with the ILS algorithm in a rolling horizon framework. The proposed RHA converts the original runway allocation, and scheduling problem into several smaller problems so as to reduce the computational burden. The objective function consists in minimizing the total penalty cost due to the deviation of landing times of aircraft from the respective TLTs. The algorithm that is proposed outperforms existing approaches in the literature (e.g., the work of Salihpour et al. [164]) in terms of solution quality and computational time. In fact, the RHA proved its high performance to a set of *Benchmark* instances involving up to 500 aircraft and 5 runways.

Besides the focus on the routing procedure as previously declared, Ma et al. [48] are also considering the issues of the ASEP on a single runway. They have also factored the resolution of conflicts that are detected at the level of each aircraft predefined terminal route network. Using a new TDA, the goals were, on the one hand, to minimize the total number of conflicts that could occur respectively in the nodes, in the links, and at the RT for each flight, and on the other hand, to minimize the number of flights that must implement changes in management's decision. The aircraft speed when entering the TMA with its changing when necessary is treated via an optimization methodology. This case that may occur in reality aims to resolve conflicts and maintain separation between aircraft with regard to the wake turbulence constraints and the runway occupancy time. Moreover, the landing time window is considered as a constraint. In addition to the TDA, an adapted SA heuristic is proposed to solve the corresponding sub-problems. The experimental of the approaches are performed on three 24-hour cases of arrival traffic at *Paris-Charles De-Gaulle airport*. In fact, the hybrid algorithm against the pure SA algorithm successfully reached conflicts-free solutions within less computational time and aircraft deviations. Moreover, there is an incentive to the benefits of the sliding window TDA. In subsequent works, Ma et al. [18, 171] re-utilized the methods proposed previously (i.e., SA combined with TDA) to address the tightly connected of airport and terminal airspace management

issues at a macroscopic level. It entails both the management of arrival, surface, and departure problems. In fact, the goal was to resolve conflicts in the airspace and to reduce airside capacity overload. This involves runways, terminals, and taxiway network evaluation. The proposed TMA route model in managing the landing from the IAF to the RT meters used the same classical node-link network introduced previously by Ma et al. [48]. For the experiment in [171], it is envisaged to schedule a large instance (1116 flights including 562 departures and 554 arrivals belonging to *Heavy* and *Medium* performance) to the four parallel runways of *Paris Charles De-Gaulle airport*, one of the busiest passenger airports in *Europe* operating at heterogeneous mixed mode (two for departures and the other for arrivals).

As mentioned above, Ji et al. [129] have been geared to minimize the total makespan of all landing flights when optimizing the ASS sequence in the dynamic case, respecting the constraints of the landing time window and the separation requirement between consecutive landings. Then, an evolutionary approach, namely Dynamic Sequence Searching and Evaluation (DSSE), is proposed to seek the optimal landing sequence of flights. The DSSE divides the ASS issues into a number of sub-problems with smaller scale and tackled them one by one. The EDA is adapted iteratively to search for a candidate scheduling scheme by directly sampling the permutation of flights for each sub-problem solving. Besides, a heuristic search approach is also proposed to seek the optimal landing sequence of flights. In this note, additional experiments have been separately conducted for the global search EDA and the heuristic search of DSSE in order to verify the strongest of each component. Subsequently, Ji et al. [129] compared the performance of the proposed DSSE with related dynamic ASS works (e.g., Zhan et al. [94], Vadlamani and Hosseini [138], Hu and Chen [154]). In fact, the outcomes of their approach in terms of solving several test instances including an instance constructed from the real data of the BCIA, illustrates its manifold improvements as compared to the latter related algorithms.

Rodriguez-Diaz et al. [85] seek to minimize delays when setting the scheduled times for upcoming and bygone flights at a heterogeneous runway under WV separation and CPS restrictions. They use SA algorithm with CPS rule based on public test bed as well as real data obtained from *London Gatwick airport* to do the experiment. The data considered range up to 200 flights and 2000 instances. The results of the SA algorithm substantially improve those reported in [47] and [164]. In this case, computational time is very important as good results are achieved in less than one second that allows this approach be used in a real configuration of TMA.

An attempt is made by Ng et al. [1] to carry out the dynamic fashion of scheduling and sequencing landing and take-off operations at a multiple heterogeneous and parallel runways. The aim of these latter is to advance the search pertaining to sequencing and scheduling of landings and taking offs, by better exploiting specific structural knowledge problems. First, the issues are treated with uncertainties which lead them to be compatible with real-time applications. The objective of this research was to minimize the maximum makespan using a MILP formulation in the min-max regret approach. An iterative relaxation procedure is incorporated to obtain a robust solution to the min-max regret optimization model. In order to minimize arrival and departure delays, several practical constraints are considered in the model (e.g., the limited number of runways, the resource capacity constraints, the air traffic regulation, and the separation requirements caused by the runway's physical properties). Then, a new earning algorithm, namely the Artificial Bee Colony (ABC) which has been successfully applied for various practical problems, is developed to conduct the robust scheduling in regard to one-hour

flight traffic planning horizon. Preliminary results obtained by the ABC were not satisfying in terms of the objective value and the computation time. Consequently, making it more attractive has involved its improvement by two novel components, a proposed constructive heuristic and an objective-guided updating mechanism in the scout bee phase so as to give the Efficient Artificial Bee Colony (EABC). Indeed, the EABC was evaluated in terms of optimality gap regret, average robust cost, relative percentage increase, least significance difference, and number of best value. The experimental results indicate outcomes effectively and efficiently of the EABC algorithm in comparison with the modified ABC algorithm (i.e., Original Artificial Bee Colony (OABC), the Modified Artificial Bee Colony (MABC), the Hybrid Artificial Bee Colony (HABC) algorithms), and other meta-heuristics, such as the B&B algorithm, the GLS, the commercial MILP solver, and the FCFS sequencing policy proposed in [4]. The instance sets represented the approximate number of air traffic scenarios during the peak hours at *Hong Kong International airport* taking into account different aircraft classes (*Small, Medium and Large Sizes*) are adopted as instance in this research.

It is of crucial importance to cite the four *Liang et al.* works [Liang et al. [8,13,51,69]]. These latter are characterized by the establishment of specialized DST based on utilizing integrated models in order to assist air traffic controllers in making dynamic decision at both the pre-tactical and tactical levels at BCIA. Liang et al. [8] present a brief look at the continent of the latter works. First of all, it is useful to report that the papers are a series of novelties starting with the improvement of the classic PMS in order to make it adaptable to the segregated parallel runway configuration. We are talking about the Multi Level-Point Merge²⁷ (ML-PM) route network that has proved its capability for realizing a shorter flying time and near Continuous Descent Approach²⁸ (CDA) for arrival aircraft. In addition, it provides an economical climbing for departure aircraft and an easier runway allocation for both arrivals and departures. First, Liang et al. [51] are developed an autonomous system with PMS. This design included the ability to perform the sequencing, merging, and spacing for arrival aircraft in order to make the trajectories of arrivals. We note a novel PM-based route network for the BCIA, in which aircraft from different departure airports will follow different arrival procedure to land on the right runway. Maximizing landing rate, minimizing the average delay, and minimizing the average conflict, are determined as a multiple objectives function for the aircraft scheduling. The operational constraints of the MPS in the sequence, arrival time windows, and minimum aircraft separation are placed in the model. The proposal of Liang et al. [51] is taken into account the conflict detection at out of merge zone, link conflict, and node conflict. Furthermore, two possible maneuvers modify the entry time at TMA or adjust the speed, were considered for solving the conflicts and controlling the rate of arrivals flows. And for optimizing the landing sequencing they are considered the control of the turning aircraft on sequencing leg. The RHC strategy is applied into the autonomous system and SA is proposed for solving three cases: low density situation, normal density situation, and high density situation. Then, Liang et al. [69] enhanced the previous study. They made a brief evaluation of the potential operational benefits of the ML-PM based autonomous arrival management system in terms of flight efficiency and runway throughput. The objective of the optimizing mathematical model was to generate a conflict-free and efficient arrival trajectory for each of the

two runways at BCIA, by considering conflict detection mode for merging zone and speed change profile. The same hybrid optimizing algorithm, which combines RHC with SA, is proposed for generating the near-optimal trajectories. In fact, the sliding window was implemented to transfer the total 24 h static optimization problem into several sub-optimization problems, and the SA algorithm is then introduced to generate the optimal trajectories in each sliding window. The case study is made by simulating arrival flows under three different operational modes: baseline (i.e., without PMS, traditional PM, and the ML-PM). By numerical simulation (93 aircraft to land at BCIA), the ML-PM shows good performance on three flight efficiency performance indices: flight time, fuel consumption, and CO₂ emission. The results on fuel consumption, flight time, flight distance, landing interval, delay to the different cases, as well as advantages of ML-PM system are detailed. In addition, the analysis is made in view of both ATC (the conflict resolution performance and the position-shift sequencing technique), and airlines (the CDA of aircraft in ML-PM scenario). As an extension of the previous works and as we have shown, the management in real-time of arrival trajectories from the TMA-EF to the RT was the basis for the subsequent work by Liang et al. [13]. The management process involves the optimization of aircraft merging, sequencing, and assigning to the parallel runways, along with the automation in conflict resolution and advanced avionics capabilities. For clarity, it is necessary to note that the control of trajectories is based on several factors, such as the entry time, the entry speed, the turning time on the sequencing leg, and the landing runway allocation factors. These latter aspects therefore confirm the accuracy of the proposed approach since we observe that the airspeed and the fuel consumption are considered, which have not made in [69]. Moreover, a suitable mathematical optimization problem based on multi-objective functions (e.g., collision avoidance, average delay minimization, optimization of fuel consumption, and balancing the runway landing rate) under hard constraints, such as time window and speed change, is built up. The achievement of these objectives is established in accordance with the ML-PM system, an economical decent approaches procedure (i.e., CDA), and a hybridized algorithm combining the SA with the RHC. The hybrid algorithm mechanism consists in splitting the entire 24 h traffic into several sub-problems by the application of the RHC technique. Then, the SA algorithm is used to search the best solution to execute for aircraft. In addition, Liang et al. [13] support the computation of 4D trajectory for each aircraft. The algorithm illustrates the efficiency in creating good and dynamic sequencing at the BCIA-TMA. This is justified by the fact that dense arrivals with *Heavy* and *Medium* aircraft types will be significantly handled, with none conflicts, and while guaranteeing high throughput at the airport. Liang et al. [8] is the last work where they re-utilize the RHC and SA developed previously in [13] for the finding of complete and efficient trajectories in order to meet the requirements of arrivals and departures. Indeed, the experimental results conducted on the BCIA study case, show performance according to the arrivals and departures management, while also pointing that the methodologies defined in [13] and [8] could be easily applied to airports worldwide.

The desire to simultaneously achieve safety, efficiency, and equity in the ATM system emerged recently in [96]. A solution for the static case, of landings and departures sequencing and scheduling, on a single runway functioning under both segregated and mixed mode, is presented. The 0–1 MIP model of Beasley et al. [127] was enhanced with the inclusion of CPS constraints to formulate the problem. Indeed, the particular version of RSP defined in [33] as it is the only work that incorporate all practical constraints including CPS, wide time windows, precedence constraints, and early landings/departures, was basically agreed

²⁷ Multi Level-Point Merge (ML-PM): an advanced Point Merge (PM) route network to meet the requirements of parallel runway operations [8].

²⁸ CDA: procedure from on-board systems provide a great probability for commercial aircraft to follow optimal descent profile on a direct flight route to any desirable waypoints [69].

in [96]. The maximization of the total throughput is fixed as a goal subject to the ALP operational constraints, safety separation standards, and wide time windows constraints. A first *data splitting algorithm* to optimize the flight sequences is introduced as a method for resolving the problem. The algorithm design is based on the using of the divide and conquer paradigm, where a given set is optimized using 0–1 MIP while ensuring the optimality of the entire set. To the best of our knowledge, the *data splitting algorithm* uses a first LP-based-B&C approach to solve MIPs. Computational results show that the new approach could solve real sized instances in real-time. Moreover, LP-based-B&C approach outperforms DP-based approaches in handling different problem scenarios (aircraft classes, arrival traffic, mixed traffic, peak traffic conditions, and scenario with or without CPS rule).

We close this section with Riahi et al. [36] that present a solution for the static case of sequencing landings and take-offs on multiple runways. Minimizing total weighted tardiness of all flights is fixed as a goal. In this research effort, several contributions are submitted, including the use of specific structural knowledge from constraint and objectives in the sequencing process, and the implementation of compact constraint optimization paradigm in artificial intelligence. The latter consists in using the constraint guided strategies within the Explorative Perturbative Search (EPS) method which itself constitutes a contribution. The EPS is then respectively compared with the SA-based combined arrival departure aircraft sequencing problem on multiple runway algorithm in [137] and the ILS-based-ALP algorithm in [108]. The experimental analysis of the proposed model and algorithm are described for two real case studies, *Milano Linate airport* and *Doha International airport* terminal areas. The former is 12 real-world instances including 60 aircraft, extracted from [37]. The latter is the *Benchmark* set used by Farhadi et al. [4] with a 20, 25, 50 aircraft to be scheduled, and 2, 3, 4, and 5 runway available for assignment. The proposed strategies significantly outperform the SA proposed by Hancerliogullari et al. [137], and the ILS developed by Sabar and Kendall [108].

4.4.3. Simulation method

Simulation has been widely used to study the airport and airspace systems and can be done in a variety of ways. Simulations are often seen to have advantages compared with experimental results performed by several studies. They are often cheaper and can be done more quickly. Also, they are generally easily repeatable and can estimate the effects of extreme conditions without the risks that live experiments may entail [92]. But even if the simulation model does not require the programming of mathematical equations, it is rarely employed to solve the ALP missions. This method is generally used to understand and study the behavior of the real system through experimentation according to input variables and constraints. The early research work of Andreussi et al. [134] is one example. They deduced that the simulation models have a remarkable ability to handle the ATC system, which is not often the case for mathematical models. In fact, they did not give formal algorithms and concerned with developing a discrete-events simulation model to evaluate different landing and take-off sequencing strategies at *Rome* terminal area. They considered a structure based on a variable number of feeder fixes, and alternative approach path from these fixes to a multiple runway system. The proposed model was judged to be operational for any TMA for the reason that it takes into account a large number of components and may therefore effectively represent the functioning of the system in practice: (1) The capability to envisage runways for both take-off and landing operations or for only one operation; (2) The ability to avoid conflicts based on certain rules of priority (e.g., the aircraft which underwent significant delay); (3) The ability to take into account various

structures of the airspace; (4) The ability to consider any category of aircraft and their particular performances; (5) The ability of using different aircraft sequencing policies; (6) The ability to simulate multiple runways; (7) The ability of generating random arrivals into the system. The optimization model generates each iteration one or several solutions which are then assessed by the simulation model to further improve their performance until a stopping criterion. It is necessary to mention the implementation of the proposed model to assist ATM in *Fumicino airport*.

Moreover, the MC simulation approach is reported for solving especially the ASS issues in this classification. But as stated by Ahmed and Alam [5], this approach is seen unsuitable for this kind of optimization problems in view of the ASS problem difficulties (i.e., various aircraft types, runway configuration, sequencing requirements). We are finished with two recent works for this field. The former correspond to Rezaei et al. [130] who propose an Agent-Based Model (ABM) for the ALP with a smaller number of aircraft. The ABM considers a decentralized environment for the ALP with the price of not being optimal solution. The latter correspond to Liu et al. [98] who propose a two-stage stochastic mixed integer 0–1 programming model for the single RSP under uncertainty. The arrival time of flight is exactly stochastic into the ALP in order to identify an optimal schedule with maximizing runway elasticity and minimizing delay times of aircraft. Then, a Sample Average Approximation²⁹ (SAA) algorithm is developed to solve this stochastic programming and the efficacy is verified under different scenarios.

4.4.4. Software systems

The assessment of the operational system under real ATC conditions required the attention of many researchers. Furthermore, some advanced automation terminal airspace was proposed for the ALP, and was implemented as an operational system for some *international airport* ATCT. In this sub-section, we will present some examples of ATM advisory systems designing to assist terminal area air traffic controllers:

Means to Aid Expedition and Sequencing of Traffic with Research of Optimization (MAESTRO) developed by Garcia [172]: MAESTRO has been operated at *Orly airport* and *Paris airport Control Center* since December 1989. This tool was help for sequencing and merging the arrival aircraft positions. In fact, it have done the computation of the ELT for each aircraft, and, kept the HP as a safety bolt in the TRACON. Concerning the decisions and actions, they are left to the air traffic controllers. From an operational viewpoint, MAESTRO worked first with the FCFS discipline. Then, it resorted to the field of optimization so as to optimize the delay through all *Paris* centers in order to better use the ATC system capacity. MAESTRO succeeded in getting the best economical delay resorption for airlines, guaranteed an optimized supply of the runway, and improved the interface between TRACON and ATCT;

Computer Oriented Metering Planning and Advisory System (COMPAS) developed by Volckers [173] has been implemented for the *Frankfurt* main ATCT. In particular, it helps the air traffic controllers to plan and control the inbound flow of air traffic into the TMA. This tool is based on techniques that allows for unnecessary delays to be avoided and unavoidable delays to be reduced as early as possible. The technical realization of COMPAS is made up of several components, such as the interface-computer, the COMPAS main-computer, the display station computer, and

²⁹ SAA: a MC simulation based approach to solve stochastic programming problems [98].

the network communication. The COMPAS planning algorithm employs the B&B method;

Dynamic Planner was developed by Wong and Gregory [174]. The functioning of this system is based on specified inputs not previously been defined in other tools. The inputs consist respectively of the flight plans, the track updates, the estimated time of arriving aircraft along the reference points (outer meter arc, meter fix, FAF, RT), the over crossing times,³⁰ the scheduling constraints including separation distance requirements, the occupancy time, the acceptance rate at different meter fix (TRACON, meter fix, gate, airport, runway), the blocked interval and finally the sequence constraints. It should be emphasized that the provision of the estimated time of arrival is made by the TMA's *Route Analyzer* and the *Trajectory Synthesizer*. The first program computes the horizontal route of arriving aircraft as well as the various speed restrictions at various points along the route. Then, the *Trajectory Synthesizer* uses this information's and computes a complete 4-dimensional trajectory from the aircraft's current location to touchdown at the RT meter. All the data defined as inputs for the **Dynamic Planner** have been designed, implemented, and integrated to compute the feasible aircraft sequences, schedule with spaced time of landing aircraft, and runway assignments into the CTAS.³¹ This has been done to handle flow of traffic approximately between 35 and 200 NM from the arrival airport. The set of outputs were constantly updated on changing events. Note that two flight rules, the IFR and the VFR, are handled by the **Dynamic Planner**. This tool was in daily use as a flow-visualization tool at Atlanta, Denver, Los Angeles, and Miami centers;

Stochastic Terminal Area Scheduling Simulation (STASS) developed by Mulfinger and Sadovsky [82]: the STASS software is originally written by NASA for airport capacity studies. It was designed as a time-based scheduling simulation tools that models aircraft arrivals in the TMA. The scheduling is performed in two steps: first, schedule aircraft from the freeze horizon to the meter fixes and runways by using the center scheduler; second, schedule aircraft to the runways upon their arrival at the meter fixes by utilizing the TRACON scheduler. It is important to specify that the uncertainty is taken into account in this support, and is accurately modeled at the freeze horizon, meter fixes, and runways. STASS presented several case studies conducted for the Atlanta, Dallas Fort Worth, and Los Angeles airports.

5. Comparative study

In view of the analysis of the major differences between the works, it is possible to remark here that some of them have been distinguished by a scope. In this last section, we present Table 2 decorticating the originality (Column 3) that can characterize the problem studied in particular (Column 2) raised by researchers (Column 1) compared to another.

In this regard, we seem to make a comparative study of computational resources, but we have identified that it is difficult. That was because different researchers will inevitably use different computer resources and do not always report factors such as the operating systems, language used, compilers, and the skill level of the programmer.

³⁰ The over crossing time refers to the time that the aircraft crosses the meter fix.

³¹ CTAS is under development at NASA Ames Research Center in Collaboration with the FAA's Terminal ATC Automation Program Office. It provides automation assistance to air traffic controllers in achieving acceptable aircraft sequencing and separation [69].

6. Discussion of published literature and conclusions

6.1. Discussion of published literature

From a practical standpoint: Regarding the landing operation, a failure with respect to the management of this procedure can be blown out of a causal chain wherein the accumulating bottleneck affects lead to the propagation of flight delays to forthcoming aircraft, unused runway capacity, and brutal increase of air traffic controller burden. This is in turn result in associated financial and environmental costs. While fully observing air traffic controllers' tasks, we highlight in the operational environment that more generally it is necessary to conduct the decisions related to aircraft waiting to land for a radar period in a few seconds. This leads us to conclude that within a short period of time, the ALP is already challenging, even with only a few aircraft. Need to pay close attention that the current tools and procedures used to manage the arriving flows are struggling to keep pace with normal conditions. Furthermore, the current environment is marked by the lack of effective accessorial decision-making tools able to support the air traffic controllers in implementing the intuitive schedules. On the basis of these observations, it is legitimate that the FCFS is still using in the real-world airports in most of its operation, and is regarded as the practical scheduling rule the most used by air traffic controllers'. Thus, there is an urgent need for developing robust real-automation terminal airspace that provides useful support for air traffic controllers for making decisions. These DST has to take into account diverse and often competing considerations of efficiency, safety, and equity among the principal stakeholders (i.e., airlines, ATC, governorate). With frequent use, the direction of these tools needs to be introduced into the ATCT. Indeed, there should be an even more efficient method of scheduling than the use of FCFS rule. It would be unsurprising to find that these advanced approaches do not meet the real-time requirements, even for a moderate number of aircraft. There are reasons for this are the same as those previously described: the ALP is still handled by human air traffic controller in an environment characterized by dynamic status with a high level of uncertainty and disturbances (e.g., aircraft speed, runway accidents). However, these factors are seldom taken into account in research. It is therefore desirable that the future advanced approaches design is no more straightforward, and is realized with the assistance of external experts. We would expect that the survey paper will appreciate scientist and would be a significant contribution to the development of new advanced approaches for the ALP.

From a methodological standpoint: By learning from a variety of solid research works that have been widely addressed in the literature regarding the ALP (more than 80 scientific publications have been provided in the past four decades), we are going to highlight a set of distinctions:

- The analysis of the related literature shows the existence of two angles: studies focusing on the case when a single runway may be used for landing and others on the case when multiple runways operate simultaneously for landing. Most models were only suited to single runway and only a little to multiple runways. This latter distinction is necessary because, in accordance with rule of ATC, the problem is less severe in the case of a single runway, and is reduced to take the relevant decisions concerning the next aircraft which has to land. Furthermore, the problem becomes more complicated if there are multiple runways to manipulate all aircraft, and to this should be added the problem of deciding which runway the aircraft should be assigned. The problem of scheduling operations on multiple

Table 2

A classification of several works for the ALP and the originality of each of them.

Works	Problem resolved type	Original contribution of the proposed work
Dear [68]	ASP on a single runway	He is the developer of the simple heuristic CPS, which is considered as a basic component in the ALP and was commonly applied.
Bianco et al. [152]	ASEP on a single runway	Bianco et al. [152] count among the first ones who used the TSP for modeling the ALP. They conclude that the difficulty of this method lies in the representation of the goal.
Lee and Davis [50]	A support tool for landing process	The main goal was to incorporate the air traffic controller into the development of advanced automation for ATC process. We talk about FAST which is the result of cooperation between the controller and the developer.
Ciesielski and Scerri [166]	ALS for an airport which has two runways	The setting of a preferred runway for some arriving aircraft is a major constraint that is considered within the sub-problem of assignment runway in this paper. Consequently, some aircraft are constrained to use specific runways.
Ernst et al. [61]	ALS on a single and multiple runway	According to [Yu et al. [99]], Ernst et al. [61] might be who first observed that the ALS problem involves two decision problems: – The sequencing problem that determines the sequence of aircraft landings; and – The scheduling decision problem that determines the precise landing time for each aircraft in the sequence, subject to the separation constraints.
Beasley et al. [127]	ALS on a single and multiple runway	The work of [Beasley et al. [127]] is considered as a relevant <i>Benchmark</i> in different works related both to the ALP and the ATP and has been improved several times.
Beasley et al. [75]	ALS at one of the two mixed runway of London Heathrow airport	– These writers made a discussion on the range of factors that the objective function may contain (e.g., European flights or not, the number of passengers per aircraft). This discussion is justified by the fact that National Air Traffic Services (NATS) deal differently with the European flights and the non-European ones. – The ability of the proposed <i>Population Heuristic</i> to solve a linear function and a non linear one.
Fahle et al. [28]	ASEP on a single runway	You can found here six different exact and heuristic solution methods for the basic ASEP. The aim of this diversity is to test the suitability of the proposed approaches for additional extensions that arise in practice. They conclude that: – Both heuristic optimization approaches (Hill Climbing (HC) and the SA algorithm) are able to make the ASEP work well. The HC is faster, the SA produces solutions with better quality, and both are flexible in modeling additional requirements; and – Integer Programming (IP) is the best compromise of exact solution methods, at least if the predefined time windows of the aircraft are not too large.
Capri and Ignaccolo [139]	Landings and Take-offs Sequencing on a runway	The introduction of a dynamic model of departing flights into the aircraft landing sequence.
Hu and Chen [154]	ASS on a single runway	Despite the fact that the GA represents the most used meta-heuristic approach for the ALP resolution, it has been combined with the RHC in [Hu and Chen [154]]. This is explained in part by the fact that the GA which was conventionally used can hardly meet the need of real-time properties in practice because of the increase of the air traffic demand. Yet, it is combined with the RHC strategy which can effectively improve its real-time properties and reduce the influence of unreliable information.
Pinol and Beasley [47]	ASS on multiple runways	In [47], the BA is used for the first time for solving the ASS, and the widest instance is presented compared with those mentioned in the previous works: 500 aircraft and 5 runways, before the recently work of Girish [74] dedicated for the ASS on single and multiple runways.
Bianco et al. [34]	Landings and Take-offs Scheduling and Sequencing on multiple runways	In this paper, the recommended model is quite general and suitable to coordinate traffic flows entering and leaving TMA. It is the first integrated model intended for two important sub-systems usually analyzed separately: the complex runway and the TMA structured in inbound and outbound paths, fixes, and holdings.
Moser [132]	ALP on a single runway	They came to a very important conclusion about the hybrid and exact algorithms. They mention that it is not impossible for a stochastic approach, like EO hybridized with a deterministic scheduler, to outperform an exact algorithm (equally hybridized with a different deterministic scheduler) in a dynamic environment. This phenomenon was already observed in [Beasley et al. [43]].
Artiouchine et al. [82]	Single RSP	The mechanisms recommended by air traffic controllers to hold aircraft before landing, such as VFS and HPs are modeled. It would be useful to know that these issues are not addressed in any of the papers above.

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runways is more complex, because it depends greatly on the relative orientation, geometry, and the configuration of the runways. These latter issues are almost neglected by researchers, which really make the approaches proposed to the ALP non suitable for use in the real-world cases. Let us hope that the subsequent researches make a sense in the future, and are pushing closed loop decision support systems for an airport with its own configuration. For example, the china researchers have to take the airports

configuration of their country into consideration. It is a well-known that in *China*, the most multi-runway airports use parallel runway configuration and there are some airports using other configurations like new airport in *Chengdu*, using two parallel runways and a lateral runway;

- Obviously, the ALP is divided into a set of decisions, decisions relating to the management aspect and one related to the control. Separating the two aspects of the procedure has been claimed that is beneficial by many operation research

Table 2 (continued).

Works	Problem resolved type	Original contribution of the proposed work
Soomer and Franx [71] Soomer and Koole [161]	<i>ASS on a single runway</i>	– The models are planned in a way that information of stakeholders (mainly airlines) in some way used to improve the decision process. This issue is not addressed in most of the previously mentioned papers. – It is primordial to note that in 2018, Serhan et al. [162] have regard to different stakeholder's preferences, even if the proposed approach integrated arrival, departure, and surface operations. The motivations are in fact twofold: minimizing airline and passenger delay cost (i.e., passenger delay, missed connection, fuel, crew, maintenance costs). In addition, several interesting assumptions (e.g., route, speed, separation, gate conflicts, HP) with an observation of arrival and departure operations at <i>John F.Kennedy International airport</i> are presented. – The tactical arrival planning (between 24 and 1 h before landing) is done. The choice of this strategy is justified by the possibility to devote flight delays prior departure.
Hu and Paolo [113]	<i>ASS on multiple runways</i>	This is probably the first presentation of a new GA which uses the relationship between aircraft for making chromosomes (uniform crossover). The main advantage of the proposed uniform crossover operator is its efficiency and profitability on substitution, and preservation of common traffic subsequences without sacrificing the capability of diversifying chromosomes. That is been proven in extensive comparative simulation study.
Ben Cheikh et al. [149]	<i>ALS on multiple runways</i>	We show several priority rules for testing the heuristic, like giving priority to the aircraft which has the soonest earliest time and causes, as well as giving the priority to the aircraft which has the soonest latest time and results in the most important lateness penalty.
Yu et al. [99]	<i>ALS on a single runway</i>	– In this paper, the CA model is characterized by using the state of cellular to describe the position and the velocity of each aircraft. This feature makes the air traffic controllers the ability to accelerate and decelerate the aircraft speed. – The solutions fulfill perfectly the real-world requirements of the ATC system in China.
Samà et al. [153]	<i>Landings and Take-offs Scheduling on a single and multiple runway</i>	These writers conclude that their basic model for scheduling landings and take-offs could be formulated as a single/ parallel machine scheduling problem, while detailed models for these procedures could be formulated as JSS problem.
Salehipour et al. [164]	<i>Multiple RSP</i>	This is the first paper that reports the optimal solution for instances with 100 aircraft before the papers of Lieder et al. [15], Girish [74], Zhou et al. [130], and Vadhvani and Hosseini [138].
Farhadi et al. [4]	<i>Single and multiple RSP for the arrivals and departures</i>	They contrasted the case of a single runway airport (as <i>Doha International airport</i>) with the two runways newly constructed at <i>Hamad International airport</i> , under mixed vs. segregated mode.
Faye [175]	<i>Single and multiple RSP</i>	– It is quite an interesting paper, because it outlines a new approach based on the approximation of the separation time matrix with the B&B algorithm and the time-indexed MIP formulations for solving the ALP. – The drawback of the important number of variables induces by the time-indexed MIP is counterbalanced by a very good LP relaxation. The discretization methods proposed in [Beasley et al. [127]] do not have this property.
Murça and Muller [62]	<i>ASS and the mixed operations case by the insertion of departure flights for the single runway case</i>	– They have the first opportunity in the literature to propose an optimization approach that can satisfy real-time requirements and assist the ATFM establishment. It boils down to the reason that Murça and Muller [62] tie the problem of controlling traffic flow during the approach toward an airport through specific terminal arrival routes. – Alternative approach routes are used as a means of shortening or lengthening the path to the runway. – It is important to notice that the dynamic approach designed to run the optimization model is based on the concepts of scheduling window and freeze horizon.
D'Ariano et al. [140]	<i>Routing, Scheduling and Sequencing of landings and take-offs on several air segments and runways</i>	In addition to their present collaborative and non collaborative models, D'Ariano et al. [140] increase the accuracy of the existing models constraints for the routing, scheduling, and sequencing of landings and take-offs by combining the following modeling features: holding circle constraints, scheduled time interval, constraints for air segments, blocking constraints for runways, and time window constraints for the aircraft travel time in air segments.
Ghoniem and Farhadi [136]	<i>Landings and Take-offs Sequencing over multiple runways</i>	– The benefit of augmenting a base MIP with valid inequalities, preprocessing routines, and symmetry defeating hierarchical constraints in order to improve the performance of B&B cut techniques, were examined in this study. – The present study addresses two algorithmic schemes, namely the Interior Point Dual Stabilization (IPDS) technique and the Complementary Column Generation (CCG). The development of a specialized <i>Column Generation</i> approach which is accelerated by incorporating IPDS, have not been empirically compared in the literature, nor the synergistic benefit of jointly using them been examined, until Salehipour [19] who improves the implementation of the <i>Column Generation</i> made by Ghoniem and Farhadi [136] and deliver satisfactory solutions with an experiment including larger set of aircraft.

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Table 2 (continued).

Works	Problem resolved type	Original contribution of the proposed work
Jaquillat and Odoni [86]	<i>Landings and Take-offs approaching, scheduling and Sequencing issues over multiple runways</i>	- A descriptive model is developed leading to integrate the airport capacity utilization and the airport demand management areas. This model consists of three different components (a queuing model for airport congestion; a DP model for capacity utilization control, and an IP model for scheduling interventions). The originality of this approach lies in jointly optimizing the airport's flight schedule at strategic level and the airport capacity utilization at tactical level subject to the capacity constraints and the queue length reduction constraints.
Chen and Solak [65]	<i>RSP</i>	A stochastic DP framework is attempted to manage the flights sequencing and the separation between them during the Optimized Profile Descent (OPD) operations. Knowing that the OPD is an operating procedure being used by airlines to improve fuel and environmental efficiency during the arrival and descent to runways. It was proved that the expected annual savings can be around 29 million dollars if such decision rules are adapted by major airports in the <i>United States</i>. Indeed, flight tests at several airports have shown that OPD can save between 25 and 50 gallons of fuel during descent, while reducing noise and emissions levels by around 30%.
Heidt et al. [39] Heidt et al. [176]	<i>Single RSP of landings and take-offs</i>	- The challenge in these papers is the consideration of the uncertainty in the RSP, which is rarely studied. This strategy enables the initial computation of the plans to be robust with respect to changes in the data, leading to better utilization of resources and more efficient support for ATM controllers and stakeholders. - According to our knowledge, there is no literature on the mixed operations of the RSP which tackles the dynamic Time Indexed Model (TIM). To clarify the purpose of the TIM, it varies the size of time slots depending on the remaining flight time for every aircraft.
Rosenthal and Eisenstein [144]	<i>Re-scheduling of Landings on a single runway</i>	- The cancellations commonly occur during landing and can significantly contribute to airport getting its landing schedule back normal. On the basis of the above, the proposed model assumes a new feature that allows for flight cancellations. - Rosenthal and Eisenstein [144] indicate that there are positive spillover effects in terms of cost that take place when arrivals are re-scheduled. In this respect, they propose a new approach-based on soliciting monetary bids from arrivals to pay to move up or to be paid to move down in the arrival sequence.
Kapolke et al. [111]	<i>Single RSP</i>	- A specific optimization approach where uncertainties are incorporated is developed for the pre-tactical planning phase. The analyses are based on the real-world data of the large <i>German airport</i> in order to obtain realistic schedules in view of the delay distributions. - The main novelties in the model lie on the assignment of time windows to aircraft "Slots" for the pre-tactical planning phase. This has led in better utilization of resources, more stable plans, and efficient support for ATM controllers and stakeholders.
Wu et al. [90]	<i>ASEP on a single runway in the deck</i>	- The above studies are regarding the landing process of the carrier aircraft. This study belongs to one of the key technologies of the automatic landing of carrier aircraft, and it is the first issue to be addressed in the whole process of landing [168]. - The state of aircraft, e.g., remaining fuel, is always changing during flight. It is understood that these changes are not taken into account by existing literatures before this reference.
Rodriguez-Diaz et al. [85]	<i>Scheduling of Arrival and Departure flights on a single runway</i>	A new condition that consists to obtain online results is presented. In fact, this requirement which have not been presented above responds to the requests of operations in real-time.
Kwasiborska [76]	<i>ASEP on a single runway</i>	- The proposed approach outperformed the <i>Longest Processing Time</i> and <i>Shortest Processing Time</i> algorithms, respectively. This is what is meant by <i>letter sequencing algorithms</i> which are classified as basic deterministic sequencing algorithms. - Three main criteria were made in respect of optimization, the length of the schedule, the average flow time, and the maximum delay.
Zhou et al. [130]	<i>ALS on multiple runways</i>	This was the first time that the FPA was used. It was embedded first with <i>Runway Balance</i>. Indeed, this approach succeed to get the optimal objective values with large number of aircraft ranges 100 to 500, and generate better solutions compared with a main stream of scheduling algorithms, such as the SS in [47], the ACA in [BenCheikh et al. [149]], and the FCFS algorithm.
Salehipour [19]	<i>ALP on a single and multiple runway</i>	He made a simple approach, but with a set of distinguishing factors. First of all, we note the separation between the scheduling and sequencing aspects, taking into account the possibility of re-ordering. Second, the allocations of aircraft for the multiple runway case are considered. Finally, Salehipour [19] documented with details that the ALP with single runway is more difficult to solve than with multiple runways.

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Table 2 (continued).

Works	Problem resolved type	Original contribution of the proposed work
Ma et al. [18]	<i>Multiple RSP of landings and take-offs</i>	An example of how flight delays can be significantly reduced in peak hour by landing and take-off runway assignments is given in this pioneering work. It is necessary to state that runway change did not create many more controllers' workload.
Rodriguez-Diaz et al. [146]	<i>ALS on multiple runways</i>	– Despite the fact that the control of noise around airports is a complex matter since it needs the contribution of various stakeholders, they optimize runway and fuel consumption of the flights that minimizes the noise impact of the final approach route without proposing any change in the existing approach routes. – They prove that there is a margin for improvement in terms of noise reduction and fuel consumption just by deciding on the most appropriate landing runway and the order in the landing sequence.
Zhang et al. [21]	<i>ALP on a single runway</i>	– This is the first study to carry out objectives selection for multi-objective ALP from perspective of theoretical analysis. Zhang et al. [21] illustrate different categories of ALP objectives and the inner relationship between them. – They prove by observations, that the min-sum objectives need more time to be solved than the min-max objectives. – They apply a hybrid simultaneous optimization, the Imperialist Competitive Algorithm, to solve instances involving 10 to 500 aircraft. It is important to note that Zhang et al. [21] introduce two metrics to indicate the tightness of instances and test the hypothesis when the arrival flight bunch is tight and there is less flexibility for maneuvering the arrival flights.

models. So it is better to study each aspect separately. It is also noticeable that a majority of the models put more focus on the sequencing and scheduling landings decisions rather than the control problem which is virtually leaving aside and it is the responsibility of the air traffic controller. We notice that there is a lack of projects which assist air traffic controllers in establishing control actions during the approach/landing to the airport;

- As far as the theoretical models are concerned, the proposed methods are most appropriate and promise to provide satisfactory solutions. It must make clear that the majority of the methods proposed for the ALP have never been implemented in practice. The reasons for this situation could be explained, on the one hand, by the fact that developers did not properly understand the operational needs of the users (e.g., air traffic controllers). In other words, there exists a huge gap between what exists in papers in terms of dynamic techniques and what requires the real-world requirements. As a result, they have made several simplifications of the problem to make it computationally tractable (e.g., the characteristics of the airport layout are drastically simplified; certain simple scenarios are considered and many actual factors such as the weather and ATC rules are ignored; the methods which have unreasonable algorithm runtime may relax or dismiss hard (critical) operational constraints). It is further recalled that most of the related studies consider the static treatment, whereas the proposed dynamic cases are simple extensions of the main results associated with static. It is understood that a solution of the static case is not of interest from an operational perspective as the ALP should match the requirement of dynamic situation. That is all because changes in data problem (slightly) from time to time (i.e., unexpected departure delays, new aircraft, time windows, separation criteria change, etc.). On the other hand, the implementation of the proposed methods is not possible for real-time purposes. Because in situations which meet air traffic controllers in practice, the proposed algorithms cannot satisfy real-time requirements, did not meet the development trend of the airport, ignore the various stakeholders' requirements, or depend on features of a specific airport and dismissing the safety issues. There are few solutions that handle all of these concerns simultaneously;
- Based on the above observations, it is clear that the ALP has been solved in various ways. As we have seen, a variety of

exact and heuristic algorithms have been explored for the ALP on a single or multiple runways. All these approaches have proved significant improvements compared with the commonly used air traffic controller rules, the FCFS rule. We can conclude that a few optimization techniques, e.g., LP-based approaches and evolutionary algorithms, have been shown good solver for the ALP with a small number of aircraft, because their relatively huge computational cost that makes their application to the real-world difficult. We can also worth pointing out that the best results reported in the related studies are not produced by a single algorithm, but a subset of algorithms. Therefore, to date there is no universal algorithm, and existing algorithms still find low quality solutions or struggle with large problems. In fact, it seems extremely difficult to know the most appropriate algorithm to tackle the ALP with a single or multiple runways. Consequently, additional research is needed in this area. This means, methods that contain idea of optimization, and are practical when the number of flights increased, must be established. Furthermore, optimization goals need to be improved in order adequately to meet actual needs of the different stakeholders. Meanwhile, the promising approaches must take into account factors like airline gate use agreements, departure fix assignments, missed approach paths, and taxi routes;

- In the light of this we can see that, in our particular context, it is quite clear why the ALP is usually more complicated and has been studied extensively more than the ATP in the literature. It is reasonable to believe that the principal reasons are inextricably linked to the fact that numerous aircraft approaching the TMA from different directions, need to land within a short time. Remember also that a landing aircraft cannot wait in the sky. However a departing aircraft can wait at ground level with fewer risks. Moreover, it is about twice as cheap to delay an aircraft on the ground instead of in the air. It is quite clear the importance of the survey regarding the ALP. The ATP in airport RSP will be addressed in a future study. In addition, the multi-airport terminal area which refers to the terminal area including two or more airports, and arrivals and departures of these airports interrelate; will be an ongoing challenge for us.

6.2. Conclusions

An obvious underlying premise of the above is that the ATM system is becoming extremely important and extremely relevant today with the growth of air traffic congestion worldwide. Whereas it is important to note that major differences exist in this respect between ATM systems in different countries. For instance, even with an improved technology, the current ATM system in Europe has more or less stagnating that it does not allow it to respond with serenity to the user expectations in the coming years. The reasons for this are that the capacity proposed by the *Air Navigation* does not meet the demand, knowing that the available theoretical capacity is not fully exploited. In addition, the current ATM systems, infrastructure, and technology are not sophisticated enough. Therefore, underpin a sustainable and high-quality ATM system, with well-trained and experienced personnel, is a prerequisite for achieving high airport resources capacities. It is also necessary and practical to introduce for the future ATM systems, not only automated procedures to maintain adequate safety levels, but also planning functions to increase system capacity and to improve system efficiency. This might involve the reduction of the total cost, the commensurate adjustment of aviation infrastructure, the use of the existing capacity more efficiently, and the improvement of the cooperation between CDM and the ATM process.

The future ATM (e.g., Next Gen launched in the United States, SESAR launched in Europe) is based on new technologies and improved procedures that are expected to the creation of a more efficient and integrated global ATM system. The automation in ATM will have a role to play in reducing terminal inefficiencies and aviation fuel consumption. Indeed, we are facing extremely dense operations in complex TMA. For the ALP topic, we can consider the realizing of an automation for a large portion of routine operations including scheduling, sequencing, spacing, and separation, and leverage the complex problem solving of human abilities to manage recovery from off-normal events. Therefore, the role of air traffic controllers in such a future system would change from handling tasks, such as routine separation assurance and arrival sequencing, to responding to exceptional traffic situations. However, to the best of our knowledge, full automation in ATM is still a challenge due to safety consideration.

In view of the above discussion, we note that the literature investigation considers the aircraft landing operation as an important and hard issue for solving. We also believe that the investigation for a model that operates in a dynamic environment, takes into account several performance indicators, and considers more of the constraints that arise in practice (e.g., priority, overtake, and weather uncertainties) is still an open problem in the related literature. Indeed, there is a clear need to incorporate an increasing level of detail and realism in the models while keeping the computation time of algorithms at an acceptable level. The approaches need to be tailored to each airport as they depend on the runway system layout and require that have run for practical use one or two seconds for run.

Concerning the subject of security, it is necessary to ensure a high uniform level of safety in the provision of air navigation services, as well as an optimal use of the airspace. Air traffic controllers make an invisible but essential work. They aim to prevent collisions and keep the efficiency in the landing operation. We note that the security concept was neglected throughout most of the works that tried to solve the ALP. Some critical operational constraints in the modeling are often ignored, e.g., the priority of flights would be interesting to explore;

Finally, we conclude by saying that it could be fruitful to well integrate in a holistic approach the three major fields of air transport research, i.e., ATFM, airport TMA operations, and en-route airspace operations for future work, for an optimum air

transportation system performance. And, since it is difficult to describe the real-time quantitatively and the judgment of a single part is inaccurate in the TMA, it is necessary to develop a comprehensive decision making method which integrates the judgment of all parties to a fully automated terminal management system.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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